Check In and Billing System with POS Integration: A Smart Operations System for IdeaHub Co-working Space + Café

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ABSTRACT

This study developed **"Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café,"** a web-based system designed to support and centralize the operational and financial processes of IdeaHub Co-working Space + Café. The study addressed challenges associated with manual and separate processes for customer check-in and check-out, workspace and session monitoring, time-based billing, food and beverage ordering, inventory monitoring, reservations, financial records, and reporting. A developmental research design was employed, while the Agile Software Development Methodology guided the iterative development of the system. Requirements were identified through client interviews, direct observation, consultation, and review of existing operational records. The developed system integrated customer session and workspace management, Point-of-Sale (POS) ordering, automated time-based billing, inventory and menu management, space booking, checkout records, expenses, payables, receivables and debt monitoring, daily sales balancing, sales and business analytics, reporting, user management, authentication, and role-based access within a centralized web-based platform.

The system was evaluated through functional testing, integration testing, and User Acceptance Testing (UAT). Functional testing verified that the major system features produced their expected outputs, while integration testing confirmed the interaction and flow of information among interconnected modules. For UAT, the client directly tested the implemented features based on the operational requirements established during the study rather than through a quantitative survey instrument. The client accepted and approved the implemented system features after testing and feedback. The results demonstrated that the developed system fulfilled the identified functional requirements and provided an integrated and organized approach to managing IdeaHub's operational and financial information. The study therefore established the developed system as a functional solution for supporting the identified operational requirements of IdeaHub Co-working Space + Café within the defined scope of the project.

Keywords: Co-working Space, Point-of-Sale (POS), Check-In and Billing, Operations Management System, Inventory Management, Financial Management, Web-Based System.

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

Digital transformation has become an important component of modern business operations as organizations increasingly integrate digital technologies into their processes, information management, and service delivery. For small and medium-sized enterprises (SMEs), digital transformation involves not only the adoption of technology but also the alignment of information systems, organizational processes, and managerial practices with business requirements. Successful digital transformation therefore requires organizations to consider their existing capabilities, limitations, processes, and strategic objectives [1], [2].

Information systems contribute to digital transformation by supporting the collection, processing, storage, and retrieval of operational information. For SMEs, integrating digital infrastructure with organizational processes can provide a more structured approach to managing and accessing business information [2]. Rather than maintaining related activities through fragmented records and separate procedures, integrated information systems can support the coordination of operational data and provide information for managerial decision-making.

The need for effective information management is also relevant to coworking spaces, which provide flexible working environments based on shared access to physical resources and collaborative spaces. Coworking environments are characterized by resource sharing, community, collaboration, and entrepreneurial activity [3]. In the Philippine context, coworking spaces have also developed in relation to local socioeconomic conditions and emerging forms of digital labor [6]. These characteristics create an operational environment in which shared workspaces, resources, and services must be organized for different users.

The operational environment becomes more interconnected when coworking services are combined with café operations. In addition to managing workspace usage, such establishments may need to process food and beverage transactions, monitor customer sessions, maintain inventory records, manage reservations, calculate customer charges, and organize operational and financial information. Digital transformation research emphasizes that technological solutions should be aligned with existing organizational processes and business requirements rather than implemented as isolated technologies [1], [2].

IdeaHub Co-working Space + Café operates within this type of hybrid service environment by providing shared workspaces together with café-related services. Based on the researchers' initial interviews, consultations, and observations, several daily activities of the establishment relied on manual recording and separate operational procedures. Customer arrivals were recorded for time monitoring, workspace usage was tracked throughout the customer's stay, food and beverage purchases were handled separately, and customer charges were computed during check-out. Inventory and other operational records also required manual monitoring and consolidation.

These existing procedures presented challenges in customer session monitoring, workspace utilization, billing, POS transaction integration, inventory management, reservations, financial monitoring, and reporting. Manual check-in and check-out procedures made active customer sessions and workspace usage more difficult to monitor through a centralized source. The separation of café purchases from workspace sessions required transaction information to be consolidated during billing. Similarly, manual inventory monitoring limited immediate visibility of recorded stock quantities, while preparing operational and financial information required records from different activities to be reviewed and consolidated.

These challenges are relevant to the broader digitalization of SMEs. Research indicates that digital transformation involves interconnected organizational, technological, and information-system considerations [1]. The integration of digital infrastructure and organizational processes must therefore correspond to the actual needs and conditions of the organization [2]. In addition, information technology, employee capabilities, and digital strategy have been identified as factors associated with SME digitalization [9]. These findings support an approach in which technology is developed around existing operational requirements rather than introduced independently from business processes.

In response to the identified operational requirements, this study developed the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. The system was designed as a centralized web-based platform integrating customer check-in and check-out, workspace and session monitoring, automated billing, point-of-sale transactions, inventory management, reservations, financial management, sales and analytics, reporting, and user management. Through this integration, information generated from different operational activities could be maintained and processed within a common system rather than through fragmented records and separate procedures.

The development of the proposed system therefore represents a digitalization initiative designed around IdeaHub's existing operational processes. Rather than automating individual activities independently, the study sought to establish an integrated platform through which related operational and financial information could be recorded, processed, retrieved, and monitored. This approach is consistent with the principle that SME digital transformation should align technological solutions with organizational processes and specific business requirements [1], [2].

1.2 Statement of the Problem

The existing operational processes of **IdeaHub Co-working Space + Café** rely heavily on manual recording, monitoring, and computation. This approach creates several operational challenges that affect the efficiency, accuracy, accessibility, and management of business information:

1. **Inefficient Customer Session and Workspace Monitoring:** Customer check-in and check-out activities are manually recorded, making it difficult to monitor active customers, occupied workspaces, and session durations accurately and in real time.
2. **Manual and Error-Prone Billing:** Customer charges are manually computed based on workspace usage and other transactions, increasing the possibility of calculation errors and making transaction histories and billing records difficult to retrieve and monitor.
3. **Fragmented Point-of-Sale Transactions:** Food and beverage transactions are processed separately from customer session records, requiring repetitive manual recording and preventing customer purchases and workspace charges from being managed within a unified transaction process.
4. **Inefficient Inventory Monitoring:** Inventory records rely on manual updates, making stock tracking time-consuming and susceptible to inaccuracies. The lack of automated inventory updates also limits the ability of staff and management to identify stock availability and replenishment requirements efficiently.
5. **Manual Reservation Management:** Workspace and boardroom reservations are manually managed, increasing the possibility of scheduling conflicts and making it difficult to organize, track, and monitor customer bookings through a centralized process.
6. **Limited Reporting and Decision Support:** Operational, sales, inventory, and financial reports require manual preparation, increasing administrative workload and limiting management's access to timely information needed for monitoring, planning, and decision-making.
7. **Limited Security and User Management:** Existing manual and separate operational processes provide limited control over user access and data management, increasing the possibility of inconsistent records, unauthorized access, and loss of important business information.

Without an **integrated web-based management system**, IdeaHub Co-working Space + Café lacks a centralized platform capable of consolidating **customer session and workspace management, automated billing, point-of-sale transactions, inventory monitoring, reservation management, user access control, financial monitoring, and reporting**. These limitations create the need for a system that can centralize business information, automate essential processes, and provide management with more accessible and reliable operational data.

1.3 Objectives of the Study

The primary objective of this study was to **design, develop, and implement a Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** that centralized and automated the establishment's major operational processes. The system aimed to improve the efficiency, accuracy, accessibility, and management of customer sessions, billing, point-of-sale transactions, inventory, reservations, user access, financial records, and operational reports.

Specifically, this study aimed to:

1. **Automate Customer Session Management:** Develop a digital check-in and check-out process that records customer sessions and provides real-time monitoring of active customers, workspace usage, and session durations.
2. **Automate Billing Management:** Implement an automated billing process that calculates customer charges based on workspace rates and usage duration while maintaining centralized billing records and transaction histories.
3. **Integrate Point-of-Sale Transactions:** Develop a POS module that integrates food and beverage orders with customer sessions and billing records to provide a unified and efficient transaction process.
4. **Improve Inventory Management:** Develop an inventory monitoring module that tracks stock quantities and product availability and provides automated inventory updates and notifications to support replenishment planning.
5. **Centralize Reservation Management:** Implement a reservation module that organizes workspace and boardroom bookings, monitors reservation schedules, and reduces potential scheduling conflicts.
6. **Provide Reporting and Decision Support:** Generate operational, sales, inventory, and financial reports and provide management with timely business information to support monitoring, planning, and decision-making.
7. **Strengthen Security and User Management:** Implement role-based access control and centralized data management to regulate system access and protect operational and financial information according to user responsibilities.
8. **Develop an Integrated Web-Based Platform:** Integrate customer session management, billing, POS transactions, inventory monitoring, reservation management, financial monitoring, user management, and reporting into a centralized web-based system designed to improve the overall operational efficiency of IdeaHub Co-working Space + Café.

1.4 Scope and Limitations

1.4.1 Scope

The study focused on the design, development, and implementation of a **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. The system was developed as a centralized web-based platform designed to automate and integrate the establishment's major operational and financial processes.

- **Customer Session Management:** The system automated customer check-in and check-out, recorded customer information, monitored active sessions and workspace usage, tracked session duration, and maintained customer session records.
- **Billing and Transaction Management:** The system automatically calculated customer charges based on workspace rates, session duration, and additional food and beverage purchases while maintaining centralized billing and transaction records.
- **Point-of-Sale (POS) Management:** The system processed food and beverage orders and integrated customer purchases with their corresponding sessions and billing records.
- **Inventory Management:** The system maintained product and stock information, updated inventory quantities based on transactions, monitored stock availability, and provided inventory notifications to support replenishment.
- **Reservation Management:** The system manages workspace and boardroom reservations by allowing authorized users to create, update, organize, and monitor booking schedules.
- **Financial Management:** The system provided modules for **Payables, Receivables, Debt Collection, Expenses, and Daily Soft Balancing** to assist management in recording and monitoring the financial activities of the establishment.
- **Reporting and Analytics:** The system generated operational, sales, inventory, reservation, transaction, and financial reports and provided analytics to support business monitoring and decision-making.
- **User Management and Security:** The system implemented user authentication and role-based access control to regulate access to system functions based on the responsibilities of administrators and staff members.
- **Database Management:** The system utilized a centralized relational database for storing, retrieving, and managing customer, transaction, inventory, reservation, financial, and user information.
- **System Platform:** The system was developed as a **web-based application** accessible through supported web browsers on desktop and laptop computers connected to the appropriate network environment.

1.4.2 Limitations

The study was specifically designed around the internal operational requirements of **IdeaHub Co-working Space + Café**. The following functionalities were intentionally excluded from the developed system:

- **Customer Self-Service Portal:** Customers cannot independently create accounts, manage their own reservations, access transaction histories, or perform other self-service activities.
- **Native Mobile Application:** The study did not include the development of dedicated Android or iOS applications; the system was developed primarily as a web-based application.
- **Online Ordering and Delivery:** The system does not support online food ordering, delivery management, or integration with third-party food delivery platforms.
- **Online Payment Gateway:** The system does not directly integrate with electronic wallets, credit/debit card processors, online banking services, or other external payment gateways.
- **Multi-Branch Management:** The system was specifically developed for the current operations of IdeaHub Co-working Space + Café and does not provide centralized management for multiple branches or locations.
- **Advanced Predictive Analytics:** The system provides operational reports and analytics but does not include machine learning, artificial intelligence, demand forecasting, or other predictive analytics capabilities.
- **Comprehensive Supplier and Procurement Management:** Inventory monitoring is included; however, supplier evaluation, automated purchasing, purchase order processing, and comprehensive procurement management are outside the scope of the system.
- **Human Resource Management:** Payroll processing, recruitment, comprehensive employee attendance management, and other full human resource management functions are not included.
- **External Business System Integration:** The system does not directly integrate with third-party accounting, Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), or similar external business platforms.

- **Advanced Enterprise Security:** The system implements authentication, authorization, and role-based access controls; however, enterprise-level security mechanisms such as biometric authentication, advanced intrusion detection, and comprehensive security auditing are outside the scope of the study.

1.5 Definition of Terms

To facilitate a better understanding of the study, the following terms are conceptually and operationally defined as they are used in the development of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**.

- **Administrator:** Refers to the authorized user responsible for managing staff accounts, monitoring system operations, accessing administrative functions, and generating reports.
- **Authentication:** Refers to the process of verifying the identity of a user through valid login credentials before access to the system is granted.
- **Automated Reporting:** Refers to the system's capability to generate organized operational, sales, inventory, reservation, transaction, and financial reports using stored system data.
- **Billing Management:** Refers to the automated calculation and recording of customer charges based on workspace usage duration, applicable rates, and additional food and beverage purchases.
- **Business Process Automation:** Refers to the use of the developed system to automate repetitive operational activities that were previously performed manually within IdeaHub.
- **Centralized Database:** Refers to the unified data repository used by the system to store and manage customer sessions, transactions, inventory records, reservations, financial information, and user accounts.
- **Check-In:** Refers to the process of registering a customer upon arrival, assigning the selected workspace, and recording the starting time of the customer's session.
- **Check-Out:** Refers to the process of ending an active customer session, recording the time-out, and initiating the calculation of the customer's final charges.
- **Customer Session:** Refers to the recorded period between a customer's check-in and check-out while using the services or workspace facilities of IdeaHub.
- **Daily Soft Balancing:** Refers to the system functionality used to review and reconcile recorded daily sales, payments, expenses, and other financial activities to assist management in identifying discrepancies and verifying daily financial records.
- **Dashboard:** Refers to the centralized system interface that presents operational information, summaries, statistics, and access to the major functions of the system.
- **Data Integrity:** Refers to the accuracy, consistency, and reliability of information stored and processed within the system.
- **Debt Collection:** Refers to the system functionality used to record, monitor, and manage outstanding debts, payments made, and remaining balances.
- **Inventory Management:** Refers to the system functionality used to record products, monitor stock quantities, update inventory levels, and identify stock availability and replenishment requirements.
- **Operational Efficiency:** Refers to the ability of IdeaHub to perform its daily business activities with reduced manual effort, processing time, and repetitive record-keeping through the use of the developed system.
- **Payables:** Refers to the financial obligations or amounts owed by IdeaHub to suppliers, service providers, or other entities that are recorded and monitored through the system.
- **Point-of-Sale (POS):** Refers to the system component used to record and process food and beverage purchases made by customers.

- **POS Integration:** Refers to the integration of food and beverage transactions with customer sessions, billing, inventory, and transaction records within the developed system.
- **Real-Time Monitoring:** Refers to the capability of the system to display updated operational information, such as active customer sessions, workspace usage, and other recorded activities as they occur.
- **Receivables:** Refers to amounts owed to IdeaHub that are recorded and monitored through the system until they are collected or settled.
- **Reservation Management:** Refers to the system functionality used to create, organize, update, and monitor workspace and boardroom reservations.
- **Role-Based Access Control:** Refers to the security mechanism used to provide or restrict access to system functions according to the assigned role and responsibilities of each authorized user.
- **Sales and Analytics:** Refers to the system functionality that organizes and presents sales and transaction information through summaries and visual representations to support business monitoring and decision-making.
- **Session Duration:** Refers to the total amount of time recorded between a customer's check-in and check-out and used as the basis for calculating workspace usage charges.
- **Smart Operations Management System:** Refers to the developed integrated web-based system designed specifically for IdeaHub Co-working Space + Café to manage customer sessions, billing, POS transactions, inventory, reservations, financial records, users, analytics, and reports.
- **Staff:** Refers to authorized IdeaHub personnel who use the system to perform assigned operational activities, such as managing customers, processing orders, handling transactions, and monitoring business activities.
- **System Integration:** Refers to the connection of different system modules to allow information to be shared and processed across customer sessions, POS transactions, inventory, billing, financial management, and reporting functions.
- **Transaction:** Refers to a recorded business activity involving workspace usage, customer purchases, billing, payments, or other financial activities processed through the system.
- **User Interface:** Refers to the visual components, pages, controls, forms, and navigation elements through which administrators and staff interact with the system.
- **User Management:** Refers to the administration of system user accounts, roles, access privileges, and account status.
- **Web-Based Application:** Refers to a software application that is accessed and operated through a supported web browser rather than requiring a separately installed desktop application.
- **Workspace:** Refers to the designated areas offered by IdeaHub for customer use, including the **Regular Lounge, Premium Lounge, and Boardroom**.
- **Workspace Utilization:** Refers to the monitoring of the occupancy, usage, and duration of use of the available workspaces within IdeaHub.

1.6 Significance of the Study

The development of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** provided an integrated approach to improving the establishment's operational processes. The system centralized customer session management, billing, point-of-sale transactions, inventory monitoring, reservations, financial management, and reporting to improve operational efficiency, data accuracy, and accessibility of business information.

The study was significant to the following stakeholders:

- **IdeaHub Management:** The system provided management with centralized access to operational, sales, inventory, reservation, and financial information. Real-time monitoring and automated reporting supported more efficient business supervision, planning, and data-driven decision-making.
- **Staff and Employees:** The system reduced repetitive manual tasks associated with customer check-in and check-out, billing computation, food and beverage transactions, inventory monitoring, reservations, and record management. This allowed staff to perform daily operations more efficiently while minimizing errors caused by manual processing.
- **Customers:** Customers benefited indirectly through faster transaction processing, accurate billing, organized reservation management, and improved service coordination. Automated session tracking also supported accurate computation of workspace usage charges.
- **IdeaHub Business Operations:** The integration of major operational functions into a centralized platform reduced fragmented record-keeping, redundant processes, and manual documentation. The system improved the consistency and accessibility of business records and supported more organized daily operations.
- **Information Technology Students and System Developers:** The study provided a practical reference for the application of web development, database management, system integration, role-based access control, business process automation, and software development concepts in addressing actual organizational requirements.
- **Future Researchers:** The study may serve as a reference for future research involving coworking space management, POS integration, inventory systems, reservation systems, financial monitoring, and business process automation. Its system design, development methodology, and findings may provide a basis for further studies and system enhancements.
- **Academic and Information Technology Community:** The study demonstrated the application of information technology to the digitalization of operational processes within a small service-oriented business. Research on SME digital transformation emphasizes the relationship between digital technologies, organizational processes, information systems, and business requirements [1]. The developed system provided an applied example of these concepts through the integration of multiple operational functions within a centralized information system.

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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of related literature and studies concerning information systems, business process automation, point-of-sale (POS) systems, billing systems, inventory management, reservation management, coworking spaces, financial reporting, usability, and digital transformation. These sources provide the theoretical and practical foundation for the development of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café.

The review focuses on studies that demonstrate how information technology can be used to organize business activities, reduce dependence on manual procedures, improve transaction accuracy, monitor resources, and provide timely information for operational decision-making. Particular attention is given to systems that combine two or more business functions because the proposed IdeaHub system is intended to connect customer session management, time-based billing, POS transactions, inventory management, reservation management, financial monitoring, and reporting.

The reviewed materials are organized into foreign studies, local studies, and related literature. Foreign studies provide broader evidence concerning information systems, digital transformation, coworking environments, POS technology, and system success. Local studies provide Philippine examples of computerized business systems and are particularly useful because they demonstrate how similar technologies have been applied in local organizations. Related literature provides the conceptual and theoretical foundation for understanding why integration, automation, information quality, usability, and digital process improvement are important to the proposed system.

The final sections synthesize the findings of the reviewed sources and identify the research gap addressed by the present study. The purpose of the synthesis is not to claim that the individual technologies used by IdeaHub are entirely new, but to determine how previous research supports their application and integration within the specific operational environment of a coworking space combined with café services.

2.1 Foreign Studies

2.1.1 Point-of-Sale and Billing System Efficiency

Adityo and Miawarni [4] developed a billing Point-of-Sale (POS) system for restaurant operations that incorporated cloud computing and Internet of Things (IoT) technology. The study addressed the need to synchronize transaction information with a centralized reporting process. The system used devices such as the Raspberry Pi and cloud infrastructure to transmit transaction information so that sales data could be monitored and processed more efficiently. The study therefore treated POS not only as a tool for recording customer purchases but also as a source of information for reporting and monitoring business transactions.

The importance of the study is found in its demonstration that transaction processing can be connected with other information-management functions. In a conventional manual environment, a transaction may first be written on a receipt or record and then separately consolidated for reporting. Such separation can create additional work and may introduce opportunities for inconsistent or delayed records. By automating the transfer of transaction information, the system studied by Adityo and Miawarni [4] illustrated how digital POS technology can reduce the distance between the point where a transaction occurs and the point where information about that transaction is used for monitoring and reporting.

The study is relevant to IdeaHub because café transactions are one of the major sources of operational data in the proposed system. When a customer purchases food or beverages, the transaction needs to be recorded accurately and later included in the customer's final billing information and in the business's sales records. The proposed system follows the general principle of connecting transaction processing with reporting, although it does not use the same IoT and cloud architecture. Instead, IdeaHub connects POS transactions with customer sessions, billing, inventory records, and operational reports within a web-based system.

A major difference is therefore the scope and architecture of the systems. The study of Adityo and Miawarni [4] concentrated on restaurant billing and real-time transmission of sales information, while the proposed IdeaHub system must handle two related service environments at the same time. The customer may consume a workspace for a period of time and may also purchase café products during that session. Consequently, the proposed system needs to coordinate service-duration information and product transactions before producing the final bill. The foreign study provides support for the POS and reporting components, while the present study adapts the concept to a hybrid coworking-space-and-café environment.

2.1.3 Integrated Information System for a Coworking Space

Putra, Pamudji, and Kartin developed an Integrated Accounting Information System with Point-of-Sale (POS) and online booking for a coworking space [12]. The study was conducted for a coworking-space company that provided event spaces, meeting rooms, private offices, and coworking spaces. The researchers identified difficulties in recording sales transactions and managing financial information because the existing free POS system had limitations and accounting activities were partly performed using manual or semi-computerized procedures.

The researchers proposed an Accounting Information System that integrated POS and online booking. They used the Rapid Application Development (RAD) method and evaluated the resulting system through User Acceptance Testing using the Blackbox method. Interviews with front-desk personnel, finance staff, and customers were also used to determine whether the system addressed user requirements. The system was intended to reduce transaction-recording errors, accelerate financial-report preparation, and improve operational management while also providing online booking functionality.

The reported results are particularly relevant because the system demonstrated improvements in several areas that are also important to IdeaHub. The authors reported that the developed system minimized inaccuracies in transaction recording, shortened the preparation of financial reports, improved operational efficiency, and enhanced customer experience through online booking [12]. The study also demonstrated the value of User Acceptance Testing in determining whether a system's functions correspond to actual

user needs.

The study is one of the closest foreign studies to the present research because both systems are designed for coworking-space operations and combine POS-related functions with other management activities. It provides strong evidence that an integrated system can be more useful than isolated manual or semi-computerized procedures when transaction and financial information must be coordinated.

Despite this similarity, the proposed IdeaHub system has a different operational scope. The present system focuses on customer check-in and check-out, time-based workspace billing, café POS transactions, inventory management, boardroom reservation management, financial monitoring, and reporting. The customer-session component is particularly important because the amount charged for workspace use depends on the duration of the customer's stay. Therefore, the proposed system extends the general integration principle demonstrated by Putra, Pamudji, and Kartin [12] to a setting in which workspace usage and café purchases are combined in a single customer transaction.

2.1.4 Coworking Spaces and Shared Work Environments

Bouncken and Reuschl examined coworking spaces as a development associated with the sharing economy and new forms of work [3]. Their work explains coworking spaces as environments where individuals and organizations can share physical resources while maintaining flexible work arrangements. Coworking therefore differs from a traditional office because users can access shared facilities without necessarily being permanent employees of one organization.

The significance of coworking research to the present study is that a management system must reflect the operational characteristics of the environment it serves. A coworking space may have multiple users occupying different spaces for different periods, and the establishment may offer different workspace types or services. This creates a need for accurate records of customer usage, workspace availability, reservations, and transactions.

For IdeaHub, the coworking environment is further combined with café services. Customers may enter the establishment primarily to study, work, meet, or use the available facilities, while also purchasing food and beverages. This creates an operational relationship between space utilization and product sales. The literature on coworking therefore supports the need to treat workspace use as an important business process rather than viewing the café and coworking services as completely independent operations.

However, the coworking literature does not by itself provide a complete solution for transaction processing or billing. It explains the nature of shared work environments and their operational requirements, but additional research concerning POS, billing, inventory, reporting, and information systems is necessary. The present study combines these areas in order to create a system suited to the specific hybrid environment of IdeaHub.

2.1.5 Inventory Management and Digitalization

Fang, Zhou, Jiang, and Li examined the relationship between digital transformation and inventory management and discussed how digital development can influence inventory-related practices [7]. Their

research emphasizes that digital technologies can change the way organizations obtain, process, and use information related to inventory and operational resources. This is especially relevant to small and medium-sized enterprises because inventory information must often be managed alongside limited personnel and financial resources.

Inventory management is not limited to counting products. It involves maintaining information about available quantities, stock movements, purchases, sales, and the timing of replenishment. When inventory information is updated manually, the recorded quantity may not always represent the actual quantity available for sale or consumption. Digital systems can help by providing a consistent location for storing inventory records and by linking stock information with operational transactions.

The relevance of this study to IdeaHub is particularly clear in the café component. Food and beverage items are sold or consumed as part of normal business operations, meaning that product availability can change throughout the day. A POS transaction can therefore provide information that is useful for inventory monitoring. In the proposed system, the inventory module is designed to work alongside POS transactions so that management can monitor product records and identify stock conditions more easily.

The present study does not claim that digital inventory management alone guarantees accurate stock levels. Actual accuracy also depends on proper encoding, correct transaction recording, appropriate stock adjustments, and user compliance. The literature instead supports the broader principle that digital inventory information can improve visibility and provide a more organized basis for monitoring and decision-making [7].

2.1.6 Information-System Success and Quality

DeLone and McLean proposed an updated model of information-systems success that identifies system quality, information quality, service quality, system use, user satisfaction, and net benefits as important dimensions of information-system success [5]. The model is relevant to the present study because a business information system cannot be evaluated solely by determining whether its buttons, forms, and database functions operate correctly.

System quality concerns characteristics of the system itself, while information quality concerns the quality of the information produced by the system. User satisfaction and system use consider the relationship between the system and its intended users. Finally, net benefits consider the benefits that result from the use of the system. These dimensions provide a useful framework for thinking about IdeaHub because the proposed system must both perform its intended functions and provide information that staff and management can use in their daily activities.

For example, a billing module may technically calculate a total, but the system would still provide limited value if the result is difficult to understand, cannot be connected to the customer's session, or does not contribute to the business's transaction records. Similarly, an inventory module may store product quantities, but its usefulness is reduced if users cannot easily update or interpret the records.

The model therefore supports the evaluation approach of the present study by encouraging attention to functionality, usability, information accuracy, and operational benefit. The model does not establish that the proposed IdeaHub system will automatically achieve a particular level of success. Instead, it provides a

theoretical basis for evaluating whether the developed system satisfies its intended users and operational requirements.

2.1.7 Technology Acceptance and User-Centered Design

Davis introduced the Technology Acceptance Model (TAM), which explains user acceptance of information technology through concepts including perceived usefulness and perceived ease of use [8]. Perceived usefulness refers to the degree to which a person believes that using a system will improve job performance, while perceived ease of use refers to the degree to which a person believes that using the system will require minimal effort.

These concepts are important for the proposed IdeaHub system because the system is intended primarily for staff and management who must perform operational tasks efficiently. A technically capable system may still be difficult to adopt if common activities require unnecessary steps or if important information is difficult to locate. Conversely, a system that presents relevant information clearly and supports familiar business processes may be easier for users to incorporate into daily operations.

The present study therefore considers the relationship between system functions and actual operational tasks. Customer check-in, session monitoring, order entry, billing, inventory management, reservations, and reporting are organized according to the workflow of IdeaHub rather than as unrelated software features. This user-centered approach is intended to reduce unnecessary duplication and make the system more understandable to the people who will use it.

The TAM framework does not mean that usability is the only determinant of successful implementation. User training, organizational practices, system reliability, and the quality of stored information may also influence adoption. For this reason, the proposed system is evaluated through its functional performance and user acceptance rather than relying on a theoretical model alone.

2.1.8 Digital Transformation of Small and Medium-Sized Enterprises

Eller, Alford, Kallmünzer, and Peters examined the antecedents, consequences, and challenges of small and medium-sized enterprise digitalization [9]. Their research considered factors associated with the digitalization of SMEs and emphasized the importance of technological resources, employee capabilities, and organizational conditions. The study also connected digitalization with business outcomes, showing that digital transformation should be considered as an organizational process rather than simply the installation of software.

Sagala and Őri [1] conducted a systematic literature review concerning successful SME digital transformation. Their review identifies factors related to digital infrastructure, organizational processes, management, and the broader conditions required for digital transformation. Similarly, Ben Slimane, Coeurderoy, and Mhenni [2] reviewed SME digital transformation and emphasized the importance of organizational and managerial mechanisms in addition to technology.

These studies are relevant to IdeaHub because the proposed system is intended to improve existing business processes rather than introduce technology without regard to the organization's actual needs. The system was designed around operational activities such as customer monitoring, billing, POS transactions, inventory, reservations, and reporting. This approach is consistent with the broader concept of digital

transformation, where technology is used to change and improve existing processes.

The reviewed digital-transformation literature also provides an important limitation for the present study. Developing a system does not by itself mean that an organization has completed digital transformation. Sustainable transformation requires appropriate processes, users, management support, and continued adaptation. Therefore, the proposed IdeaHub system should be understood as a technology intervention intended to support operational digitalization rather than as a claim that the business has fully transformed all of its activities.

2.2 Local Studies

2.2.1 Coworking Spaces in the Philippine Context

Tintiangko and Soriano examined coworking spaces in the Global South and focused particularly on their local articulation in the Philippine context [6]. Their research examined coworking spaces in Metro Manila in relation to digital labor, emerging knowledge economies, and Filipino online freelancers. Through a multi-sited ethnographic approach, the researchers explored how coworking environments are organized and experienced by their users.

The study is important to the present research because it establishes that coworking spaces in the Philippines should be understood within their local social and economic environment. The way users interact with coworking spaces is influenced by the type of work they perform, their need for flexible facilities, and the services provided by the establishment. This provides context for IdeaHub, which serves students, professionals, remote workers, and other customers who require a shared working or studying environment.

The study also helps distinguish the present research from purely technical research. Tintiangko and Soriano [6] were primarily concerned with coworking as a social and economic phenomenon, whereas the present study examines how information technology can support the operational management of a coworking business. The two perspectives complement one another: understanding the local nature of coworking helps establish the business context, while the proposed system addresses the information-management needs that arise from operating such an environment.

The limitation of the study in relation to IdeaHub is that it does not develop a POS, billing, inventory, or customer-session management system. Therefore, it cannot directly provide a technical model for the proposed application. Its contribution is instead the establishment of a Philippine coworking context within which an operations-management system can be considered relevant.

2.2.2 Technology Infrastructure for Collaborative Workspace Businesses

Buladaco, Necio, Pilongo, and Timosan proposed an ideal network design for collaborative workspace businesses in the Philippines [10]. The study recognized collaborative workspaces as an emerging business model and focused on the technological infrastructure required to support such environments. Because coworking spaces depend heavily on connectivity and shared digital services, reliable network infrastructure is an important part of their operations.

The proposed network design illustrates that the technology requirements of a collaborative workspace extend beyond the physical work areas. Customers may require network connectivity for work, study, communication, and access to online services. At the same time, the establishment may require technology for internal management, monitoring, and service delivery. Reliable infrastructure therefore contributes to the overall ability of the organization to provide consistent service.

The study is relevant to IdeaHub because the proposed information system will operate within the same type of technology-dependent environment. However, network infrastructure and application software serve different purposes. The study of Buladaco et al. [10] concentrates on network architecture, whereas IdeaHub's proposed system operates at the application level by managing customer sessions, billing, POS transactions, inventory, reservations, users, and reports.

The relationship between the two studies is therefore complementary. A reliable network can provide the technical environment through which a web-based management system is accessed, while the management system provides the operational functions needed to process and organize business information. The present study contributes at the application level by focusing on operational integration.

2.2.3 Computerized Restaurant Sales, Billing, and Inventory Management

Chicano and Huervana developed a Computerized Restaurant Sales and Stock Monitoring System for a selected specialty restaurant in Iloilo City [13]. The researchers examined a business that relied on manual operations and identified manual billing and unmonitored stocks as major day-to-day problems. The study also considered the advantages of POS technology and the importance of stock monitoring to restaurant operations.

The researchers gathered information by interviewing the restaurant owner and managers and studying the existing system. Based on their observations, they designed a computerized POS and stock-monitoring system intended to address the identified operational problems. The proposed system also considered security and privacy of encoded data, demonstrating that the researchers did not treat transaction automation as merely a calculation problem.

The study is highly relevant to IdeaHub because the café component of the proposed system involves many of the same operational activities. Café employees need to record customer purchases, calculate charges, monitor product availability, and maintain transaction records. If these activities are handled through separate manual records, staff may have to repeat information or reconcile records at a later time.

Chicano and Huervana's [13] study therefore supports the use of POS and inventory functions as integrated components of café operations. The study also provides a local example from Iloilo City, which

makes it particularly relevant to the geographical context of the present research.

Nevertheless, the scope differs from IdeaHub. The earlier study was designed for a restaurant, while IdeaHub combines restaurant-like café services with coworking-space services. In IdeaHub, the customer may incur a workspace charge based on session duration in addition to food and beverage charges. The proposed system therefore extends the transaction-management concept by connecting café sales with customer-session and workspace billing information.

2.2.4 Inventory, Point-of-Sale, and Sales Reporting

Bernasol, Comprendio, Ganancial, Henson, and San Felix developed the Tinukib Inventory Management System with Online Profiling of Supplier's Information at Central Philippine University [14]. The system was designed for Tinukib Café and Souvenirs and included a supplier-record module, inventory management, point-of-sale transactions, sales reports, product statistics, and an online module for product advertising.

The inventory module was intended to manage product stocks, returned items, and product sales. The POS module handled transactions and sales, while the sales-report module stored recorded sales information. The statistics module provided information concerning saleable products during a specified period. The researchers used an iterative life-cycle model involving analysis, specification, design and development, coding, and testing [14].

The reported results indicated that the system helped management organize product inventory and keep sales reports on track. Suppliers could also view product inventory and order requests through the web module. The study consequently demonstrates the benefit of connecting inventory information with sales and transaction processing rather than treating stock records as an entirely separate activity.

The relationship to IdeaHub is direct because IdeaHub also requires inventory and POS functions for its café operations. Product information, sales transactions, and inventory quantities are related data that can provide management with a clearer picture of daily operations when they are maintained within a coordinated system.

At the same time, the proposed IdeaHub system differs in its purpose and scope. The Tinukib system included supplier profiling and online product advertising, while IdeaHub emphasizes customer-session management, workspace billing, reservations, financial monitoring, and integrated checkout. Thus, the study supports the POS and inventory components while the present research adds workspace-oriented operations.

2.2.4 Workspace Reservation, Billing, Payment, and Monitoring

Cang, Gemanga, Macahilig, Pauya, and Taunan developed a Building Management Information System at Central Philippine University [15]. The system was designed to manage several building-related activities, including registration, billing, payments, report generation, account monitoring, room reservation, maintenance records, and tenant monitoring.

The study used a prototyping model consisting of planning, prototyping, design, implementation, testing, and maintenance. The system included a billing module that computed power consumption and other charges, a payment module for processing transactions, and a report-generation module for producing summaries. It also provided account monitoring for financial information and tenant monitoring for details and transaction history [15].

This study is relevant because it demonstrates how information systems can coordinate the management of physical spaces with financial transactions. Although a building-management system is different from a coworking-space system, both environments require records of people using physical facilities, charges associated with those facilities, payments, and historical transactions.

The reservation function is particularly relevant to IdeaHub's boardroom and workspace services. A reservation record must identify the customer, the facility, and the intended period of use. Billing information can then be associated with the relevant service. The proposed IdeaHub system follows the general principle demonstrated by Cang et al. [15], while adapting it to short-term coworking sessions and workspace rates.

The difference is that the proposed system calculates workspace charges according to session duration and combines those charges with café POS transactions. The building-management study instead addresses tenant-oriented charges such as rentals, utilities, maintenance, and power consumption. Its contribution to the present study is therefore the demonstrated relationship among registration, reservation, billing, payment, monitoring, and reporting.

2.2.5 Usability of Centralized POS and Inventory Systems

Bienvenido examined the usability of a centralized Point-of-Sale system used for inventory management in the rotisserie operation of Bounty Agro Ventures, Inc. [16]. The study was conducted as an undergraduate thesis at the University of the Philippines Los Baños and focused on the adoption of a centralized POS system in response to challenges associated with manual inventory management and inventory inaccuracies.

The study is relevant because centralization changes the way transaction information and inventory information are handled. When sales and stock information are maintained within a coordinated system, employees can use the same operational source when performing related tasks. However, the usability of the system remains important because employees must be able to interact with the POS and inventory functions correctly for the resulting records to be useful.

For IdeaHub, usability is particularly important because staff may perform several tasks during a single customer interaction. A staff member may need to register a customer, select a workspace, monitor the session, enter café orders, process payment, and complete checkout. A system that technically contains all

of these functions but presents them in a confusing workflow may increase rather than reduce the operational burden.

The study therefore supports the inclusion of usability considerations in the development and evaluation of IdeaHub. The present study differs because its POS and inventory functions are integrated with coworking-space operations rather than being limited to a retail or food-service environment.

2.3 Related Literature

2.3.1 Point-of-Sale and Billing Systems

Point-of-Sale systems are designed to record and process transactions involving products or services. In a conventional business environment, a POS system can serve as the point where information about a customer purchase is entered, a total is calculated, and a transaction record is produced. When the POS is connected with other business modules, transaction data can also become an input for inventory updates, sales reports, financial monitoring, and other management activities.

The literature reviewed in this study demonstrates that the usefulness of POS systems increases when transaction information is not isolated from other operational records. Adityo and Miawarni [4], for example, connected restaurant billing with centralized reporting. Local studies by Chicano and Huervana [13] and Bernasol et al. [14] similarly demonstrate the relationship between POS transactions and inventory or sales reporting.

For IdeaHub, POS is therefore treated as one component of an integrated system rather than a standalone cashier application. Café orders contribute to the customer's final bill and provide information for sales and inventory monitoring. This relationship allows one transaction to serve several operational purposes without requiring staff to manually reproduce the same information in multiple records.

2.3.2 Coworking Space Management

Coworking-space management involves the coordination of shared physical resources, customers, services, and usage schedules. Research on coworking spaces describes them as flexible environments that allow different users to share resources and facilities [3]. In the Philippine context, Tintiangko and Soriano [6] examined coworking spaces in relation to local digital labor and the needs of Filipino users.

From an operational perspective, the shared nature of coworking means that management must know which customers are using which facilities and for what period. This is particularly important when services are priced according to time or when certain facilities require advance reservations. Without accurate session and reservation records, management may have difficulty determining current occupancy and calculating customer charges.

The proposed IdeaHub system translates these operational requirements into customer-session management, workspace monitoring, check-in and check-out, and reservation functions. The system consequently treats coworking management as an information-management problem as well as a physical-space-management activity.

2.3.3 Inventory Management Systems

Inventory management concerns the monitoring of products and stock movements so that an organization can maintain adequate resources for its operations. For a café, inventory may include food ingredients, prepared products, beverages, packaging materials, and other items required for daily service.

Digital inventory systems can centralize product information and provide a more organized method for recording changes in stock. When inventory is linked to sales transactions, the organization can use transaction information to support stock monitoring. The literature concerning digitalization and inventory management supports the broader role of information technology in improving the visibility and coordination of inventory-related information [7].

Local studies provide practical examples of this principle. Chicano and Huervana [13] developed a restaurant POS and stock-monitoring system, while Bernasol et al. [14] incorporated inventory, POS, sales reporting, and product statistics into a single system. These studies provide support for the proposed integration of IdeaHub's café transactions and inventory records.

2.3.4 Reservation and Workspace Management

Reservation management involves recording, organizing, updating, and monitoring advance requests for facilities or services. Within a shared workspace, reservations can help management allocate rooms or facilities according to expected customer demand. Reservation information can also become part of a customer's operational record and may later be associated with billing or payment information.

Cang et al. [15] demonstrate the usefulness of combining reservation with registration, billing, payment, and monitoring within a building-management system. Putra, Pamudji, and Kartin [12] likewise incorporated online booking into an accounting and POS system for a coworking-space company.

The proposed IdeaHub system applies this concept to boardroom reservation and workspace use. The purpose is not simply to record that a reservation exists, but to provide management with organized information that can be used when monitoring facility availability and processing related transactions.

2.3.5 Financial Information and Reporting

Financial reporting systems collect and organize transaction information so that management can evaluate business activity. Reports can summarize sales, expenses, receivables, payables, collections, and other financial information. The usefulness of such reports depends heavily on the quality of the underlying transaction records.

DeLone and McLean [5] identify information quality as an important component of information-system success. This concept is relevant to financial reporting because reports generated from incomplete or inaccurate records can provide misleading information. Digital transformation research also emphasizes the importance of aligning technology and organizational processes so that digital tools contribute to actual business objectives [1], [2], [9].

The proposed IdeaHub system includes sales and analytics, reports, payables, receivables, debt collection, and daily soft-balancing functions as part of its operational management scope. These functions are

intended to organize information generated by daily transactions. They should not be interpreted as a replacement for a full professional accounting system; rather, they provide operational information that management can use for monitoring and decision-making.

2.3.6 User-Centered Information Systems

Information systems are used by people, and therefore system design must consider how users understand and interact with the available functions. Davis [8] explains perceived usefulness and perceived ease of use as important concepts in technology acceptance. DeLone and McLean [5] similarly include use and user satisfaction among the dimensions associated with information-system success.

For IdeaHub, user-centered design is relevant because the staff members who operate the system are responsible for time-sensitive activities. Customer check-in and checkout, order entry, billing, and payment processing occur during actual business operations. Excessive steps or unclear information may slow service even if the underlying database and calculations are correct.

The proposed system therefore organizes related functions according to the expected operational flow. Customer information is recorded during check-in, session information is monitored while the customer uses the facility, orders can be associated with the session, and final charges can be consolidated during checkout. This workflow-oriented approach is intended to make the system correspond more closely to actual staff activities.

2.3.7 Digital Transformation and Business Process Integration

Digital transformation involves changes in how organizations use technology, information, people, and processes to improve business activities. The literature reviewed in this study indicates that digital transformation should not be reduced to the simple conversion of paper records into digital records.

Ben Slimane et al. [2] and Sagala and Őri [1] emphasize organizational and process considerations in SME digital transformation. Eller et al. [9] likewise identify relationships among information technology, employee capabilities, digital strategy, and organizational digitalization. These studies support the idea that a system should be aligned with the organization's actual operational conditions.

The proposed IdeaHub system applies this principle by integrating activities that are already part of the establishment's operations. Instead of creating separate applications for customer sessions, billing, POS, inventory, reservations, and reports, the proposed system brings these functions into one platform. The integration is intended to reduce repeated data entry, improve information consistency, and make operational information more accessible.

System integration is particularly important because the functions of a hybrid coworking-space-and-café business are interconnected. A customer's workspace session affects the time-based charge. The same customer may place food orders that affect the total bill and inventory. A reservation may affect workspace availability. Sales transactions may contribute to daily reports. These relationships justify an integrated approach rather than isolated modules.

2.4 Synthesis of Reviewed Literature and Studies

The reviewed foreign and local studies collectively demonstrate that information systems can improve business operations by organizing transactions, automating repetitive procedures, connecting related records, and providing information for decision-making. However, the specific value of each system depends on the operational environment in which it is implemented.

The POS-related studies demonstrate the value of computerized transaction processing. Adityo and Miawarni [4] connected restaurant billing with centralized reporting, while Chicano and Huervana [13] addressed manual restaurant billing and stock monitoring through a computerized POS and stock system. Bernasol et al. [14] extended this concept by combining inventory, POS transactions, sales reports, statistics, and supplier information. These studies indicate that transaction processing becomes more useful when it is connected with other business records.

The coworking-related studies provide a different but complementary perspective. Bouncken and Reuschl [3] describe coworking as a shared and flexible work environment, while Tintiangko and Soriano [6] demonstrate how coworking spaces are expressed within the Philippine context. Buladaco et al. [10] address the technological infrastructure required by collaborative workspace businesses. Together, these studies establish the operational and technological context in which a coworking-management system can be applied.

Studies involving reservation, billing, and physical-space management further demonstrate the usefulness of centralized information. Cang et al. [15] integrated registration, reservation, billing, payment, account monitoring, and reporting in a building-management environment. Putra, Pamudji, and Kartin [12] developed a coworking-space system that integrated accounting, POS, and online booking. These studies provide particularly strong support for the idea that multiple business processes can be coordinated within one information system.

The reviewed literature also emphasizes the importance of information quality and users. DeLone and McLean [5] identify system quality, information quality, use, user satisfaction, and net benefits as important dimensions of information-system success. Davis [8] emphasizes perceived usefulness and ease of use. These concepts support the inclusion of usability, accuracy, and user acceptance considerations in the development and evaluation of IdeaHub's system.

Digital-transformation studies provide a broader organizational perspective. Sagala and Őri [1], Ben Slimane et al. [2], and Eller et al. [9] demonstrate that digitalization is influenced by organizational processes, technological resources, management, and employee capabilities. This supports the approach of developing the IdeaHub system according to actual operational requirements rather than simply adding technology for its own sake.

The local studies are particularly useful because they demonstrate that the concepts discussed in the foreign literature can also be applied within Philippine organizations. The studies conducted at Central Philippine University provide examples of computerized systems for restaurant sales, stock monitoring, inventory, POS, sales reporting, building management, reservation, billing, and payment [13]–[15]. The University of the Philippines Los Baños study further demonstrates the importance of usability when centralized POS technology is applied to inventory management [16].

Despite these contributions, the studies differ in scope. Some focus on restaurants, others on inventory management, building management, coworking environments, or accounting and POS systems. Even when two or more functions are integrated, the operational environment is different from IdeaHub's combination of coworking services and café operations.

The reviewed literature therefore supports the proposed system from several complementary perspectives. Coworking research supports the business environment; POS research supports transaction processing; inventory research supports stock monitoring; reservation research supports facility scheduling; financial-reporting research supports operational monitoring; information-system success research supports quality and evaluation; and digital-transformation research supports the integration of technology with organizational processes.

2.5 Research Gap

The reviewed foreign and local studies demonstrate that computerized information systems can improve specific business activities such as POS transactions, billing, inventory management, reservation management, coworking operations, and financial reporting. The literature also demonstrates that some systems can integrate multiple functions. Therefore, the research gap of the present study should not be stated as a lack of previous work on POS, billing, inventory, or coworking systems individually.

Instead, the gap concerns the specific combination and adaptation of these functions within a hybrid coworking-space-and-café environment. Existing coworking studies provide knowledge about shared work environments and their users [3], [6], while Buladaco et al. [10] address infrastructure for collaborative workspace businesses. Restaurant and inventory studies address POS, billing, sales, and stock monitoring [13], [14]. Building-management research addresses reservation, billing, payment, and monitoring [15]. Putra, Pamudji, and Kartin [12] provide particularly relevant evidence by integrating accounting, POS, and online booking in a coworking-space context.

However, the operational requirements of IdeaHub combine several of these areas simultaneously. A customer may check into a workspace, remain for a period of time, purchase food or beverages, and then settle a final bill that includes both workspace usage and café transactions. At the same time, the business needs to monitor inventory, reservations, payments, sales, and operational reports. These processes are related through the customer's session and therefore benefit from being coordinated within a centralized system.

The gap is therefore not that previous researchers have failed to develop integrated systems. Rather, there is limited attention in the reviewed literature to a system specifically adapted to the combined operational requirements of a coworking space and café in which time-based workspace charges and product transactions must be managed together. This distinction is important because the billing logic, workflow, and information requirements of a hybrid establishment differ from those of a conventional restaurant, retail store, standalone coworking space, or building-management environment.

The proposed study addresses this gap through the development of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. The system integrates customer check-in and check-out, workspace and session monitoring, time-based billing, POS transactions, inventory management, boardroom reservation management, financial monitoring, sales and

analytics, reporting, and user management.

The contribution of the study is therefore the adaptation and integration of established information-system functions to the operational requirements of a specific hybrid business environment. The system does not claim that POS, billing, inventory, reservation, or reporting technologies are individually new. Instead, it seeks to demonstrate how these functions can be coordinated through one platform to reduce manual work, improve transaction and operational information, and support management of a coworking space combined with café services.

This research gap also provides a basis for the system development methodology presented in the succeeding chapter. The identified functions and relationships among them guide the analysis of requirements, system architecture, database design, interface development, testing, and evaluation. In this way, the reviewed literature and studies do not merely provide background information but also establish the rationale for the structure and scope of the proposed IdeaHub system.

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CHAPTER III

METHODOLOGY

This chapter presents the methodology employed in the design, development, and evaluation of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. The study adopted a developmental research approach focused on developing an information system that addressed the operational requirements identified within the establishment.

The methodology involved analyzing existing processes, including customer check-in and check-out, workspace and session monitoring, billing, point-of-sale (POS) transactions, inventory management, reservations, financial management, and reporting. The identified requirements served as the basis for defining the system's functionality and development process.

This chapter also presents the system development methodology, system architecture and design, database structure, development tools and technologies, testing procedures, and evaluation methods employed throughout the study. These methods provided a structured framework for developing and validating the system according to the operational requirements of IdeaHub Co-working Space + Café.

3.1 Research Design

The study employed a **developmental research design** focused on the systematic design, development, and evaluation of a functional information system. Developmental research provides a structured approach to the creation and evaluation of products, tools, processes, and models designed to address identified needs [17].

This research design was selected because the study aimed to develop a technological solution to the operational requirements identified at **IdeaHub Co-working Space + Café**. Existing processes and operational requirements were initially identified through observation and consultation with the client. The findings from these activities served as the basis for defining the requirements and functionalities of the proposed system.

The research process involved **problem identification, requirements analysis, system design, development, testing and evaluation, and refinement**. Through these stages, the researchers translated the identified operational requirements into system functionalities and evaluated the developed system according to its intended functions and the requirements established with the client.

The developmental research design therefore provided a structured approach for developing the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** and evaluating whether its implemented functionalities addressed the operational requirements identified during the study.

3.2 System Development Methodology

The **Agile Software Development Methodology** was adopted in developing the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. Agile emphasizes iterative development, customer collaboration, working software, and responsiveness to changing requirements throughout the development process [18].

Agile was selected because the operational requirements of IdeaHub evolved as the researchers conducted interviews, observations, system demonstrations, and consultations with the client. The methodology allowed identified changes and additional requirements to be incorporated into succeeding development iterations without restarting the entire development process.

The development process consisted of six major phases, as illustrated in Fig. 3.1.

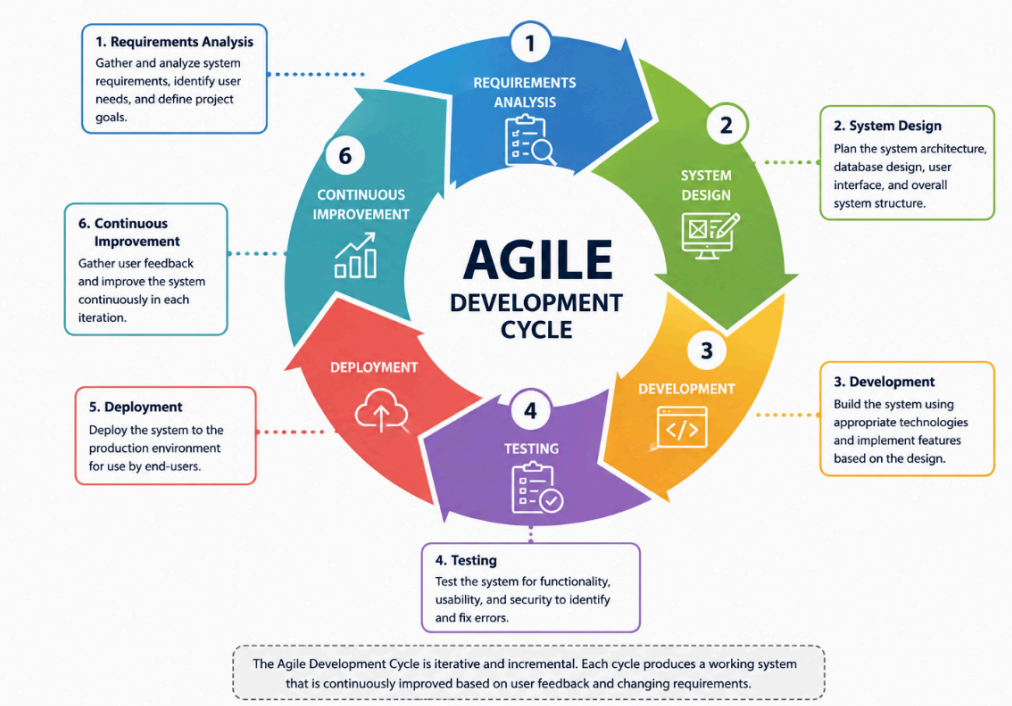


Fig. 3.2. Agile Development Methodology

- **Requirements Analysis.** The researchers gathered system requirements through interviews, consultations, and observations of the existing operations of IdeaHub. Existing procedures involving customer check-in and check-out, workspace usage, billing, POS transactions, inventory monitoring, reservations, financial records, and reporting were examined. The identified operational problems and client requirements were then translated into functional and non-functional system requirements.
- **System Design.** The identified requirements were transformed into the technical and functional design of the system. This phase included the development of the system architecture, database

structure, process flows, and user interfaces. Design models such as the Entity-Relationship Diagram (ERD), Data Flow Diagram (DFD), system workflow, and interface designs were prepared to guide system implementation.

- **System Development.** The system was developed incrementally by implementing individual functional modules and subsequently integrating them into a centralized web-based platform. The developed modules included customer check-in and check-out, workspace and session management, POS and billing, inventory management, reservations, financial management, sales and analytics, reports, user management, and other administrative functions. Individual modules were reviewed and tested during development before their integration into the complete system.
- **Testing and Evaluation.** Functional and integration testing were conducted to verify whether individual features and interconnected modules operated according to their intended requirements. The completed system was also directly tested by the client as part of **User Acceptance Testing (UAT)**. Feedback and identified issues were documented and incorporated into succeeding revisions. Following the testing and revisions, the client approved the implemented functionalities.
- **Prototype Implementation.** Following testing and revisions, the completed system prototype was prepared for use within the intended IdeaHub operational environment. The researchers demonstrated the major system functions and provided guidance on the use of customer session management, POS and billing, inventory management, reservations, financial management, and reporting functionalities.
- **Continuous Improvement and Iteration.** Feedback obtained throughout development and client testing was used to refine the system. Necessary modifications were applied to system functionality, interface design, workflow, and data processing through succeeding iterations. This process is consistent with Agile principles that emphasize responding to changing requirements and regularly adjusting development practices based on feedback [18].

The iterative nature of Agile allowed the researchers to continuously refine the system according to identified requirements and client feedback. Through this approach, the final prototype was developed in accordance with the operational processes and requirements established with **IdeaHub Co-working Space + Café**.

3.3 System Architecture

The Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café utilized a Three-Tier Client-Server Architecture consisting of the Presentation Layer, Application Layer, and Data Layer, as illustrated in Fig. 3.2. A three-tier architecture separates the user interface, application processing, and data management into distinct logical layers, allowing each layer to perform specific responsibilities within the system [19].

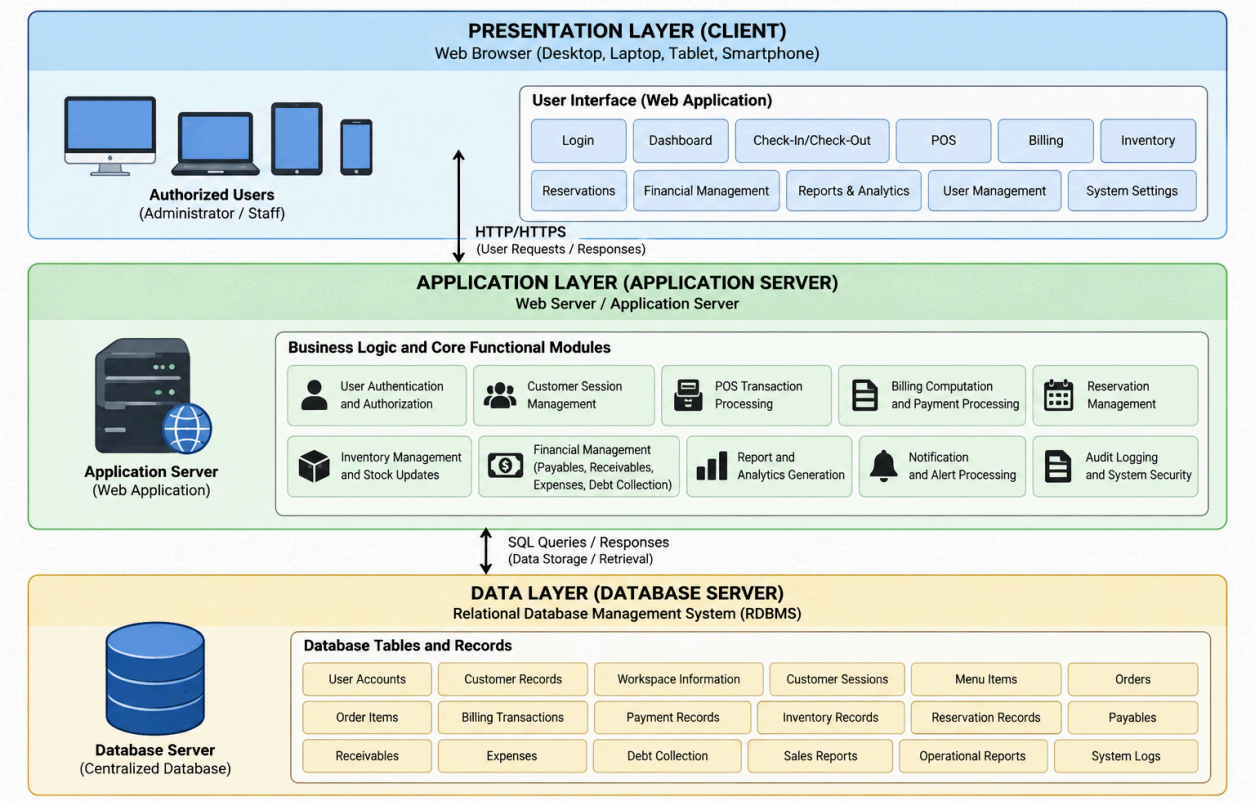


Fig. 3.2. System Architecture Diagram

- **Presentation Layer (Client).** The Presentation Layer served as the primary interface between authorized users and the system. Administrators and staff accessed the web application through supported web browsers. The interface provided access to the login, dashboard, customer check-in and check-out, workspace and session management, POS and billing, inventory management, reservations, financial management, sales and analytics, reports, user management, and system settings. User requests and system responses were communicated between the Presentation Layer and Application Layer through web-based requests.

- **Application Layer (Application Server).** The Application Layer contained the system's core business logic and processed requests received from the Presentation Layer. It handled authentication and authorization, customer session management, POS transaction processing, billing and payment computation, reservation management, inventory updates, financial management, report and analytics generation, notifications, and other system functions. The Application Layer validated and processed information before communicating with the Data Layer, preventing users from directly interacting with stored database records.
- **Data Layer (Database Server).** The Data Layer served as the centralized repository for the system's operational data. It stored and managed records related to user accounts, customers, workspace types, customer sessions, menu items, orders, order items, billing transactions, payments, inventory, reservations, payables, receivables, expenses, debt collection, and other operational records. The Application Layer communicated with the database to perform the required storage, retrieval, and updating of system information.

The architecture followed a controlled data flow in which requests initiated from the Presentation Layer were processed by the Application Layer. When information needed to be stored, updated, or retrieved, the Application Layer communicated with the Data Layer. The processed results were then returned through the Application Layer and presented to the authorized user through the Presentation Layer.

Through this separation of responsibilities, the Three-Tier Client-Server Architecture provided a structured foundation for organizing the different components of the developed system. The separation of presentation, application logic, and data management also supports maintainability and the independent modification of system components [19]. The architecture enabled IdeaHub's different operational modules to function within a centralized web-based system while maintaining clear separation between user interaction, business processing, and data storage.

3.4 Tools and Technologies

The development of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** utilized a combination of web development technologies, frameworks, database technologies, development tools, and testing tools. These technologies were selected according to the functional and architectural requirements of the system, including web-based accessibility, interface design, server-side processing, relational data management, data visualization, version control, and system testing.

A. Frontend Technologies

- **HTML, CSS, and JavaScript.** HTML, CSS, and JavaScript were used to implement the Presentation Layer of the system. HTML provided the structural components of the web pages, CSS controlled their visual presentation and layout, while JavaScript supported client-side interactions and dynamic interface behavior. Responsive web design allows web interfaces to adapt their presentation according to different screen sizes and resolutions [20].
- **Bootstrap.** Bootstrap was utilized to provide responsive layouts and reusable interface

components throughout the system. Research on Bootstrap-based web development demonstrates its application in developing responsive interfaces for different browsing devices and screen dimensions [21]. In the proposed system, Bootstrap supported the development of forms, navigation elements, cards, tables, modal components, and responsive page layouts.

- **Chart.js.** Chart.js was utilized in the dashboard and analytics components to visually present selected operational and sales information. It was used to display summarized system data through graphical representations to support the monitoring of relevant operational information.

B. Backend Technologies

- **Python.** Python served as the primary server-side programming language of the system. It was used to implement application logic, data processing, billing calculations, database operations, and other backend functions required by the system.
- **Flask.** Flask was used as the primary web application framework for the backend. It handled application routes, HTTP requests and responses, user sessions, authentication, server-side processing, and communication between the Presentation and Data Layers. Flask is a Python web framework designed to facilitate the development of web-based applications [22]. In the proposed system, Flask provided the framework through which the different backend functions and application modules were organized.

C. Database Technologies

- **MySQL.** MySQL was utilized as the relational database management system of the proposed system. It served as the centralized repository for records involving users, customers, workspace types, customer sessions, menu items, orders, transactions, inventory, reservations, and financial information. The relational database structure allowed related operational information to be organized through interconnected tables and relationships.
- **SQLAlchemy.** SQLAlchemy was used as the Object-Relational Mapping (ORM) component between the Flask application and the relational database. In the developed system, SQLAlchemy models were used to define database entities, relationships, and operations required by the different application modules.
- **PyMySQL.** PyMySQL served as the database driver used to establish communication between the Python application and the MySQL database. It supported database connectivity required by the backend application for storing, retrieving, and updating system information.

D. Development and Collaboration Tools

- **Visual Studio Code.** Visual Studio Code was used as the primary development environment for writing, organizing, testing, and debugging the system's source code. Its integrated development features supported the technologies used throughout the implementation of the system.
- **Git.** Git was utilized as the distributed version control system for tracking modifications to the system's source code throughout development. It allowed the researchers to maintain versions of the source code and manage changes made during the development process.
- **GitHub.** GitHub served as the remote repository and collaborative development platform for the project. It was used for source-code storage, version management, backup, and collaboration among the researchers.
- **XAMPP.** XAMPP provided the local database environment used during development and

testing. Its MySQL-compatible database service and phpMyAdmin interface supported local database creation, administration, inspection, and testing.

E. Testing Tools

- **Thunder Client.** Thunder Client was used during backend development to send HTTP requests to selected application endpoints and inspect their corresponding responses. It assisted the researchers in verifying the behavior of backend functions involving customer sessions, orders, billing, and other application endpoints before complete frontend integration.

Collectively, these technologies supported the Three-Tier Client-Server Architecture of the developed system. HTML, CSS, JavaScript, Bootstrap, and Chart.js supported the Presentation Layer; Python and Flask implemented the primary business logic within the Application Layer; while MySQL, SQLAlchemy, and PyMySQL supported data persistence and database communication within the Data Layer. Visual Studio Code, Git, GitHub, XAMPP, and Thunder Client supported the development, version control, database administration, and testing processes.

3.5 Data Gathering Techniques

The researchers employed multiple data gathering techniques to obtain information regarding the existing operations of IdeaHub Co-working Space + Café, identify operational challenges, and determine the functional requirements of the proposed system. The techniques included stakeholder interviews, direct observation, review of existing operational records, and collection of menu and pricing data. Information gathered through these methods served as the basis for requirements analysis, system design, and development.

A. Stakeholder Interviews

The researchers conducted interviews and consultations with the owner and relevant staff of **IdeaHub Co-working Space + Café**. These activities were used to gather information regarding the establishment's existing procedures for customer check-in and check-out, workspace usage, food and beverage ordering, billing, inventory monitoring, reservations, financial records, and reporting.

The interviews also provided information regarding the operational requirements expected from the proposed system. These included the need for a simple check-in process, centralized monitoring of customer sessions, integration of workspace charges and café transactions, inventory monitoring, financial management, and accessible operational reports. The information gathered was used to identify and refine the functional requirements of the system.

B. Direct Observation

The researchers directly observed the existing operations of IdeaHub to understand how activities were performed within the actual business environment. The observation focused on customer check-in and check-out, workspace usage, order processing, billing, payment, inventory monitoring, and other related

operational activities.

Through direct observation, the researchers were able to compare the procedures described during interviews with the actual workflow of the establishment. The process also assisted in identifying operational activities that depended on manual recording, separate records, or manual computation. These observations contributed to the development of system requirements for customer session monitoring, billing, POS integration, inventory management, and reporting.

C. Review of Existing Operational Records

The researchers reviewed available operational records used by IdeaHub, including records associated with customer sessions, orders, inventory, and sales activities. The review provided additional information regarding the types of data maintained by the establishment and how these records were organized under the existing procedures.

The information obtained from the existing records also contributed to the design of the system's database structure. Relevant data elements identified from the existing procedures were translated into appropriate database entities and attributes for customer sessions, transactions, orders, inventory, and other operational records.

D. Collection of Menu and Pricing Data

The researchers collected information regarding IdeaHub's available food and beverage items, corresponding prices, product categories, and workspace rates. These data were necessary for configuring the POS, workspace management, and billing functionalities of the proposed system.

The collected menu information was used as the basis for the initial menu records within the system, while the established workspace rates were incorporated into the computation of time-based charges. This ensured that the system's initial operational data corresponded to the pricing information provided by the client.

The information obtained through these data gathering techniques was consolidated and analyzed to determine the requirements of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. The gathered data guided the design of the system's functionalities, database structure, workflow, and user interfaces to ensure that the developed prototype corresponded to the operational requirements identified during the study.

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CHAPTER IV

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

This chapter presents the development, implementation, testing, evaluation, results, and discussion of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. It describes how the requirements identified during the research and development process were translated into functional system modules designed to support the operational processes of the establishment.

The chapter presents the developed system and its major functionalities, including customer check-in and check-out, workspace and session management, POS and billing, inventory management, reservation management, financial management, sales and analytics, reporting, user management, and other administrative functions. It also discusses the implementation of these modules and their integration within the centralized web-based platform.

Furthermore, this chapter presents the functional testing, integration testing, and User Acceptance Testing (UAT) conducted to verify the operation of the developed system. The results of the testing and direct client evaluation are presented and discussed in relation to the functional and operational requirements established for IdeaHub Co-working Space + Café.

Finally, the chapter discusses the findings obtained from the development and evaluation of the system, including how the implemented functionalities addressed the operational requirements identified during the study. These findings provide the basis for the conclusions and recommendations presented in the succeeding chapter.

4.1 Overview of the Developed System

The Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café is a web-based system developed to integrate and manage the establishment's primary operational processes within a centralized platform. The system was designed in response to the operational requirements identified during the study, particularly those involving customer session monitoring, workspace management, billing, point-of-sale (POS) transactions, inventory management, reservations, financial records, sales monitoring, reporting, and user management.

The developed system enables authorized users to record customer information during check-in, assign the appropriate workspace, monitor active customer sessions, and record customer check-out. The duration of each customer session is determined by the recorded time-in and time-out, while the corresponding workspace charge is calculated according to the applicable workspace rate.

The integrated Point-of-Sale (POS) functionality allows staff to record food and beverage orders associated with customer sessions. During billing, the system combines the calculated workspace charge with the recorded food and beverage purchases to determine the customer's total bill. This integration

allows customer session information and POS transactions to be processed within the same system rather than maintained through separate procedures.

The system also provides inventory management for monitoring recorded stock information and updating inventory based on applicable transactions. Reservation management supports the recording and monitoring of workspace reservations, while the financial management functions provide facilities for managing payables, receivables, expenses, debt collection, and daily soft balancing.

Operational and transaction information is consolidated through the system's dashboard, sales and analytics, and reporting functions. These features allow authorized users to access information regarding active customer sessions, recorded transactions, sales activities, and other relevant operational data. Reports can also be generated from stored system records to support the review of business activities.

Access to the system is controlled through user authentication and role-based access, allowing users to access functions according to their assigned roles. System information is maintained within a centralized database, enabling related operational records to be stored and retrieved through the different system modules.

The major functions of the developed system include:

- Customer check-in and check-out;
- Customer session and workspace monitoring;
- Automated computation of workspace charges;
- Integrated POS and food and beverage ordering;
- Billing and payment processing;
- Inventory management;
- Reservation management;
- Financial management, including payables, receivables, expenses, debt collection, and daily soft balancing;
- Sales and analytics;
- Report generation;
- User authentication and role-based access;
- User management; and
- System settings and administrative functions.

The developed system consists of interconnected modules that exchange and process related operational information within a centralized web-based platform. For example, customer session information is connected with POS orders and billing, while applicable transaction information is reflected in inventory, sales, financial, and reporting functions. This integration was designed to provide IdeaHub with a more organized approach to managing its operational information and daily business processes.

Overall, the developed system provides a centralized digital platform for managing the operational requirements identified at IdeaHub Co-working Space + Café. Rather than handling customer sessions, POS transactions, billing, inventory, reservations, financial records, and reports through separate procedures, the system integrates these functions within a single web-based application.

4.2 Entity Relationship Diagram (ERD)

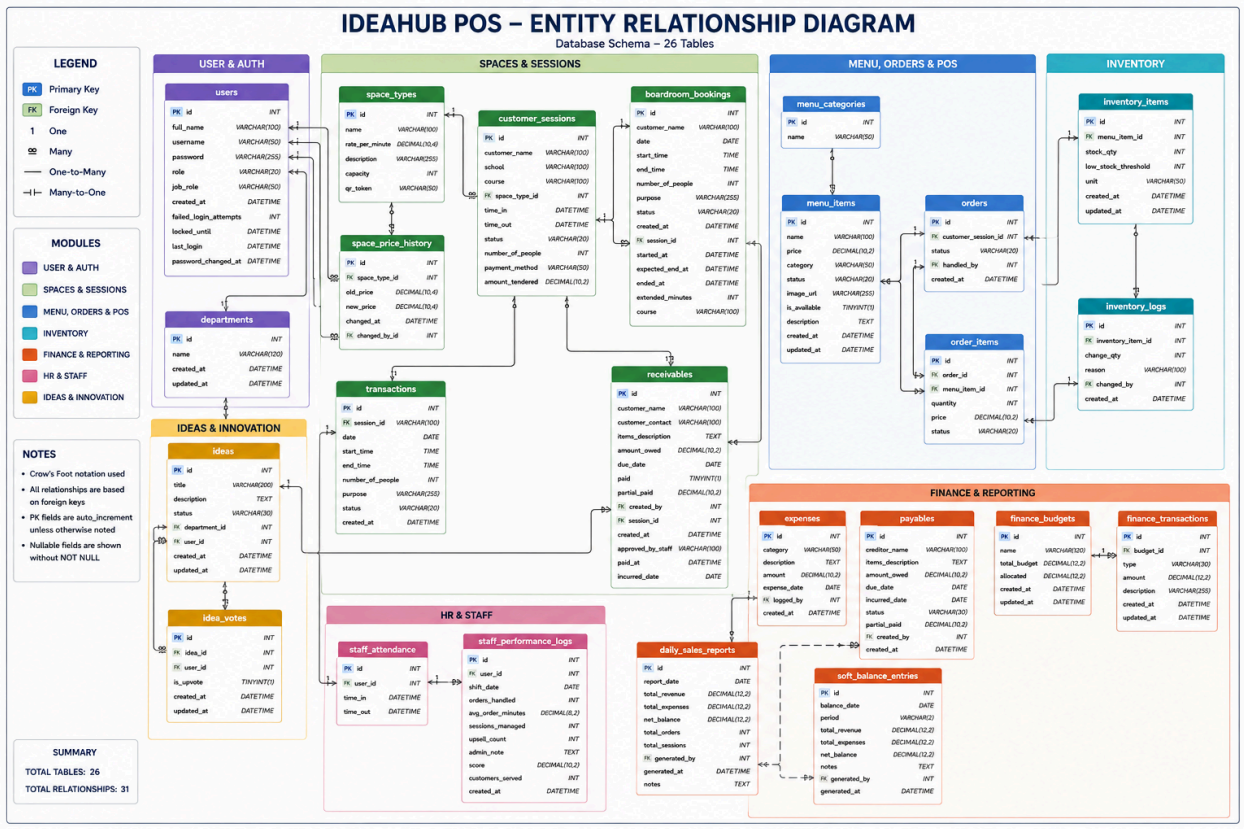


Fig. 4.1. Entity Relationship Diagram (ERD)

Figure 4.1 presents the **Entity Relationship Diagram (ERD)** of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. The diagram illustrates the relational database structure of the developed system and shows how the different entities are connected through primary and foreign keys. The database supports the major functions of the system, including customer session management, workspace management, reservations, POS transactions, inventory management, financial management, staff monitoring, reporting, and user administration.

The **users** table stores the accounts of authorized system users. It contains information such as the user's full name, username, password, role, job role, login-related information, and account timestamps. The table is connected to several other entities in which a system user is responsible for creating, generating, handling, or modifying a record. These relationships allow applicable system activities to be associated with the responsible user.

The management of customer workspace sessions is primarily supported by the **customer_sessions** and **space_types** tables. The customer_sessions table stores customer information, selected workspace, time-in, time-out, session status, number of people, payment method, and amount tendered. The space_type_id foreign key associates each session with a record in space_types, which contains the

workspace name, rate per minute, description, capacity, and QR token. The **space_price_history** table maintains records of changes made to workspace rates and identifies the user responsible for applicable price modifications.

Reservation functionality is represented by the **boardroom_bookings** and **lounge_bookings** tables. The boardroom_bookings table maintains information regarding boardroom reservations, including the customer, date, scheduled time, number of people, purpose, status, and applicable session information. The lounge_bookings table maintains corresponding reservation information for lounge spaces. These entities allow reservation records to be maintained separately from regular customer session information while supporting the workspace reservation functions of the system.

The POS component is supported by the **menu_categories**, **menu_items**, **orders**, and **order_items** tables. menu_categories provides classifications for the available products, while menu_items stores information such as product name, price, category, availability, status, image, and description. The orders table associates food and beverage orders with customer sessions and identifies the user who handled an order. The order_items table records the individual products, quantities, prices, and statuses associated with each order.

Inventory management is supported by the **inventory_items** and **inventory_logs** tables. The inventory_items table associates inventory information with individual menu items and stores the current stock quantity, low-stock threshold, and unit of measurement. The inventory_logs table maintains records of changes in inventory quantities, the reason for each adjustment, and the user responsible for the change. Together, these entities provide the database structure required for recording and tracing inventory movements.

The financial management component consists of the **payables**, **receivables**, **expenses**, **finance_budgets**, and **finance_transactions** tables. The payables table contains billing-related information such as time charges, food charges, total bill, and payment method, while the receivables table maintains outstanding customer balances, payment status, partial payments, due dates, and related session information. The expenses table records operational expenditures. The finance_budgets and finance_transactions tables support the recording of budget allocations and corresponding financial activities.

The system also maintains summarized financial and operational information through the **daily_sales_reports** and **soft_balance_entries** tables. The daily_sales_reports table stores information such as total revenue, total expenses, net balance, total orders, and total sessions for a particular reporting date. The soft_balance_entries table maintains summarized revenue, expenses, and net balance according to the applicable date and reporting period.

Staff-related information is maintained through the **staff_attendance** and **staff_performance_logs** tables. staff_attendance associates individual users with their recorded time-in and time-out, while staff_performance_logs maintains information such as orders handled, sessions managed, customers served, and other performance-related records. Both entities are associated with the corresponding records in the users table.

The database further includes the **departments**, **ideas**, and **idea_votes** tables. The departments table provides organizational classifications for submitted ideas. The ideas table associates each suggestion

with the corresponding department and user, while idea_votes records the voting activity associated with submitted ideas and participating users.

Overall, the ERD demonstrates how the different database entities are interconnected to support the operations of the developed system. Customer sessions are associated with workspace information and POS orders; orders are connected to individual menu items; menu items are associated with inventory records; and financial, reporting, reservation, staff, and administrative information is maintained through their corresponding entities. The use of primary and foreign keys establishes the relationships required for information to be shared across the different system modules.

Through this relational database structure, the developed system provides a centralized foundation for managing the operational information of **IdeaHub Co-working Space + Café**, including customer sessions, workspace usage, reservations, POS orders, billing, inventory, financial records, staff activities, reports, and user-related information.

4.3 Data Flow Diagram (DFD)

The **Data Flow Diagram (DFD)** presents the movement of information within the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. It illustrates how external entities interact with the system, how information is processed by the major system functions, and how operational records are stored and retrieved.

The DFD is presented at different levels to provide both a general and detailed representation of the system. The **Context Diagram** defines the system boundary and its interaction with external entities, while the **Main Process DFD** presents the major functional processes of the system. The **Detailed Process DFD** further decomposes these processes to illustrate the specific activities involved in processing system information.

4.3.1 Context Diagram

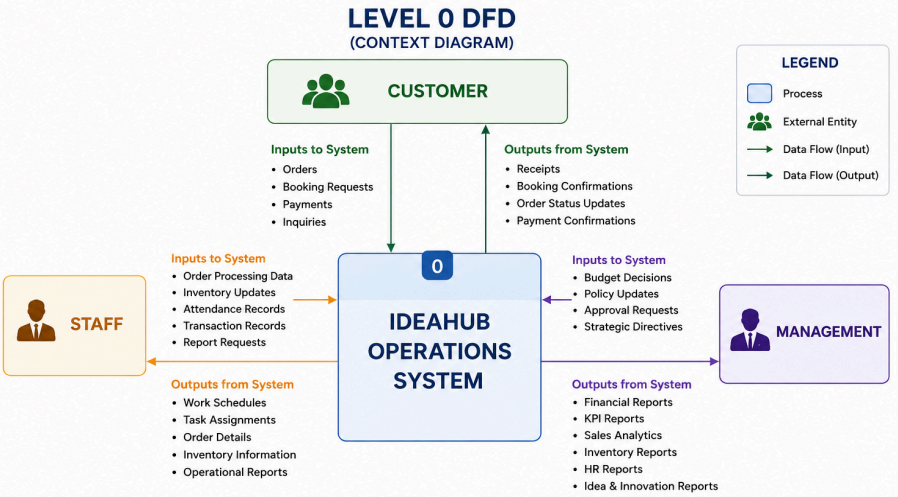


Fig. 4.2. Context Diagram of the IdeaHub Co-working Space + Café System

Figure 4.2 presents the **Context Diagram** of the developed system. At this level, the entire application is represented as a single central process interacting with its external entities. The diagram establishes the system boundary and illustrates the major information exchanged between the system and its users. The uploaded diagram description identifies **Customers, Staff, and Management** as the three primary external entities.

Customers represent the individuals who avail themselves of the workspace and café services of IdeaHub. Customer-related information processed by the system includes information required for customer sessions, workspace usage, food and beverage orders, reservations, billing, and payments. In return, the applicable transaction, billing, and reservation information is made available through the corresponding system processes.

Staff represent authorized personnel responsible for performing the daily operational activities of the establishment. Staff interact with the system to perform customer check-in and check-out, manage active sessions, record POS orders, process billing and payments, monitor inventory, manage reservations, and perform other functions permitted by their assigned roles. The system provides staff with the information required to complete these operational activities.

Management represents authorized users responsible for monitoring and administering the operations of IdeaHub. Management interacts with the system through administrative, financial, reporting, inventory, staff-monitoring, and other management functions. The system provides access to applicable sales information, financial records, inventory information, operational reports, staff-related records, and other information maintained by the system.

Overall, the Context Diagram provides a high-level representation of how information is exchanged between the developed system and its primary external entities. It establishes the scope of the system before the internal processes are further decomposed in the succeeding Data Flow Diagrams.

4.3.2 Main Process Data Flow Diagram

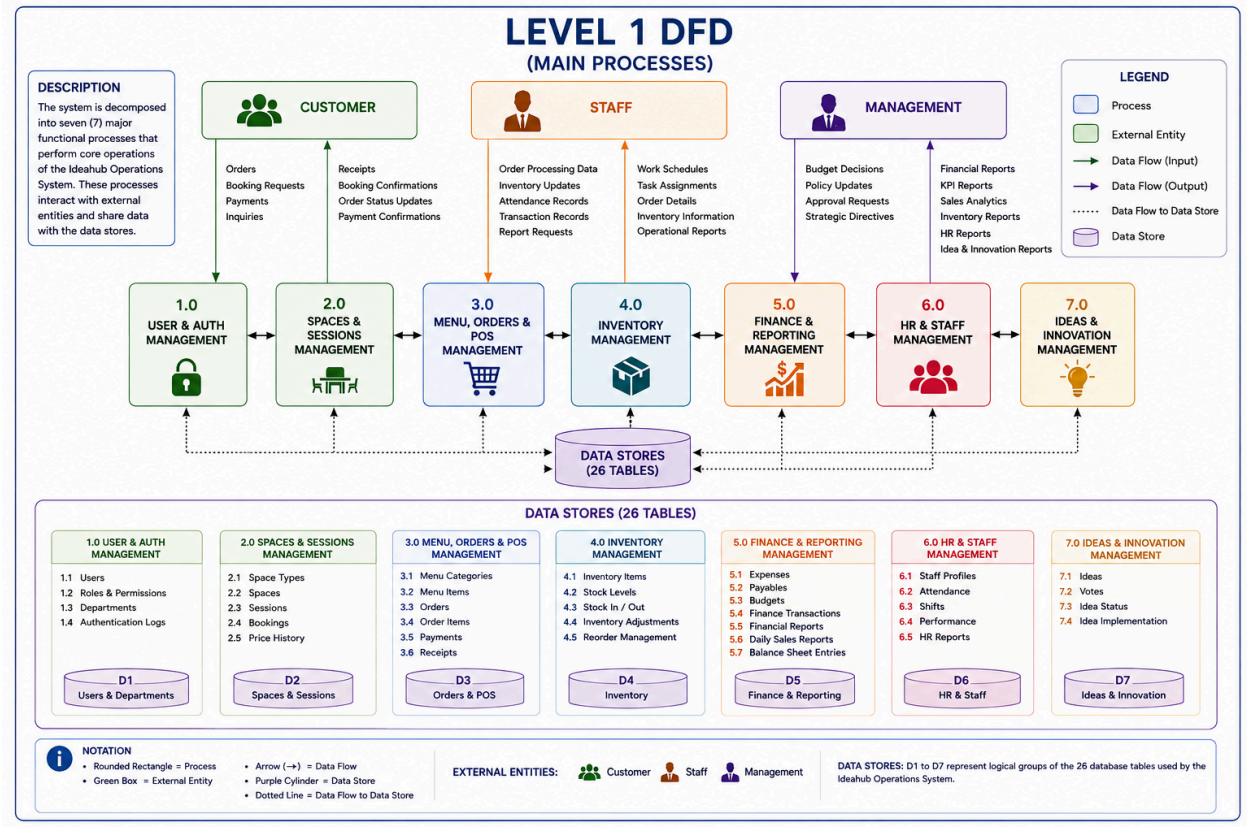


Fig. 4.3. Main Process Data Flow Diagram of the IdeaHub Co-working Space + Café System

Figure 4.3 presents the **Main Process Data Flow Diagram**, which decomposes the central system process shown in the Context Diagram into its major functional areas. Based on the developed system, these processes include **User and Authentication Management, Spaces and Sessions Management, Menu, Orders and POS Management, Inventory Management, Finance and Reporting Management, Staff Management, and Ideas Management**. These functional areas are also reflected in the major processes identified in the original DFD description.

User and Authentication Management. This process manages user authentication, account information, roles, and applicable access permissions. User credentials are validated before access to protected system functions is granted. Assigned roles determine the functions available to the authenticated user.

Spaces and Sessions Management. This process manages customer check-in and check-out, workspace selection, customer sessions, and applicable reservations. Customer information and workspace details are recorded when a session is created, while time-in and time-out information supports session monitoring and workspace billing. Reservation information for applicable workspaces is also maintained through this process.

Menu, Orders, and POS Management. This process manages menu information and customer food and beverage orders. Orders are associated with customer sessions, while individual ordered items, quantities, and prices are recorded for POS and billing purposes. This allows applicable food and beverage charges to be included with the customer's workspace charges.

Inventory Management. This process maintains inventory information associated with menu products. Stock quantities and applicable inventory adjustments are recorded, while inventory logs maintain information regarding changes made to inventory records. The system also maintains low-stock threshold information for applicable inventory items.

Finance and Reporting Management. This process manages applicable financial and reporting information, including billing-related records, receivables, payables, expenses, budgets, financial transactions, daily sales reports, and soft balancing information. It provides the system with a centralized means of maintaining financial records and generating applicable operational summaries.

Staff Management. This process maintains staff-related information associated with system users. It includes attendance records and staff performance information maintained through the corresponding database entities.

Ideas Management. This process supports the submission and organization of ideas within the system. Submitted ideas are associated with users and departments, while voting records maintain information regarding user responses to submitted ideas. This corresponds more closely to your actual ERD than the original description of a complete “innovation implementation” workflow.

Overall, the Main Process DFD demonstrates how the developed system is divided into interconnected functional areas responsible for processing operational information. These processes provide the foundation for the more detailed decomposition presented in the Detailed Process Data Flow Diagram.

4.3.3 Detailed Process Data Flow Diagram

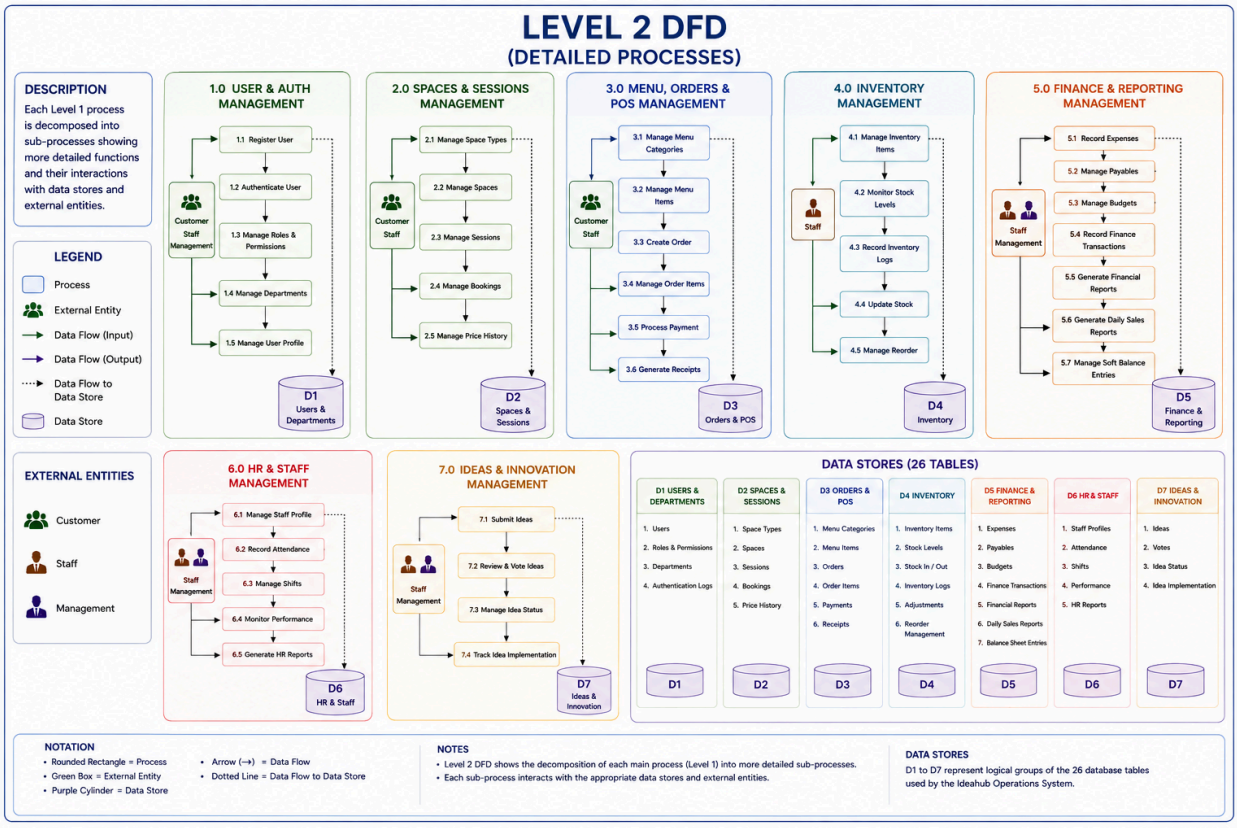


Fig. 4.4. Detailed Process Data Flow Diagram of the IdeaHub Co-working Space + Café System

Figure 4.4 presents the **Detailed Process Data Flow Diagram**, which further decomposes the major processes identified in the Main Process DFD. It illustrates the specific activities involved in processing, validating, storing, retrieving, and updating information within the developed system.

User and Authentication Management. The detailed process includes credential validation, user account management, role assignment, and access control. User information is maintained within the corresponding data store, while authentication determines whether access to protected system functions is granted.

Spaces and Sessions Management. This process includes customer check-in, workspace selection, session creation, session monitoring, customer check-out, and reservation processing. Customer session information is associated with the selected workspace, while recorded time-in and time-out information is used in determining the duration of workspace usage. Reservation information is maintained for

applicable workspace bookings.

Menu, Orders, and POS Management. This process includes menu management, order creation, item selection, quantity recording, and association of orders with customer sessions. The recorded information is subsequently used in determining applicable food and beverage charges during billing.

Inventory Management. This process includes maintaining inventory items, monitoring recorded stock quantities, updating applicable quantities, recording inventory adjustments, and identifying items that have reached established low-stock thresholds. Inventory logs maintain records of applicable stock changes and responsible users.

Finance and Reporting Management. This process includes maintaining billing and payment-related information, receivables, payables, expenses, budgets, financial transactions, daily sales information, and soft balance records. These data are processed and organized to support the financial and reporting functions available to authorized users.

Staff Management. This process includes maintaining staff attendance and performance-related information. Staff records are associated with corresponding user accounts, enabling the system to maintain attendance and applicable performance information for individual staff members.

Ideas Management. This process includes idea submission, departmental classification, and voting. Submitted ideas are associated with the user who created them and the corresponding department, while voting records identify the users who responded to each idea.

The Detailed Process DFD therefore illustrates how information moves through the specific activities of the developed system and how these activities interact with their corresponding data stores. Together, the Context Diagram, Main Process DFD, and Detailed Process DFD provide progressively detailed representations of the flow of information throughout the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café.

4.4 Flowchart

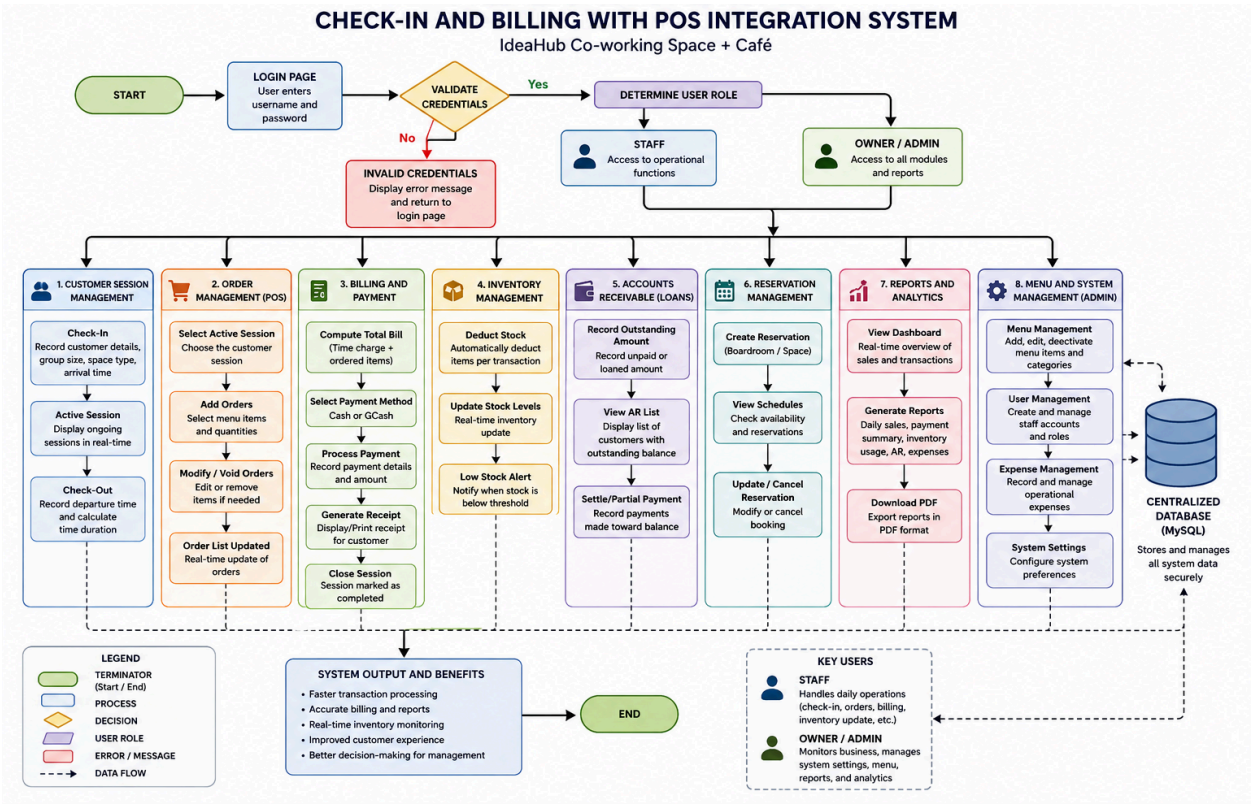


Figure 4.5. Overall System Flowchart of the Check-In and Billing with POS Integration System

Figure 4.5 presents the overall system flow of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. The flowchart illustrates the complete operational flow of the developed system, beginning with user authentication and role verification and continuing through customer session management, order processing, billing and payment, inventory management, accounts receivable and loans, reservation management, reports and analytics, and administrative management. The flowchart also demonstrates how the different modules communicate with and contribute information to the centralized MySQL database. Through this centralized structure, operational records generated from different activities are stored and made available to authorized users for processing, monitoring, reporting, and management decision-making.

The overall process begins when an authorized user accesses the **Login Page**. The user is required to enter a registered username and password before gaining access to the system. The login process serves as the first level of system access control because the system contains customer information, transaction records, payment information, inventory records, financial data, and administrative settings. After the user submits

the required credentials, the system performs credential validation. This process determines whether the entered username and password correspond to a valid account stored in the system.

If the credentials are incorrect or do not correspond to an authorized account, the system displays an **Invalid Credentials** message and redirects the user back to the login page. This prevents unauthorized users from proceeding to the operational modules. The user must then provide the correct credentials before another authentication attempt can be made. This authentication process helps protect the system from unauthorized access and ensures that operational and administrative information is accessible only to users with valid accounts.

If the credentials are successfully validated, the system proceeds to **Determine User Role**. The system identifies whether the authenticated account belongs to a **Staff** user or an **Owner/Administrator**. Role determination is important because different users have different responsibilities within IdeaHub. Staff members primarily perform daily operational activities, while the Owner/Administrator requires broader access for business monitoring, system management, reports, and administrative functions. The role-based structure therefore allows the system to provide users with functions appropriate to their responsibilities.

The **Staff** role is intended for employees who handle the day-to-day activities of the establishment. Depending on their authorized access, staff members can perform functions such as customer check-in, customer session monitoring, order processing, billing and payment, inventory-related activities, and other operational tasks. On the other hand, the **Owner/Administrator** has broader access to the system and can monitor business activities, manage users, configure menu and system settings, view reports and analytics, and access information necessary for management and decision-making.

After the user's role has been determined, the system provides access to the appropriate modules. These modules are interconnected because information produced by one operational process can become input for another process. For example, a customer's active session can be selected when creating an order; the order can contribute to the customer's final bill; completed transactions can affect inventory quantities and sales records; and transaction information can subsequently be included in reports and analytics. This interconnected flow is one of the major functions of the proposed system because it reduces the need to maintain separate records for related business activities.

Customer Session Management

The first major operational module is **Customer Session Management**. This module manages the customer's workspace usage from the time of arrival until checkout. The process begins with **Check-In**, where the staff records the required customer information, including the customer's name or details, group size, selected workspace type, and arrival time. The selected workspace determines the type of service that will be provided to the customer and serves as a basis for calculating the applicable workspace charge.

After the check-in information is recorded, the system creates an **Active Session**. The active session represents a customer who is currently using an IdeaHub workspace. Active sessions can be displayed to authorized staff so that they can determine which customers are currently present in the establishment. This provides a real-time operational view of ongoing customer sessions and reduces dependence on manually maintained attendance or monitoring records.

The active session also serves as an important reference for other modules. For example, when a customer

places a food or beverage order, the staff can select the corresponding active customer session before adding the order. This allows the food transaction to remain associated with the correct customer session and ensures that the customer's workspace charges and ordered items can later be combined during checkout.

When the customer is ready to leave, the staff initiates **Check-Out**. The system records the departure time and determines the total duration of the customer's workspace session. The calculated duration is then used together with the applicable workspace rate to determine the customer's time-based workspace charge. The session is subsequently marked as completed after the checkout process has been successfully processed. This provides a clear beginning and ending point for every customer session and creates the information required for accurate billing.

Order Management (POS)

The second major module is **Order Management (POS)**. This module handles food and beverage transactions made by customers during their stay. The process begins when staff members select an active customer session. Selecting the active session ensures that the order is associated with the appropriate customer rather than being recorded as an independent transaction.

After selecting the customer session, staff can use the POS interface to **Add Orders**. Available menu items are displayed so that the staff can select the products requested by the customer and specify the appropriate quantities. The system records the selected items and quantities as part of the customer's order.

If an order needs to be corrected before final billing, the system allows authorized staff to **Modify or Void Orders**. Items can be edited or removed when necessary, such as when a customer changes an order or when an item was entered incorrectly. This provides flexibility during transaction processing while helping maintain more accurate order records.

After changes have been made, the **Order List** is updated. The updated list provides staff with the current items associated with the customer session. This real-time update is important because the final food bill must reflect the latest valid order information. The order records also provide information that can later be used by the billing, inventory, and reporting modules.

The POS module therefore serves as the primary transaction-recording component for café operations. Rather than keeping food orders separate from customer sessions, the system connects them to the corresponding customer session. This integration allows food purchases to become part of the customer's final transaction when the customer checks out.

Billing and Payment

The third major module is **Billing and Payment**. This module consolidates the customer's workspace usage and ordered food and beverage items into a final bill. When the customer is ready to check out, the system calculates the total amount based on the customer's **time charge and ordered items**.

The time charge is determined from the customer's recorded workspace session. The system uses the recorded session duration and the applicable workspace rate to calculate the amount associated with the customer's use of the facility. The food component of the bill is based on the valid menu items and quantities recorded through the POS module. The two components are then combined to determine the

customer's total bill.

After the total amount has been computed, the staff selects the customer's **Payment Method**. The system supports the payment methods specified for IdeaHub, including **Cash and GCash**. Recording the payment method provides additional transaction information that can later be used for sales monitoring and payment reporting.

The system then proceeds to **Process Payment**, where the payment details and corresponding amount are recorded. Once the payment is successfully processed, the system can **Generate a Receipt** for the customer. The receipt serves as a record of the completed transaction and provides information about the customer's charges and payment.

After the payment process has been completed, the customer session is **Closed** and marked as completed. Closing the session prevents the completed customer transaction from continuing to appear as an active session. It also establishes the final status of the customer visit and allows the completed transaction to become part of the system's historical records.

The billing process therefore connects several operational activities that would otherwise need to be recorded separately. Customer session information provides the time charge, POS records provide the food and beverage charges, and payment information completes the transaction. By bringing these records together, the system supports more consistent and organized billing.

Inventory Management

The fourth module is **Inventory Management**, which manages product stock information associated with the café's menu items. Inventory information is connected to the POS process because products sold through customer orders affect the available stock.

When a transaction is recorded and finalized, the system performs **Deduct Stock** based on the items included in the transaction. For example, when a menu item is sold, the corresponding inventory quantity is reduced according to the recorded transaction. This provides a more direct relationship between actual sales and inventory records.

After stock has been deducted, the system **Updates Stock Levels**. The updated quantity represents the current recorded inventory level of the affected product. Real-time updating allows authorized users to monitor stock information without having to manually calculate or update the quantity after every transaction.

The system also provides a **Low Stock Alert** function. When the available quantity of an item reaches or falls below the configured threshold, the system can notify authorized users that the item requires attention. This allows management or staff to become aware of products that may need replenishment before the stock is completely depleted.

The integration between POS and inventory management is important because sales transactions provide the information required to update product quantities. This reduces the need to maintain completely separate sales and inventory records and gives management a more organized view of available products.

Accounts Receivable and Loans

The fifth module is **Accounts Receivable (Loans)**. This module is intended to support the recording and monitoring of outstanding amounts that have not yet been fully settled. Instead of limiting the system to fully paid transactions, the module provides a mechanism for recording financial obligations that remain outstanding.

The process begins with **Record Outstanding Amount**, where an unpaid or loaned amount is recorded in the system. This creates a financial record that identifies an amount that still requires payment. Recording these amounts allows the establishment to maintain information about customers or transactions with remaining balances rather than relying only on manually maintained notes.

The system then provides a **View AR List** function. This displays a list of customers or accounts with outstanding balances, allowing authorized users to identify unpaid amounts that still need to be collected. The list provides management with a centralized view of outstanding financial obligations.

When a customer makes a payment toward an outstanding amount, the system supports **Settle/Partial Payment**. A customer may either settle the entire remaining balance or make a partial payment. The amount paid is recorded and applied to the existing balance, allowing the system to maintain an updated outstanding amount.

This module provides an additional financial-management function because it allows IdeaHub to monitor receivables separately from completed cash or GCash transactions. It can therefore assist authorized users in monitoring unpaid balances and determining which accounts still require collection.

Reservation Management

The sixth module is **Reservation Management**. This module manages reservations for spaces such as the boardroom and other reservable facilities. It provides a structured process for recording and monitoring scheduled use of available spaces.

The process begins with **Create Reservation**, where the authorized user records the necessary reservation information and selects the intended space. The reservation provides the system with information about the requested schedule and space.

After reservations have been recorded, authorized users can **View Schedules**. This allows staff or administrators to check existing reservations and determine the availability of spaces. The schedule provides an organized view of reservations and helps prevent scheduling conflicts.

When changes are required, the system provides **Update/Cancel Reservation** functionality. Authorized users can modify reservation details when a customer changes the requested schedule or cancel a reservation when it is no longer needed. Maintaining updated reservation records ensures that the schedule reflects the latest information available to the establishment.

Reservation management is connected to the overall workspace-management process because it allows IdeaHub to organize scheduled use of its spaces in addition to monitoring active customer sessions. This is particularly useful for spaces such as the boardroom, where advance scheduling may be necessary.

Reports and Analytics

The seventh module is **Reports and Analytics**. This module converts stored operational information into summaries that can be used for monitoring and management. The process begins with the **Dashboard**, which provides a real-time overview of relevant sales and transaction information.

The dashboard allows authorized users, particularly the Owner/Administrator, to obtain a quick view of business activities without having to manually examine individual transaction records. The information displayed on the dashboard is derived from data stored within the centralized database.

The system also provides **Generate Reports** functionality. Reports may include information concerning daily sales, payment summaries, inventory usage, accounts receivable, and expenses. These reports allow management to examine operational information according to specific business requirements.

The generated reports can then be **Downloaded as PDF files**. PDF export allows users to retain a formal copy of generated reports for documentation, review, or other authorized business purposes. This also provides an alternative to viewing information exclusively within the system.

The Reports and Analytics module therefore serves as an information-consolidation component of IdeaHub. Data generated from customer sessions, POS transactions, payments, inventory activities, receivables, reservations, and expenses can contribute to management information. Instead of reviewing each module separately, management can use reports and dashboard information to obtain a broader understanding of the establishment's operations.

Menu and System Management

The eighth major module is **Menu and System Management**, which is primarily associated with administrative functions. This module allows the Owner/Administrator to maintain system information and configure selected aspects of the system.

The **Menu Management** function allows authorized administrators to add, edit, and deactivate menu items and categories. This is necessary because café products may change over time. New products may be introduced, existing product information may require modification, and unavailable products may need to be deactivated. Maintaining current menu information ensures that the POS interface reflects the products available for sale.

The **User Management** function allows administrators to create and manage staff accounts and assign appropriate roles. Since the system contains different types of information and functions, user accounts help establish controlled access to the system. Administrators can manage which users are authorized to access operational or administrative functions.

The module also includes **Expense Management**, which allows authorized users to record and manage operational expenses. Expense information contributes to the financial records used by the system's reporting functions and provides management with information about business-related expenditures.

The **System Settings** function allows authorized administrators to configure system preferences. These settings provide a centralized area for maintaining applicable system configurations without requiring

changes to the underlying application for every operational adjustment.

Centralized Database

All major modules interact with the **Centralized MySQL Database**. The database serves as the central storage location for information generated and used by the system. Customer records, customer sessions, orders, menu items, inventory information, payments, reservations, user accounts, expenses, and other system records are stored and managed through the database.

The centralized database is important because the modules are not independent applications with separate records. Instead, they share related information. For example, a customer session can be associated with orders; orders can contribute to billing; completed transactions can affect inventory and sales records; and transaction information can later be used for reports and analytics.

This structure allows information entered during one stage of the process to be reused during subsequent stages. It also reduces the need for repeated manual encoding of the same information. For example, staff do not need to manually calculate the customer's total food purchases separately from the workspace charge when the necessary information has already been recorded in the system.

The database also supports the system's administrative functions by maintaining user accounts, menu information, system records, and other data required for the operation of the application. Through centralized storage, authorized modules can retrieve current information when needed, supporting consistency among the different parts of the system.

Data Flow Between Modules

The flowchart also illustrates that information does not simply move from one module to another in a single direction. Instead, the modules form an interconnected operational cycle.

A customer first enters the system through the **Customer Session Management** process. The customer session establishes the information needed to identify the customer's current workspace use. The **Order Management** module then uses the active session as a reference when recording food and beverage orders. The **Billing and Payment** module combines the customer's workspace time charge and valid orders to calculate the total amount due.

Once the transaction is completed, information from the transaction contributes to other processes. The **Inventory Management** module uses finalized product transactions to update stock quantities. Transaction information also becomes available to the **Reports and Analytics** module, where it can contribute to sales summaries and other reports.

Similarly, the **Accounts Receivable** module provides a separate flow for outstanding amounts. When an unpaid or loaned amount is recorded, the information remains available for monitoring until a full or partial payment is recorded. The **Reservation Management** module operates alongside the active-session process by maintaining scheduled workspace use and allowing staff to view and update reservations.

Administrative activities support all of these processes. Menu information maintained through **Menu Management** affects the products available in the POS. User accounts and roles determine which functions are accessible to staff and administrators. Expense records can contribute to financial reports,

while system settings support the configuration of the application's operational behavior.

System Output and Benefits

After the different operational processes have been completed, the system produces several outputs that support IdeaHub's daily operations. These include completed customer-session records, order records, billing information, payment records, updated inventory quantities, receivable balances, reservation schedules, sales summaries, expense information, and downloadable reports.

The integrated process is intended to provide several operational benefits. First, it supports **faster transaction processing** by reducing repetitive manual recording and calculations. Second, it promotes **more consistent billing** by using recorded session duration and order information when determining the customer's total charge. Third, it supports **real-time inventory monitoring** by connecting completed sales transactions with stock-level updates. Fourth, it provides **centralized operational information** that can be accessed through authorized modules. Finally, reports and dashboard information can assist the Owner/Administrator in monitoring business activities and making decisions based on recorded operational data.

Overall, Figure 4.5 demonstrates that the developed system is not limited to a single POS or billing function. It represents an integrated operational management system in which customer sessions, workspace usage, food orders, billing, payments, inventory, receivables, reservations, reports, and administrative functions are connected through a centralized database. The complete flow begins with secure user authentication, continues through the operational processes performed by staff and administrators, and ends with organized transaction records and management information that can support the daily operation of IdeaHub Co-working Space + Café.

4.5 System Workflow Diagram

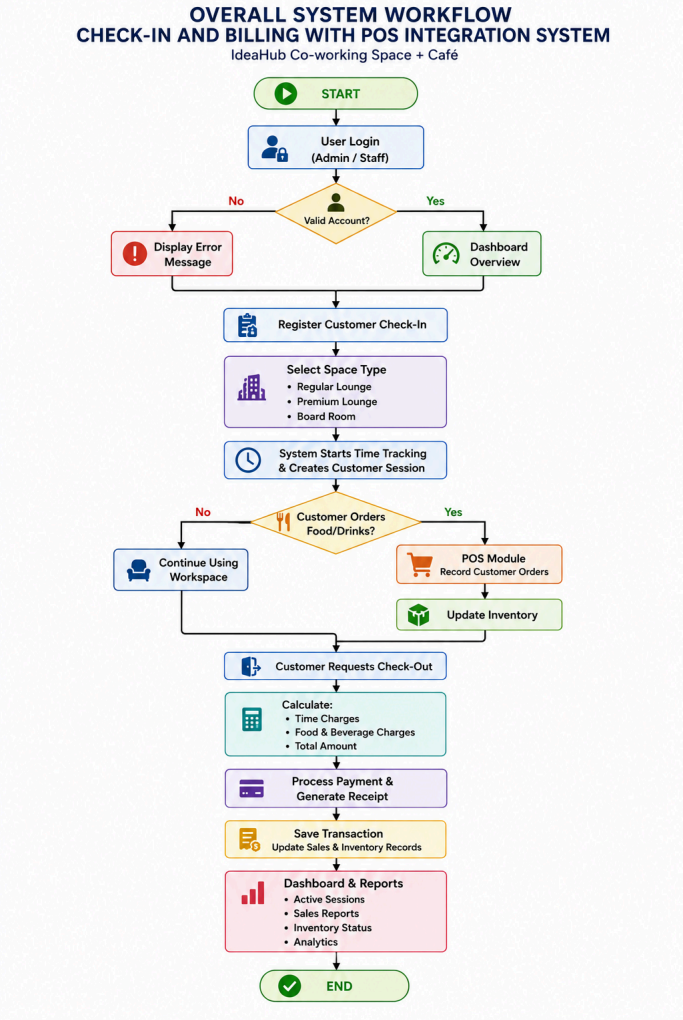


Fig. 4.6. System Workflow of the IdeaHub Co-working Space + Café System

Figure 4.6 illustrates the overall workflow of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. The workflow presents the sequence of processes from user authentication and customer check-in to session monitoring, POS ordering, billing, payment, transaction recording, and reporting. It also demonstrates how the major system modules interact within the centralized web-based platform.

The workflow begins when an authorized user accesses the system and enters the required login credentials. The system validates the submitted credentials against the registered user information stored in the database. If the credentials are invalid, access is denied and the user is prompted to enter valid credentials. If authentication is successful, the user is granted access according to the assigned role and redirected to the system dashboard.

The **Dashboard** serves as the primary interface for accessing the operational functions of the system. It provides authorized users with relevant information regarding active customer sessions, transactions, sales, inventory, and other operational records available within the system.

The customer transaction process begins with **Customer Check-In**. The staff records the required customer information and selects the appropriate workspace type. The available workspace types include the **Regular Lounge, Premium Lounge, and Boardroom**, each of which has a corresponding rate used for billing computation.

After check-in, the system creates an active customer session and records the customer's **time-in**. The session remains active throughout the customer's use of the selected workspace. The recorded time-in and eventual time-out are used to determine the duration of the session and calculate the applicable workspace charge.

While the session is active, the customer may purchase food or beverages. If an order is placed, the staff records the selected menu items and quantities through the integrated **Point-of-Sale (POS) module**. The recorded order is associated with the customer's active session so that the corresponding charges can be included in the final bill. Applicable inventory records are also updated according to the recorded transaction.

If no food or beverage order is placed, the customer may continue using the assigned workspace while the session remains active. Both workflow paths eventually proceed to **Customer Check-Out** when the customer requests to end the session.

During check-out, the system records the customer's **time-out** and determines the total session duration. The billing process then calculates the applicable workspace charge based on the session duration and workspace rate. Any food and beverage purchases associated with the session are included in the computation. The customer's total bill is determined using the following relationship:

Total Bill = Workspace Time Charge + Food and Beverage Charges

After the bill has been determined, the transaction proceeds to **payment processing**. The applicable payment information is recorded, and the completed transaction data are stored in the system database. Relevant sales, inventory, financial, and other operational records are updated according to the completed transaction.

The completed transaction is subsequently reflected in the system's **Dashboard, Sales and Analytics, and Reporting** functions. These modules allow authorized users to review recorded operational and transaction information generated by the system.

The workflow ends after the customer transaction has been completed and the corresponding information has been recorded. Through this process, customer check-in and check-out, workspace

usage, POS orders, billing, payment, inventory updates, and transaction records are integrated within a single web-based system.

Overall, the system workflow illustrates how the developed platform connects the major operational processes of **IdeaHub Co-working Space + Café**. The workflow provides a structured sequence for processing customer sessions and related transactions while maintaining the resulting information within the system's centralized database.

4.6 System Implementation

This section presents the implementation of the major functional modules of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. The developed system was implemented as a centralized web-based platform intended to support the daily operational, transactional, financial, and administrative activities of IdeaHub Co-working Space + Café. The implementation translated the identified business requirements and system design into functional modules that can be accessed and used by authorized personnel through the system interface.

The system was developed to provide a centralized environment for managing the different activities involved in the operation of the establishment. Instead of maintaining customer information, workspace usage, food orders, payments, inventory records, reservations, and financial information through separate manual records, the developed system organizes these activities within interconnected modules. This allows information entered during one operational activity to be utilized by other related processes, reducing unnecessary duplication of data entry and supporting more consistent record management.

The implemented system consists of **User Authentication and Role-Based Access, Dashboard, Customer Check-In and Check-Out, Customer Session and Workspace Management, POS and Order Management, Billing and Payment, Inventory Management, Reservation Management, Financial Management, Sales and Analytics, Reporting, Staff Management, User Management, and System Settings**. Each module was designed according to its specific operational purpose while remaining connected to the overall system. The modules collectively provide the functions required to manage customer sessions, workspace usage, café transactions, billing, inventory, reservations, financial records, and management information.

The **User Authentication and Role-Based Access** component serves as the entry point of the system. Users are required to provide valid account credentials before accessing the application. After successful authentication, the system identifies the user's assigned role and provides access to the functions permitted for that role. Staff users are primarily provided with operational functions necessary for handling daily activities, while the Owner/Administrator is provided with broader access to management, reporting, user, and system-administration functions. This implementation helps ensure that sensitive operational and administrative information is accessed only by authorized users.

The **Customer Check-In and Check-Out** and **Customer Session and Workspace Management** modules work together to manage customer visits from arrival until departure. During check-in, customer information, group size, selected workspace, and arrival time are recorded. The resulting session is maintained as an active customer session while the customer remains in the establishment. At checkout,

the system records the departure time and determines the customer's session duration, which becomes an important component in calculating the applicable workspace charge.

The **POS and Order Management** module is integrated with active customer sessions to allow staff to record food and beverage purchases under the appropriate customer session. Menu items and quantities can be selected, modified, or removed when necessary. The recorded order information is then made available to the billing process. This integration allows the customer's workspace charges and café purchases to be considered together when preparing the final bill.

The **Billing and Payment** module processes the financial settlement of completed customer sessions. The system uses the recorded workspace duration and applicable workspace rate to determine the time-based charge and combines this amount with valid food and beverage orders. The resulting total is presented for payment processing. The system supports the designated payment methods, including **Cash and GCash**, and records the payment information as part of the completed transaction. A receipt can subsequently be generated to provide a record of the customer's completed payment.

The **Inventory Management** module is connected to the POS process to maintain current information about available café products. When applicable transactions are completed, the corresponding inventory quantities are deducted according to the items sold. Updated stock levels can then be viewed by authorized users. The system also provides low-stock monitoring to help identify products that have reached or fallen below their configured stock threshold. This integration allows inventory information to be maintained alongside actual sales activity.

The **Reservation Management** module supports the scheduling and monitoring of reservable spaces, such as the boardroom. Authorized users can create reservations, view existing schedules, and update or cancel bookings when necessary. The module provides a centralized record of scheduled workspace use, allowing staff to review availability and manage reservations more systematically.

The system also includes **Financial Management** functions for recording and monitoring financial information associated with the establishment. These functions include the management of accounts receivable or outstanding amounts, partial or full payments toward balances, and operational expenses. The information recorded through these functions can be used by the system's reporting and analytical components to provide a more organized view of the establishment's financial activities.

The **Sales and Analytics** module consolidates relevant transaction information into summarized views that can assist the Owner/Administrator in monitoring business activities. Sales information, payment records, inventory usage, receivables, and expenses can be processed into management-oriented information. The **Dashboard** provides an overview of selected operational and sales information, allowing authorized users to obtain a quick view of current business activities without manually reviewing individual records.

The **Reporting** module provides mechanisms for generating formal reports from stored system data. Reports may include daily sales, payment summaries, inventory usage, accounts receivable, and expense information. Generated reports can be viewed and exported in PDF format for documentation and management purposes. This function allows the information stored in the system to be transformed into organized records that can support monitoring, evaluation, and decision-making.

The **Staff Management and User Management** functions support the administration of personnel

accounts within the system. Authorized administrators can create and manage user accounts and assign appropriate roles and access permissions. This ensures that employees can access the functions required for their assigned responsibilities while administrative functions remain restricted to authorized personnel.

The **System Settings** module provides the Owner/Administrator with functions for maintaining applicable system configurations and preferences. Menu information can also be maintained through the administrative functions, allowing menu items and categories to be added, modified, or deactivated when business requirements change. This enables the system to remain aligned with the current products, users, and operational conditions of IdeaHub.

All implemented modules operate within a **centralized database environment**. Information generated from customer sessions, orders, transactions, inventory activities, reservations, payments, expenses, users, and other system functions is stored in the MySQL database. The centralized structure allows related modules to retrieve and use existing information rather than requiring the same information to be repeatedly entered into different parts of the system. For example, an active customer session can be selected when recording an order, the order can be included in the customer's final bill, and the completed transaction can subsequently contribute to inventory updates and sales reports.

The implementation therefore establishes an interconnected workflow across the major operations of IdeaHub. Customer information begins with the check-in process, continues through workspace usage and order processing, and proceeds to billing and payment during checkout. Completed transactions can then contribute to inventory monitoring, financial records, sales summaries, and reports. At the same time, reservation, staff, user, and system-management functions provide supporting administrative capabilities necessary for the continued operation of the establishment.

Overall, the implementation of these modules provides IdeaHub Co-working Space + Café with a centralized platform for managing its major operational and administrative activities. The developed system is intended to reduce dependence on fragmented manual records, improve the organization of operational information, support more consistent transaction processing, and provide authorized personnel with timely access to relevant business information. The following subsections present the implementation and functionality of each major module, including their corresponding interfaces, primary functions, processes, and contribution to the overall operation of the system.

4.6.1 Login Page

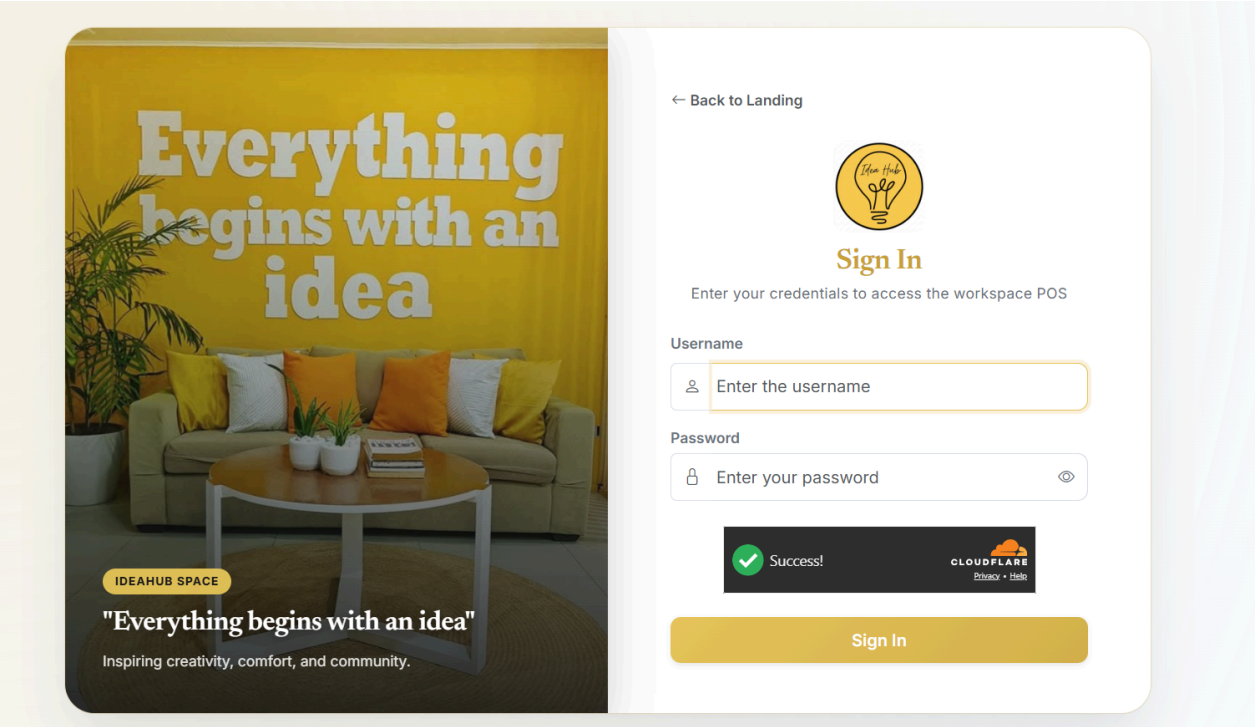


Figure 4.7. Login Page of the IdeaHub Co-working Space + Café System

Figure 4.7 presents the **Login Page** of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. The page serves as the primary authentication interface through which registered users gain access to the system.

The interface requires users to enter their registered **username and password** before authentication can proceed. The password field conceals the entered characters and provides a visibility control that allows the user to temporarily view the password when necessary. The page also incorporates a verification mechanism before the sign-in request is submitted.

When the **Sign In** button is selected, the submitted credentials are validated against the registered user account information maintained by the system. Valid credentials allow the user to proceed to the appropriate system interface, while invalid credentials prevent access. After successful authentication, the system uses the assigned user role to determine the functions and modules available to the account.

The Login Page also incorporates the visual identity of **IdeaHub Co-working Space + Café**, including its logo, establishment imagery, and the tagline *"Everything begins with an idea."* The interface was designed to provide a straightforward entry point to the system while maintaining authentication as a prerequisite for accessing protected operational and administrative functions.

4.6.2 Admin Panel

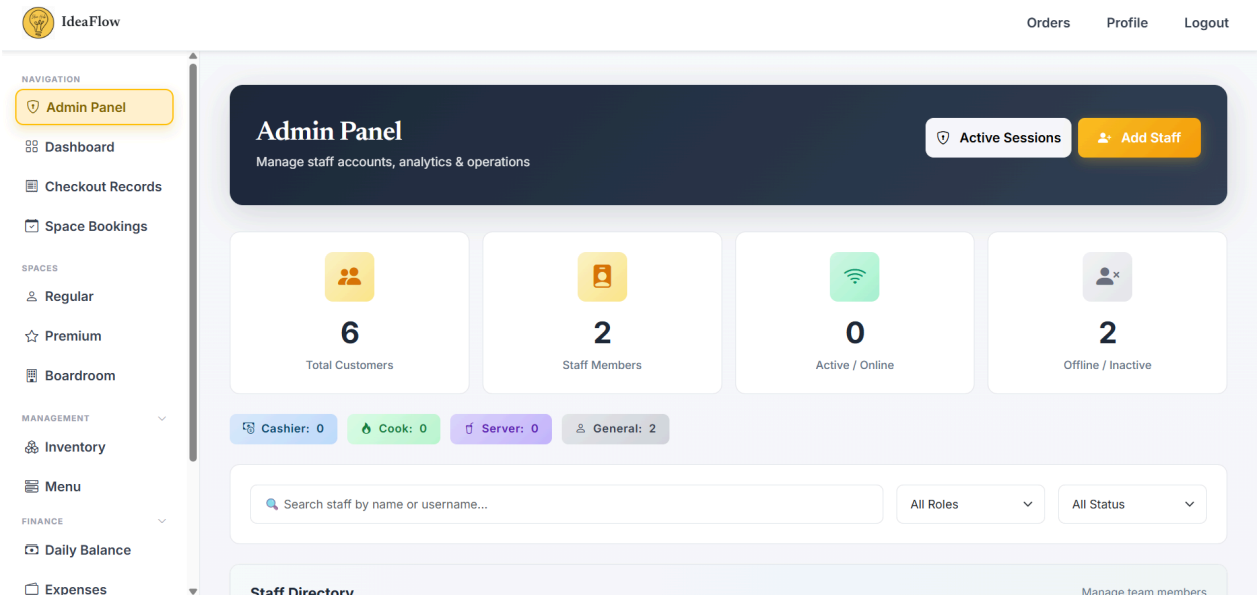


Figure 4.8. Admin Panel of the IdeaHub Co-working Space + Café System

Figure 4.8 presents the Admin Panel of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. The Admin Panel serves as the primary management interface for authorized administrator accounts and provides access to administrative, operational, and staff-management functions of the system.

The interface presents summary information regarding the current number of customers, registered staff members, active or online users, and offline or inactive users. Staff members are also categorized according to their assigned job roles, such as cashier, cook, server, and general staff. These summaries provide the administrator with an organized view of user and staff information maintained by the system.

The panel provides an Add Staff function for creating staff accounts and assigning the appropriate account information and roles. Staff records can be located using the search function and filtered according to role and account status. These functions support the management and organization of staff accounts within the system.

The navigation panel provides the administrator with access to the major operational and management functions, including the Dashboard, Checkout Records, Space Bookings, Regular Lounge, Premium Lounge, Boardroom, Inventory, Menu, Daily Balance, Expenses, and other authorized administrative functions. The Active Sessions function also provides direct access to ongoing customer sessions.

The separation of administrative and operational functions supports the system's role-based access structure. Administrator accounts are provided with broader access to management functions, while staff accounts are limited to the functions associated with their assigned permissions and responsibilities.

4.6.3 Checkout Records

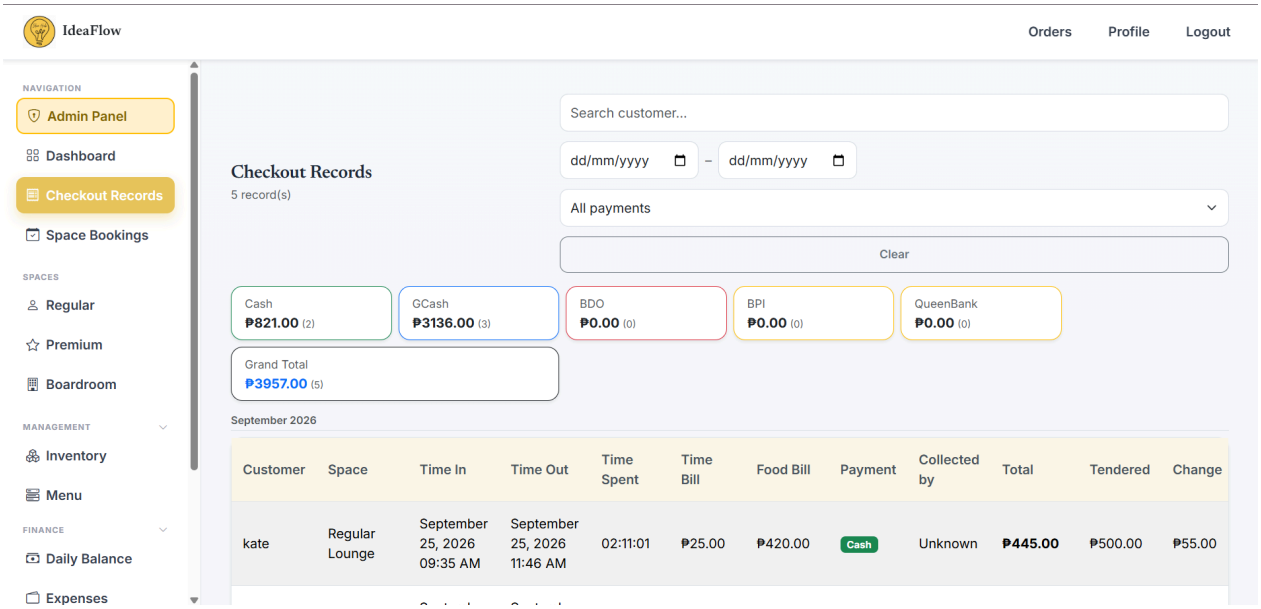


Figure 4.9. Checkout Records Page of the IdeaHub Co-working Space + Café System

Figure 4.9 presents the Checkout Records page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This interface provides authorized users with a centralized record of completed customer sessions and their corresponding billing and payment information.

The page displays transaction details including the customer name, workspace used, time-in, time-out, total time spent, workspace time bill, food bill, payment method, collecting staff, total amount, amount tendered, and change. These records allow completed customer sessions to be reviewed together with their associated workspace and POS charges.

The interface also provides search and filtering functions to assist users in locating specific checkout records. Transactions can be searched by customer, filtered according to a selected date range, and viewed according to the applicable payment method. These functions allow authorized users to review particular transactions without manually examining the entire collection of checkout records.

Payment summaries are presented according to the available payment methods, including Cash, GCash, BDO, BPI, and QueenBank, together with the corresponding recorded amounts and number of transactions. A Grand Total is also displayed to summarize the value and number of transactions represented by the current records or applied filters.

The Checkout Records page therefore serves as a historical reference for completed customer transactions. By maintaining session duration, workspace charges, food and beverage charges, and payment information within the same interface, authorized users can review the details associated with previously completed checkouts and verify the corresponding transaction records.

4.6.4 Space Booking

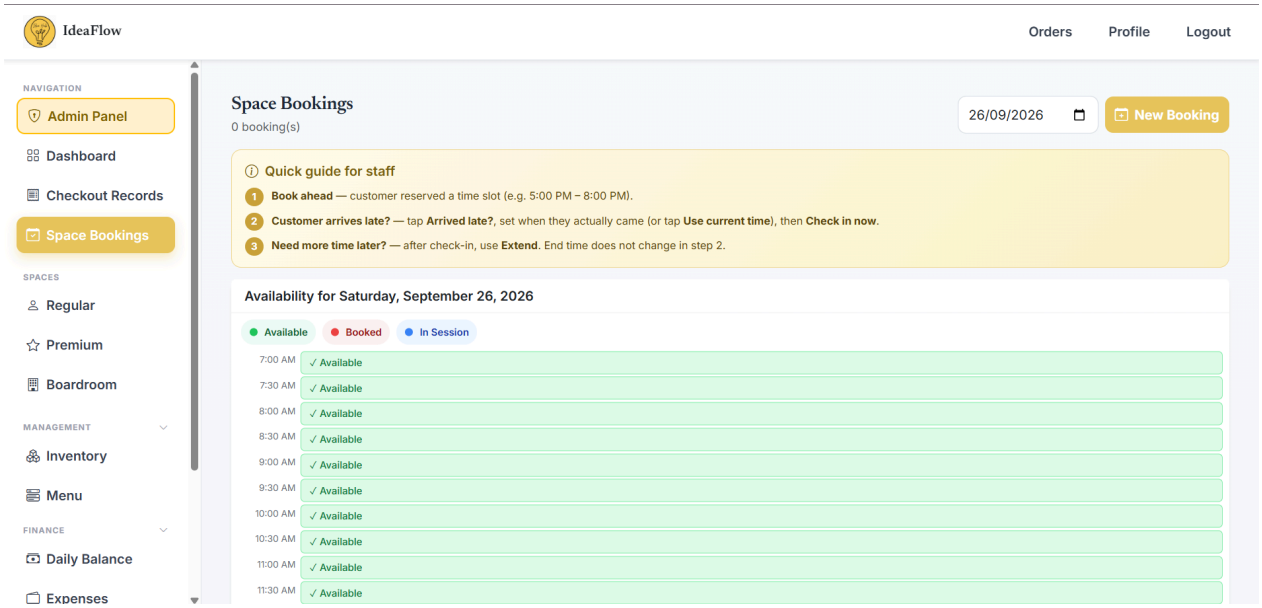


Figure 4.10. Space Booking Page of the IdeaHub Co-working Space + Café System

Figure 4.10 presents the Space Booking page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. The module was implemented to support advance reservations for the Boardroom and the entire IdeaHub space, allowing authorized users to record and manage scheduled use of these facilities.

The interface provides a date-based schedule that allows staff to review the availability of the selected space before creating a reservation. Available time slots are displayed according to scheduled intervals, while visual status indicators distinguish between Available, Booked, and In Session periods. This provides staff with a centralized view of the reservation schedule for the selected date.

A new reservation can be created through the New Booking function. The corresponding booking information, including the customer, selected space, reservation date, scheduled start and end times, number of people, and other applicable details, is recorded through the system. Maintaining these details allows reservations to be associated with a defined schedule and facility.

The module also supports situations in which the customer's actual arrival differs from the originally scheduled time. Staff can record the applicable arrival information and initiate the customer session when the customer arrives. When additional usage time is required, the reservation or active session can be extended according to the available system function.

Existing reservations can also be reviewed and managed through the booking interface. This enables authorized users to monitor scheduled reservations, identify available time periods, and make applicable changes to booking information when necessary.

The Space Booking module therefore provides a structured method for managing reservations of the Boardroom and the entire IdeaHub facility. By maintaining reservation schedules and their corresponding statuses within the system, staff can review facility availability and manage advance bookings together with the other operational information maintained by IdeaHub Co-working Space + Café.

4.6.5 Regular Lounge Customer Sessions

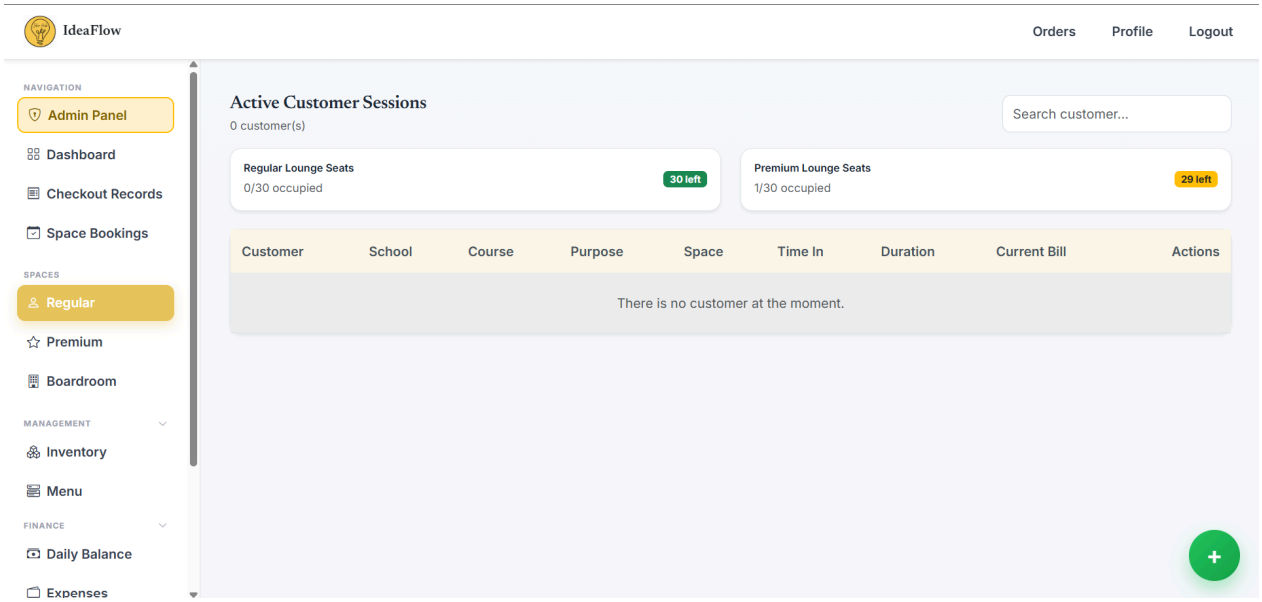


Figure 4.11. Regular Lounge Customer Sessions Page of the IdeaHub Coworking Space + Café System

Figure 4.11 presents the Regular Lounge Customer Sessions page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This interface allows authorized users to monitor customers currently using the Regular Lounge and provides information regarding workspace occupancy and active customer sessions.

The upper portion of the interface displays the current occupancy of the Regular Lounge and Premium Lounge, including the number of occupied and remaining seats. These indicators allow staff to view the recorded workspace capacity and determine the availability of lounge seating before registering additional customer sessions.

Active customer information is organized in a table containing the customer name, school, course, purpose, assigned space, time-in, session duration, current bill, and available actions. The session duration and current billing information allow staff to monitor an ongoing customer's workspace usage while the session remains active.

A search function is also provided to allow staff to locate a particular customer from the list of active sessions. When no customer is currently registered under the selected workspace, the interface displays an appropriate message indicating that there are no active customers.

The Add Customer function, represented by the add button, allows staff to initiate a new customer

check-in. During this process, the required customer and workspace information is recorded and a new active session is created. The session remains available on the page until the customer completes the check-out process.

The navigation between Regular, Premium, and Boardroom provides separate access to the applicable workspace-management interfaces. This organization allows staff to monitor customer sessions according to the type of workspace being used while maintaining the corresponding information within the centralized system.

4.6.6 Premium Lounge Customer Sessions

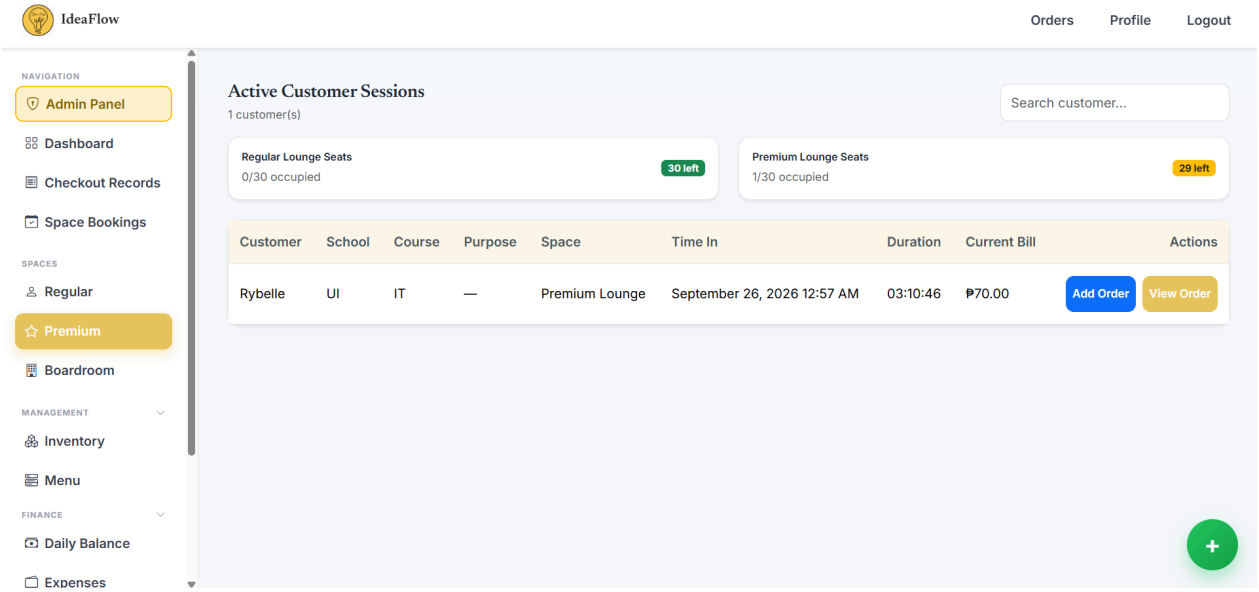


Figure 4.12. Premium Lounge Customer Sessions Page of the IdeaHub Co-working Space + Café System

Figure 4.12 presents the Premium Lounge Customer Sessions page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This interface allows authorized users to monitor customers currently assigned to the Premium Lounge and provides the corresponding session, occupancy, billing, and order information.

The interface displays the current occupancy of both the Regular Lounge and Premium Lounge, including the number of occupied and remaining seats. In the Premium Lounge view, customers with active Premium Lounge sessions are displayed in the session table, allowing staff to monitor the utilization of the workspace.

Each active session presents relevant customer and session information, including the customer name, school, course, purpose, assigned space, time-in, current session duration, and current bill. The duration is updated while the session remains active, while the current bill reflects the applicable workspace charge based on the recorded session information and Premium Lounge rate.

The Actions column provides access to POS-related functions for the active customer. The Add Order function allows staff to record food and beverage purchases and associate them with the corresponding

customer session. The View Order function allows authorized users to review the orders already associated with that session. These functions connect workspace usage with the system's POS and billing processes.

A search function is also provided for locating specific customers from the active session records. The add button allows staff to initiate a new customer check-in and assign the appropriate workspace according to the customer's selected service and available capacity.

4.6.7 Boardroom Walk-In Customer Sessions

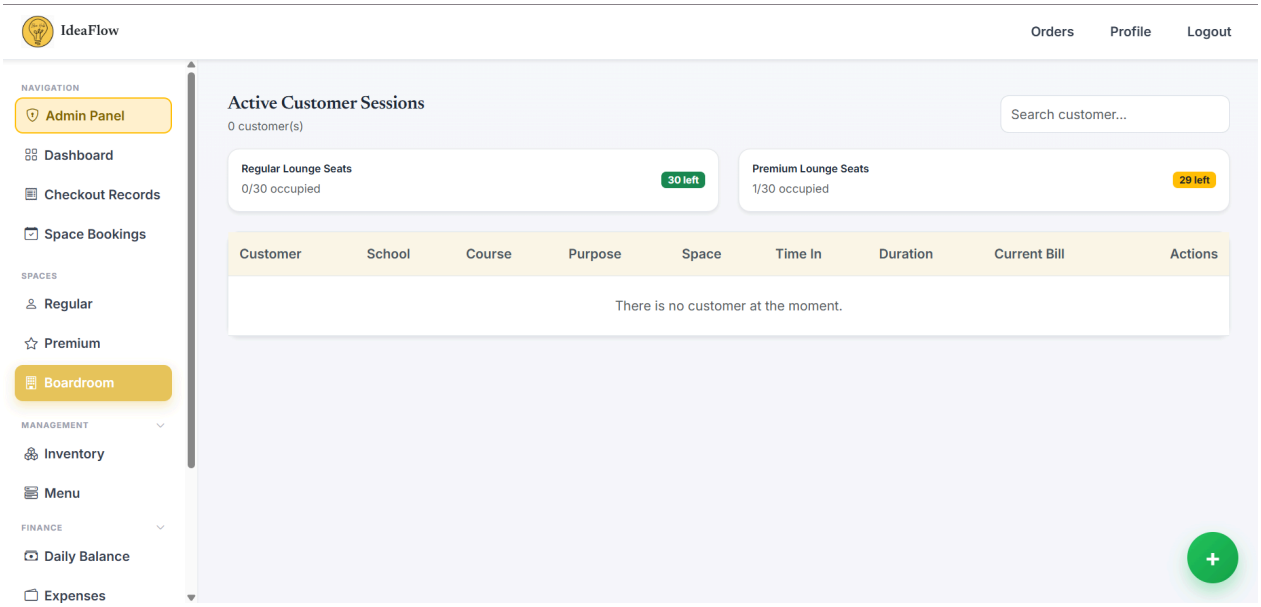


Figure 4.13. Boardroom Walk-In Customer Sessions Page of the IdeaHub Co-working Space + Café System

Figure 4.13 presents the Boardroom Walk-In Customer Sessions page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This interface supports customers who use the Boardroom without an advance reservation and allows authorized staff to manage their sessions through the same check-in and billing process used for other workspace types.

For walk-in customers, staff can initiate a new session using the Add Customer function. The required customer information is recorded and the Boardroom is selected as the assigned workspace. Once the check-in process is completed, the system records the customer's time-in and creates an active Boardroom session.

Active Boardroom sessions are displayed in a table containing relevant information such as the customer name, school, course, purpose, assigned space, time-in, session duration, current bill, and available actions. This allows staff to monitor the customer's ongoing Boardroom usage and the

corresponding workspace charge while the session remains active.

The interface also provides a search function for locating a specific customer among active sessions. When no walk-in customer is currently using the Boardroom, the system displays a message indicating that there are no active customers.

During an active Boardroom session, applicable customer orders can be associated with the session through the system's POS functions. When the customer requests check-out, the recorded session duration and applicable Boardroom rates are used to determine the workspace charge, while any associated food and beverage purchases are included in the final billing process.

The Boardroom Walk-In Customer Sessions page operates separately from the Space Booking module, which is used for advance reservations. This distinction allows the system to support both scheduled Boardroom reservations and customers who request immediate Boardroom use, subject to the availability of the space.

4.6.8 Inventory Management

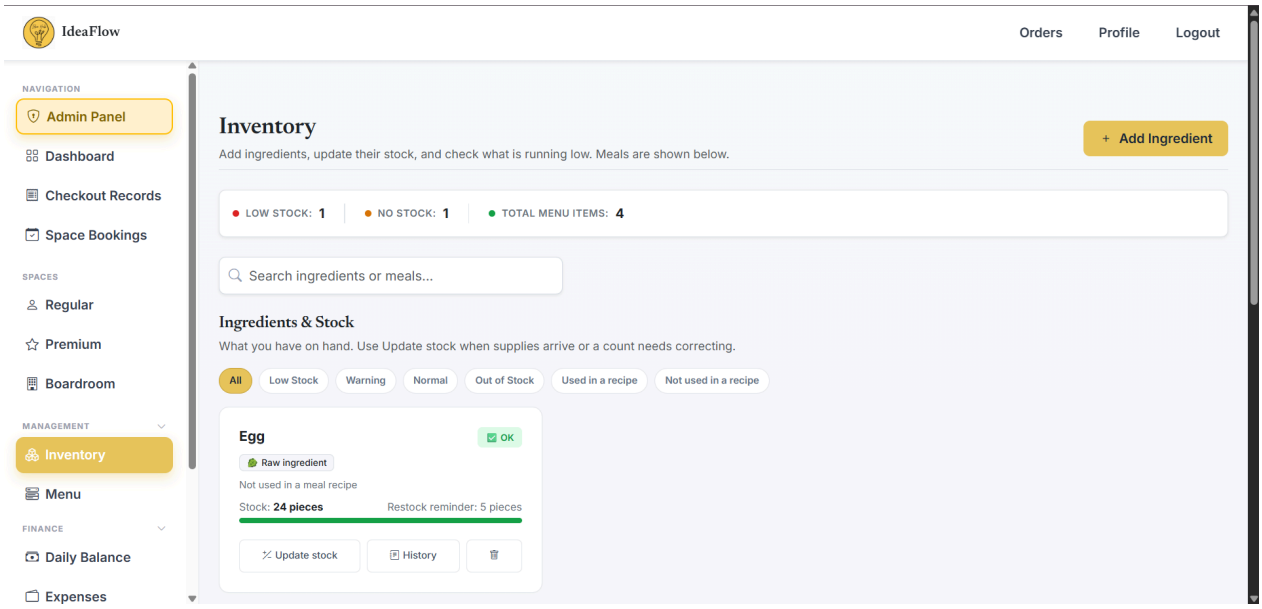


Figure 4.14. Inventory Management Page of the IdeaHub Co-working Space + Café System

Figure 4.14 presents the Inventory Management page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This module provides authorized users with an interface for maintaining ingredient and stock information used in the establishment's café operations.

The interface displays an inventory summary showing the number of items classified as Low Stock and No Stock, together with the total number of menu items maintained by the system. These indicators allow users to identify inventory items that may require attention based on their recorded stock quantities and established restock thresholds.

Individual inventory records display information such as the ingredient name, current stock quantity, unit of measurement, restock reminder level, and stock status. The system compares the recorded quantity with the established threshold to indicate whether an item has sufficient stock, is running low, or is out of stock.

The Add Ingredient function allows authorized users to create new inventory records, while the Update Stock function is used to record changes when supplies are added or when inventory quantities require correction. A History function is also provided to review recorded stock changes, supporting the tracking of inventory adjustments made within the system.

The interface includes search and filtering functions that allow users to locate specific ingredients or menu items and organize records according to conditions such as Low Stock, Warning, Normal, Out of Stock, Used in a Recipe, and Not Used in a Recipe. These functions provide a more organized method of reviewing inventory records according to their current status and use.

The inventory module is also associated with the system's menu and POS functions. Where applicable, ingredients linked to menu recipes can be reflected in inventory processing when corresponding products are recorded through the POS module. This enables stock information to remain connected with the food and beverage operations maintained by the system.

4.6.9 Menu Management

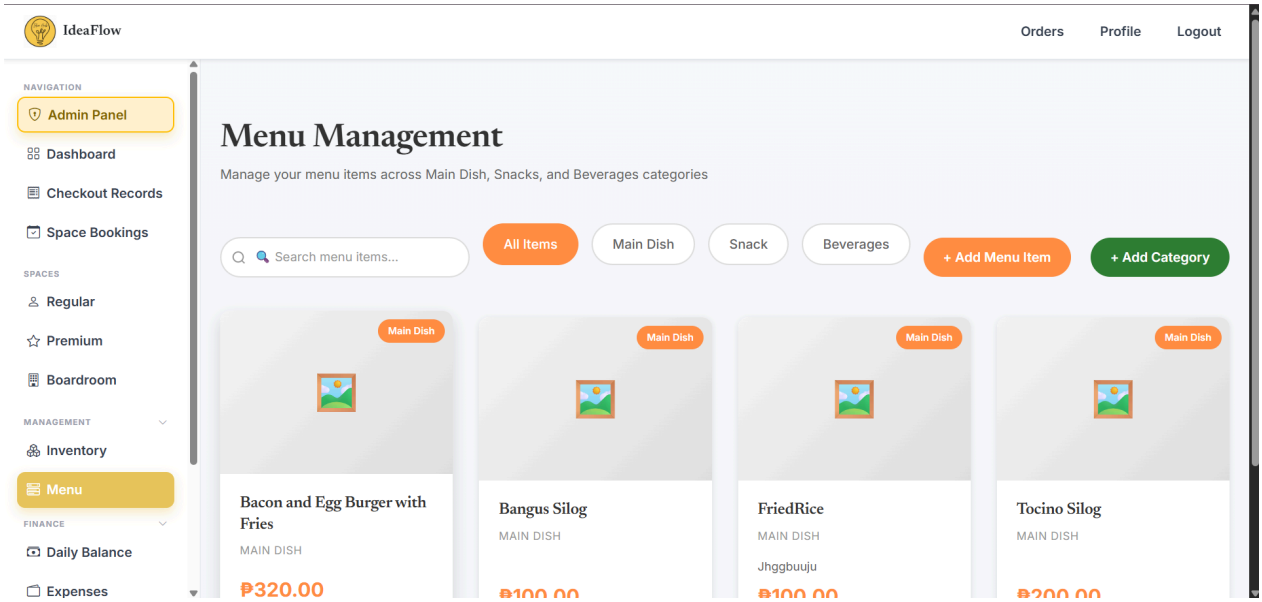


Figure 4.15. Menu Management Page of the IdeaHub Co-working Space + Café System

Figure 4.15 presents the Menu Management page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This module allows authorized users to maintain the food and beverage items offered by IdeaHub and organize them into appropriate menu categories for use in the system's POS functions.

The interface displays menu items in an organized card-based layout containing relevant product information, including the item name, category, price, description, and associated image, where available. Menu items are grouped into categories such as Main Dish, Snacks, and Beverages, allowing products to be organized according to the establishment's menu structure.

The page provides a search function that enables authorized users to locate specific menu items. Category filters are also available to display all products or limit the displayed records according to a selected menu category. These functions support the organization and retrieval of menu information as the number of available products increases.

The Add Menu Item function allows authorized users to create new menu records and provide the corresponding product information. The Add Category function enables additional menu classifications to be created when required. Existing menu information can also be maintained through the applicable management functions, allowing changes in product details, pricing, availability, and classification to be reflected within the system.

Menu records maintained through this module are associated with the system's POS and inventory functions. The menu information provides the products and prices available during order processing, while applicable ingredient and inventory relationships support the management of stock associated with café products.

4.6.10 Daily Sales Balancing

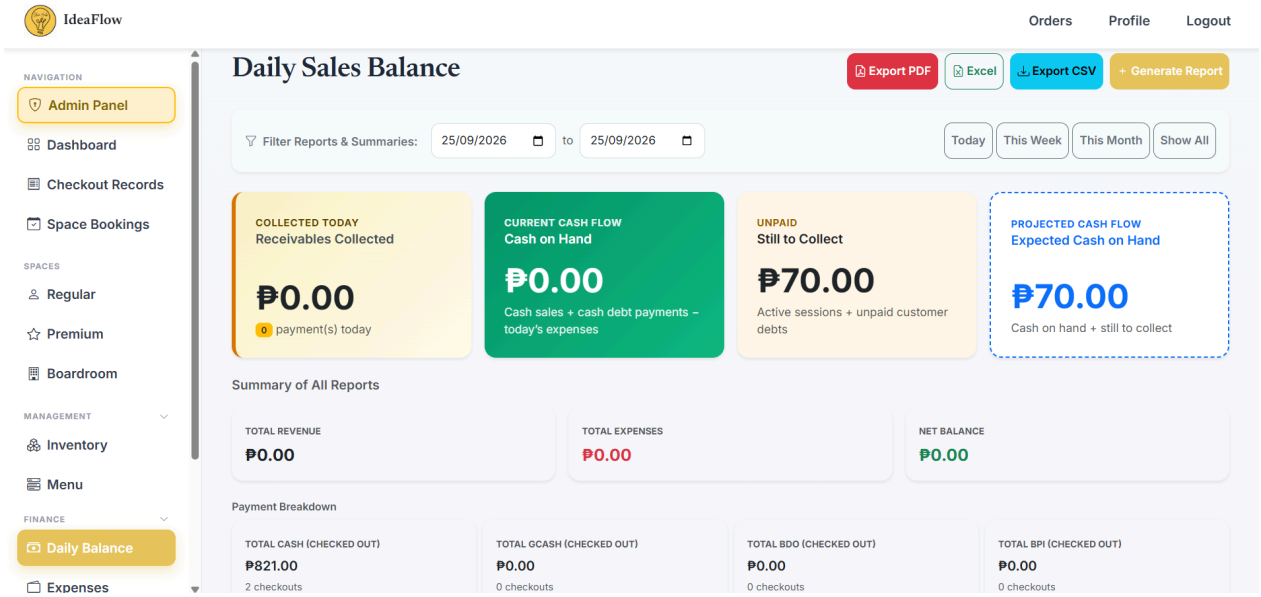


Figure 4.16. Daily Sales Balance Page of the IdeaHub Co-working Space + Café System

Figure 4.16 presents the Daily Sales Balance page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This module provides authorized users with a centralized view of recorded sales, expenses, receivables, payment collections, and other financial information used for reviewing the daily financial activities of the establishment.

The interface presents summary indicators for Receivables Collected, Cash on Hand, Unpaid Amounts, and Projected Cash Flow. Receivables Collected represents payments recorded against outstanding customer balances, while Cash on Hand summarizes applicable cash-based collections after recorded expenses. Unpaid amounts represent outstanding balances that remain to be collected, while Projected Cash Flow combines the applicable recorded cash position with outstanding amounts expected to be collected.

The Summary of All Reports section presents the recorded Total Revenue, Total Expenses, and Net Balance for the selected reporting period. These values provide management with a consolidated view of the financial records maintained by the system and support the review of daily operational transactions.

The module also provides a payment breakdown according to the payment methods recorded during customer checkout. Applicable payment channels, such as Cash, GCash, BDO, and BPI, are presented separately together with their corresponding recorded totals and transaction counts. This enables management to distinguish collections according to the payment method used.

Users can select a specific date range or use predefined periods such as Today, This Week, This Month, and Show All to review financial records for the required reporting period. The Generate Report function allows the corresponding summary to be produced from the stored records.

The interface further provides options for exporting financial information in PDF, Excel, and CSV formats. These functions allow generated financial records to be retained outside the system for documentation, review, or other applicable administrative purposes.

4.6.11 Hub and Staff Expenses

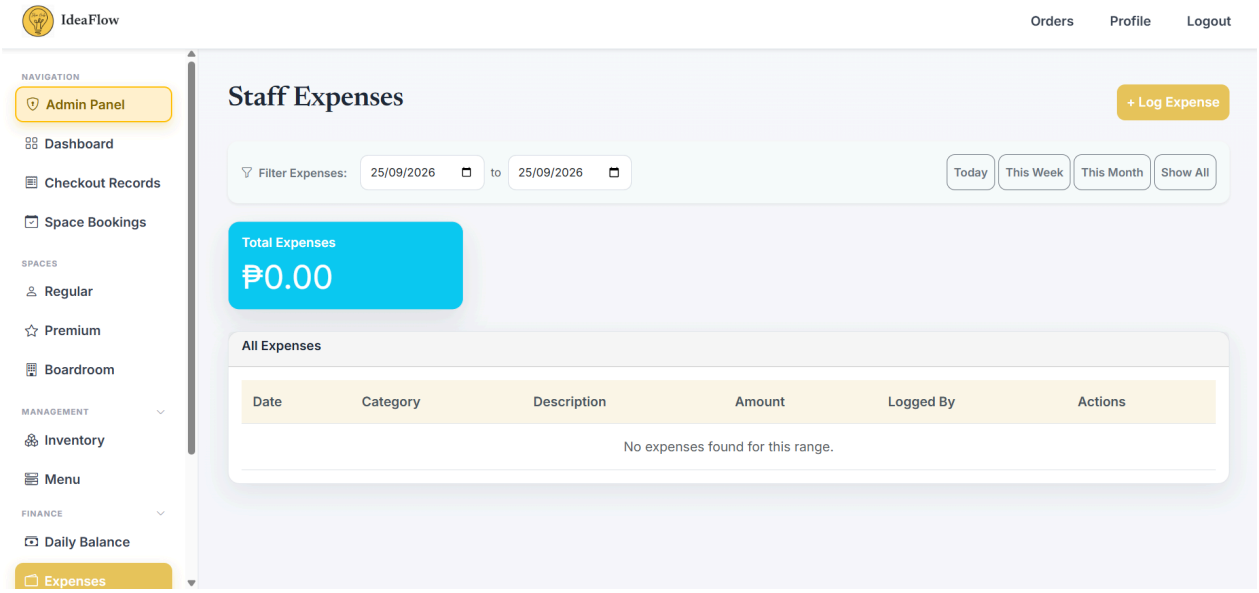


Figure 4.17. Hub and Staff Expenses Page of the IdeaHub Co-working Space + Café System

Figure 4.17 presents the Hub and Staff Expenses page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This module allows authorized users to record and monitor operational expenses incurred in relation to the daily activities of IdeaHub.

The Log Expense function allows an authorized user to create a new expense record by entering the applicable expense information. Each recorded expense contains details such as the date, category, description, amount, and staff member who logged the transaction. Identifying the user responsible for recording an expense provides a reference for the source of each entry maintained within the system.

The interface displays the Total Expenses for the selected reporting period and provides a table containing the individual expense records. Authorized users can review the recorded date, expense category, description, amount, and responsible user through the centralized expense list. Applicable actions are also provided for managing existing expense records.

Expense information can be filtered according to a specified date range or predefined periods, including Today, This Week, This Month, and Show All. These filtering options allow management to review operational expenditures according to the period relevant to financial monitoring and reporting.

The expense records maintained through this module are also used by the system's financial functions. Recorded expenses contribute to the information presented in the Daily Sales Balance, where applicable revenue and expense records are considered when determining the corresponding financial summary

and net balance for the selected period.

4.6.12 Receivables and Debt Tracking

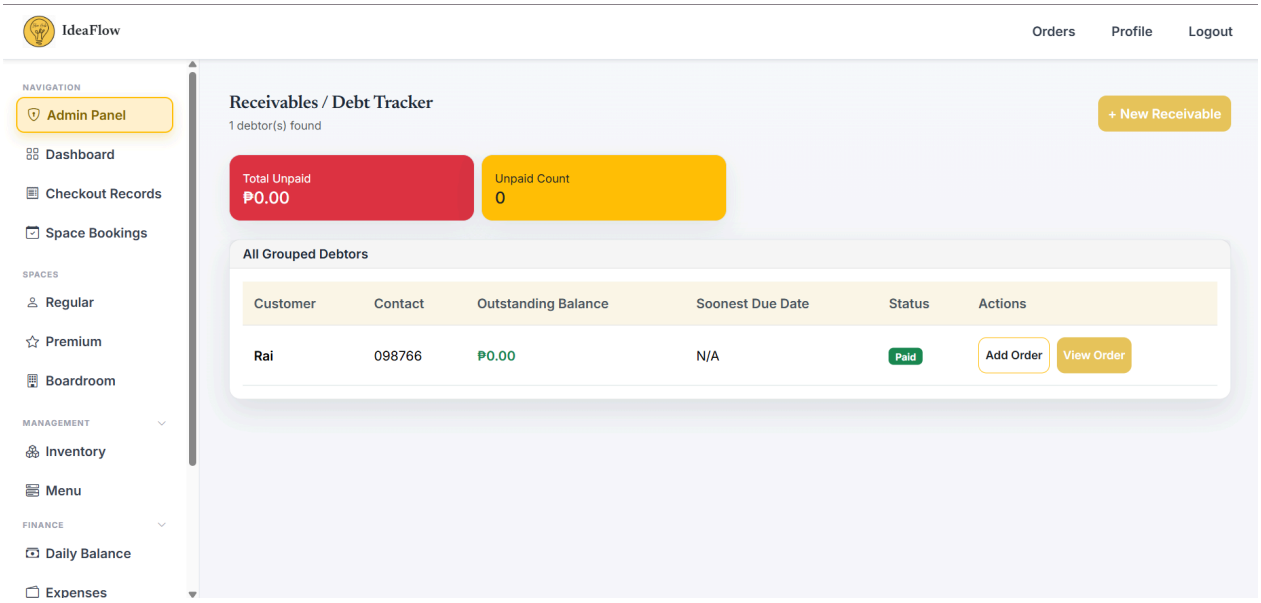


Figure 4.18. Receivables and Debt Tracking Page of the IdeaHub Co-working Space + Café System

Figure 4.18 presents the Receivables and Debt Tracking page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This module was implemented to record and monitor amounts owed by customers to the establishment and to maintain information regarding outstanding and settled balances.

The interface provides summary indicators for the Total Unpaid Amount and Unpaid Count, allowing authorized users to review the value and number of outstanding receivables recorded in the system. Customer receivables are grouped according to the corresponding debtor to provide a consolidated view of each customer's financial obligation.

The receivables table displays relevant information including the customer name, contact information, outstanding balance, soonest due date, payment status, and available actions. The status indicator identifies whether the customer's recorded balance remains unpaid or has already been settled.

The New Receivable function allows authorized users to manually create a receivable record when an amount is owed to the establishment. Relevant information, such as the customer details, amount due, description, and applicable due date, can be recorded to support subsequent monitoring and collection.

The module also provides Add Order and View Order functions for applicable customer records. These functions allow additional food or beverage transactions to be associated with the corresponding customer record and enable authorized users to review related orders maintained within the system.

As payments are made toward an outstanding balance, the corresponding receivable record can be

updated to reflect the amount collected and the remaining balance. Once the obligation has been fully settled, its status can be reflected as Paid, allowing management to distinguish completed collections from outstanding receivables.

The receivables information is also associated with the system's Daily Sales Balancing functions. Payments collected from outstanding balances can be reflected in the applicable financial records, while amounts that remain unpaid contribute to the system's recorded outstanding receivables.

4.6.13 Payables and Supplier Debt Tracking

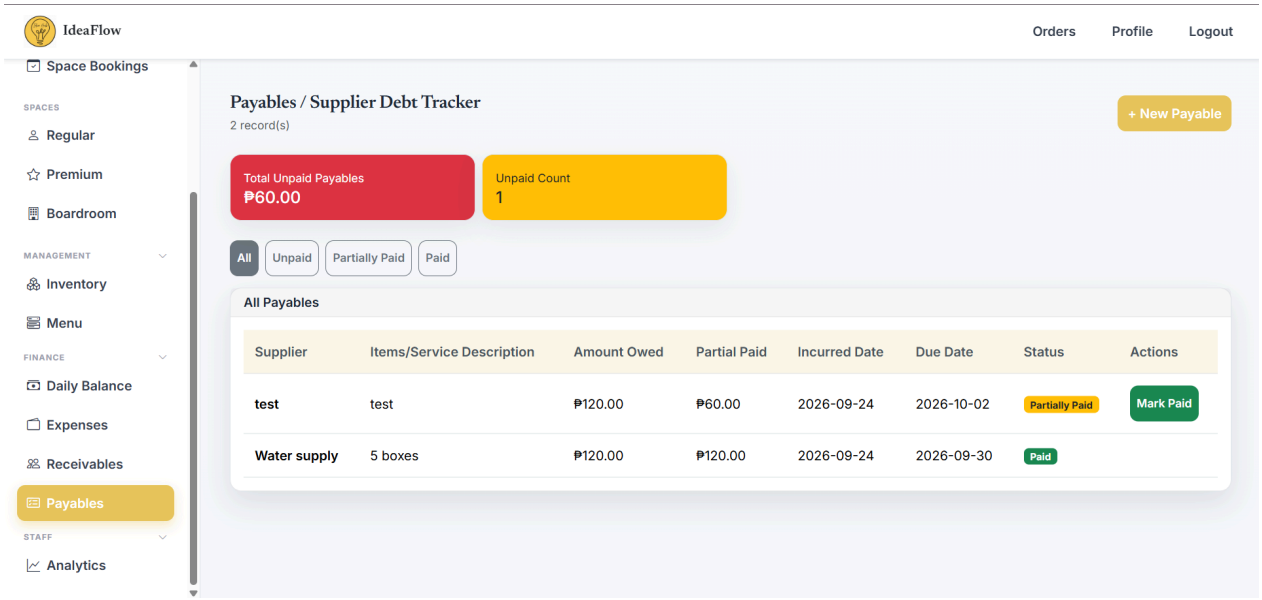


Figure 4.19. Payables and Supplier Debt Tracking Page of the IdeaHub Co-working Space + Café System

Figure 4.19 presents the Payables and Supplier Debt Tracking page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This module was implemented to record and monitor the financial obligations of IdeaHub to suppliers and service providers, including amounts that remain unpaid, partially paid, or fully settled.

The interface provides summary indicators for the Total Unpaid Payables and Unpaid Count, allowing authorized users to view the remaining financial obligations recorded in the system. These indicators provide management with a consolidated view of outstanding amounts that still require settlement.

Each payable record contains relevant information including the supplier, item or service description, amount owed, amount partially paid, incurred date, due date, payment status, and available actions. These details allow management to review the origin, amount, and payment condition of each financial obligation.

The New Payable function allows authorized users to create a payable record when the establishment obtains goods or services for which payment is due. The corresponding supplier information, description, amount, incurred date, and due date can be recorded to maintain an organized history of the

establishment's obligations.

The module classifies payable records according to their payment status, including Unpaid, Partially Paid, and Paid. Filtering options allow users to display records according to these classifications. When a portion of an obligation has been settled, the corresponding payment information can be reflected in the record, while fully settled obligations are identified as Paid.

The Mark Paid function allows an authorized user to update an applicable payable once the remaining obligation has been settled. Maintaining both active and completed payable records provides management with a reference for previously recorded supplier obligations and their payment status.

4.6.14 Business Intelligence and Analytics

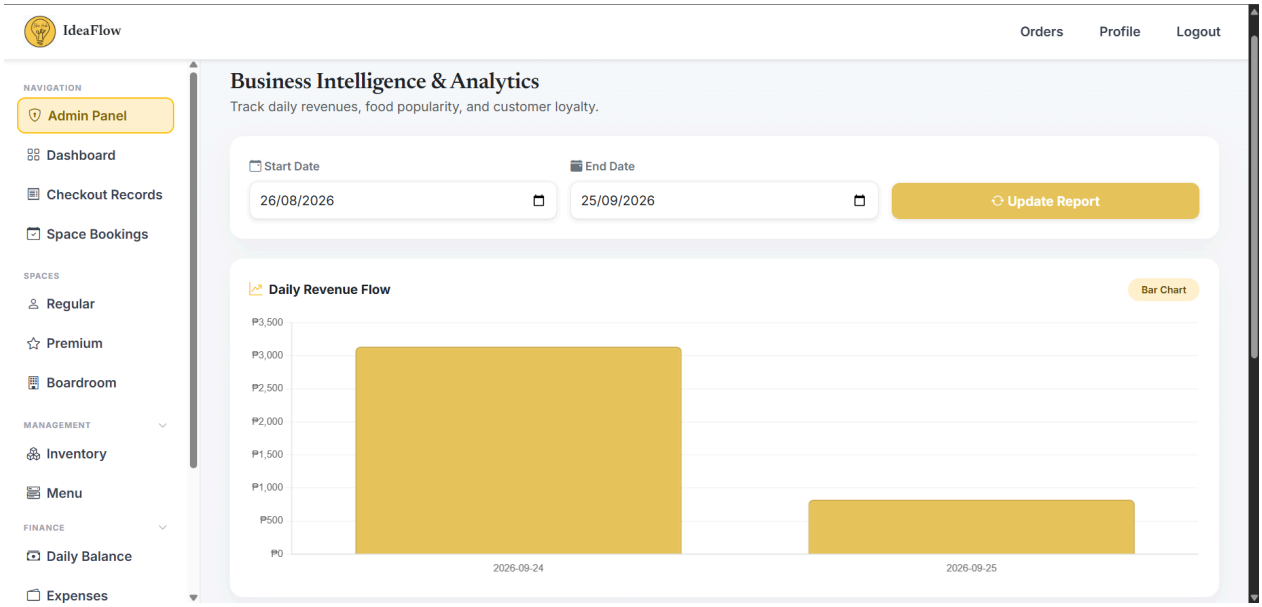


Figure 4.20. Business Intelligence and Analytics Page of the IdeaHub Co-working Space + Café System

Figure 4.20 presents the Business Intelligence and Analytics page of the Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café. This module provides authorized users with graphical and summarized views of operational and sales information recorded within the system.

The interface allows users to specify a start date and end date to define the reporting period. Once the selected dates are applied through the Update Report function, the module retrieves the corresponding records and updates the displayed analytical information. This enables management to review business data according to a specific period rather than being limited to a single reporting date.

One of the primary visualizations presented by the module is the Daily Revenue Flow, which displays recorded revenue according to date. The graphical presentation allows management to compare revenue across the selected period and observe variations in recorded sales activity.

In addition to revenue information, the analytics module provides information related to food popularity and customer activity, based on the transaction and operational records maintained by the system. These analytical views organize existing system data into summarized information that can be reviewed by management when examining business operations.

The module obtains its information from records generated through interconnected system functions such as customer sessions, POS transactions, billing, payments, and other applicable operational records. This allows analytical information to be produced from data already maintained within the centralized database rather than requiring separate manual consolidation.

4.7 System Testing

System testing was conducted to verify whether the developed **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** performed according to its specified functional and operational requirements. The testing process was conducted to identify functional issues, inconsistencies, and integration problems prior to the final evaluation of the system.

The testing activities consisted of **Functional Testing, Integration Testing, and User Acceptance Testing (UAT)**. Functional testing examined the individual features of the system, integration testing verified the interaction and flow of information among interconnected modules, and UAT determined whether the completed system satisfied the operational requirements established by the client. These testing procedures provided a structured basis for verifying the functionality and integration of the developed system.

4.7.1 Functional Testing

Functional testing was conducted to determine whether each major system feature produced the expected output according to its intended function. Each test case involved providing the required input, performing the corresponding system operation, observing the resulting output, and comparing the actual result with the expected result. A test case was considered successful when the actual behavior of the system corresponded to the defined expected result.

The functional testing covered the major operational, transactional, and financial functions of the developed system. These included **user authentication, customer check-in, workspace selection, time tracking, POS ordering, inventory updating, customer checkout, automated billing, transaction recording, payables, receivables, debt collection, daily soft balancing, sales and analytics, and report generation**.

The testing of **user authentication** verified that users with valid credentials could access the system while users providing invalid credentials were prevented from entering the system. This was important because the system contains operational, customer, transaction, and financial information that should only be accessible to authorized users.

The **customer check-in and workspace selection** functions were tested to determine whether customer information, selected workspace, and time-in information could be properly recorded. The test also verified whether the selected workspace was correctly associated with the customer's active session. This

information serves as an important basis for monitoring customer occupancy and calculating the applicable workspace charge.

The **time tracking** function was tested to verify whether the system could maintain the customer's active session and record the information necessary to determine the duration of the customer's stay. Since IdeaHub uses time-based workspace rates, accurate session information is essential to the billing process.

The **POS ordering** function was tested by adding food and beverage items to an active customer session. The test verified whether the selected items and quantities were correctly recorded. Order modification and related transaction behavior were also considered to ensure that the order information displayed by the system reflected the current transaction.

The **inventory update** function was tested to verify whether product quantities were properly adjusted after applicable sales transactions. This was important because the inventory module is connected to the POS module. A successful transaction should therefore be reflected in the corresponding inventory records.

The **customer checkout** function was tested to verify whether an active customer session could be properly completed. The test also examined whether the customer's session information could be used by the billing function to determine the appropriate time-based charge.

The **automatic billing** function was tested to verify the calculation of the customer's time bill and food bill. The system was expected to combine the applicable workspace charge and recorded food and beverage purchases to produce the customer's total bill. This function was particularly important because incorrect calculations could result in inaccurate customer charges and financial records.

The **transaction recording** function was tested to determine whether completed transactions were properly stored in the centralized database. This ensured that completed transactions could subsequently be retrieved for financial monitoring, sales analysis, and report generation.

The financial-management functions, including **payables, receivables, debt collection, and daily soft balancing**, were also tested. Payables testing verified whether financial obligations could be recorded and monitored. Receivables testing verified whether amounts owed to the establishment could be recorded and tracked. Debt collection testing examined whether outstanding balances and collection information could be properly maintained. Daily soft balancing was tested to determine whether daily financial records could be reviewed and compared for operational balancing purposes.

The **Sales and Analytics** function was tested to verify whether recorded transaction information could be presented through appropriate summaries and visualizations. This provides management with an overview of business activities and supports the monitoring of sales performance.

Finally, **report generation** was tested to determine whether the system could produce reports based on the information stored in the database. The test verified whether the generated reports contained information corresponding to the recorded system data and could be used for operational and management purposes.

Test Case	Expected Result	Actual Result	Status
User Login	Valid credentials allow the user to access the system.	The user successfully accessed the system.	Passed
Invalid Login	Invalid credentials prevent access and display an appropriate message.	Access was denied and an error message was displayed.	Passed
Customer Check-In	Customer information and time-in are recorded successfully.	The customer session was successfully recorded.	Passed
Workspace Selection	The selected workspace is correctly associated with the customer session.	The selected workspace was correctly recorded.	Passed
Time Tracking	System records and monitors the customer's session duration.	Session duration was successfully monitored.	Passed
POS Ordering	Selected food and beverage items are added to the customer's order.	Items were successfully added to the order.	Passed
Inventory Update	Inventory quantity is updated after a product is ordered.	Inventory quantity was successfully updated.	Passed
Customer Checkout	The customer session is properly ended upon checkout.	The customer session was successfully completed.	Passed
Automatic Billing	The system	The total bill was	Passed

	calculates the time bill and food bill correctly.	successfully calculated.	
Transaction Recording	Completed transactions are stored in the database.	The transaction was successfully recorded.	Passed
Payables	Payable records can be added, updated, and monitored.	Payable records were successfully managed.	Passed
Receivables	Receivable records can be recorded and monitored.	Receivable records were successfully managed.	Passed
Debt Collection	Outstanding balances and collection information can be recorded.	Debt collection records were successfully managed.	Passed
Daily Soft Balancing	Daily financial records can be reviewed and balanced.	Daily records were successfully reviewed and balanced.	Passed
Sales and Analytics	Sales information is displayed using summaries and visualizations.	Sales data was successfully displayed.	Passed
Report Generation	System generates reports based on recorded data.	Reports were successfully generated.	Passed

Table 4.1 Functional Testing Results

The results presented in Table 4.1 show that **all identified functional test cases received a Passed status**. This indicates that the tested features produced results corresponding to their defined expected outputs during functional testing. The successful results covered the system's primary operational processes, beginning with user authentication and continuing through customer session management, ordering, billing, inventory updating, financial management, analytics, and reporting.

The successful result for **customer check-in, workspace selection, and time tracking** indicates that the system was able to establish and maintain the information required for customer session management. This information is particularly important to IdeaHub because workspace charges depend on the customer's recorded usage duration and selected workspace type.

The successful **POS ordering and inventory update** results indicate that the system was able to process customer orders and update related inventory information. This demonstrates the functional relationship between sales transactions and stock monitoring. Similarly, the successful **automatic billing** and **transaction recording** results indicate that completed customer transactions could be calculated and stored for subsequent use in financial and reporting functions.

The successful results for **payables, receivables, debt collection, and daily soft balancing** indicate that the developed system was also capable of supporting the financial-monitoring functions included within the scope of the project. Meanwhile, the successful results for **sales and analytics and report generation** indicate that information stored during daily operations could be transformed into summaries and reports for management purposes.

Overall, the functional testing results provide evidence that the major tested functions of the developed system operated according to their defined requirements. However, functional testing primarily examines individual functions. Because the IdeaHub system consists of interconnected modules, an additional integration-testing process was necessary to determine whether these functions continued to operate correctly when information moved between related modules.

4.7.2 Integration Testing

Integration testing was conducted to determine whether the different modules of the developed system could properly communicate, exchange information, and work together as a unified system. Unlike functional testing, which focuses primarily on whether an individual feature produces its expected result, integration testing focuses on the relationships between two or more modules and verifies whether information generated from one process is correctly transferred, processed, and utilized by another related process.

This type of testing was particularly important for the developed system because the operations of IdeaHub Co-working Space + Café involve several interconnected activities. A customer session, for example, does not operate independently from billing. The customer's selected workspace and session duration are required to determine the workspace charge, while food and beverage orders recorded through the POS are also included in the customer's final bill. Similarly, completed transactions can affect inventory records and contribute to sales, financial monitoring, and reporting. Therefore, the successful operation of one module depends on the proper exchange of information with other modules.

The first integration flow tested was the **customer transaction process**. The process begins when a customer is recorded through the **Customer Check-In** module. The customer is assigned to the appropriate workspace, and the system records the corresponding time-in information. The customer then appears in **Active Customer Sessions**, allowing staff to monitor customers who are currently using the facility. When the customer proceeds to checkout, the system retrieves the active session information and

records the time-out. The resulting session duration is then used by the **Check-Out and Billing** process to determine the customer's workspace charge. This integration ensures that information entered during check-in is preserved and utilized during checkout rather than requiring the staff to manually re-enter the customer's session information.

The **workspace billing integration** was also tested to verify whether the selected workspace type was correctly associated with the appropriate rate. Since IdeaHub provides different workspace options with different hourly rates, the system must maintain the relationship between the selected workspace, the customer's session, the duration of use, and the resulting time charge. The integration test verified that the workspace information recorded during the customer's session could be retrieved during billing and used in calculating the applicable charge.

The **POS and inventory integration** was tested by recording food and beverage orders while a customer had an active session. When an item was selected and added to the customer's order, the order information was associated with the appropriate customer session. The corresponding inventory record was then updated based on the recorded product transaction. This integration is important because the POS and inventory modules depend on related product information. A completed sale should not only appear as an order but should also be reflected in the available stock of the corresponding product.

The **POS and billing integration** was tested by placing one or more food and beverage orders during an active customer session and subsequently proceeding to checkout. The system retrieved the valid food and beverage orders associated with the customer and included their corresponding amounts in the final billing computation. The food bill was then combined with the workspace time bill to determine the customer's total amount due. This integration verifies that customer orders recorded during the session are not treated as separate transactions but are properly incorporated into the customer's final billing process.

The **customer session and transaction integration** was also examined. After the customer completed the checkout process, the session information and corresponding billing details were used to create or update the completed transaction record. This allows the system to retain a historical record of the customer's completed activity, including applicable workspace charges, food and beverage charges, payment information, and total amount. The resulting transaction record can then be utilized by other modules that require completed transaction data.

The integration between **transactions and sales analytics** was tested to determine whether completed transactions were properly reflected in sales-related information. Once transactions were recorded, the corresponding sales data could be retrieved by the Sales and Analytics module. This allows the system to transform individual transaction records into summarized information that can be used by authorized users to monitor sales activity.

The **financial-management integration** was also tested. Information generated from transactions and payments can be associated with applicable financial records, while separate records for **payables, receivables, expenses, and debt collection** can be maintained through the corresponding financial-management functions. This integration allows the system to provide a more organized view of the financial activities recorded within the establishment. For example, outstanding amounts recorded under receivables or debt collection can be monitored separately from completed sales transactions, while operational expenses and payables can be recorded for financial monitoring.

The interaction between these financial records and **Daily Soft Balancing** was also tested. Daily Soft Balancing allows the administrator to review the financial activities recorded during a particular operating period and compare the available transaction, payment, expense, and other applicable financial information. The integration ensures that the information used during the daily review originates from the records maintained within the system rather than requiring all information to be manually consolidated from separate sources.

The **reporting integration** was tested by verifying whether information stored in the different system modules could be retrieved and presented through generated reports. Transaction records, sales information, inventory-related data, payment information, expenses, receivables, and other applicable records can contribute to the reports available to authorized users. This integration is important because the reporting module depends on the accuracy and availability of information generated by the operational modules.

The following table summarizes the integration testing conducted for the developed system.

Integration Test	**Modules Involved**	**Expected Result**	**Status**
Customer Session Flow	Check-In → Active Sessions → Checkout	Customer information and session duration are maintained throughout the process.	Passed
Workspace Billing	Workspace → Time Tracking → Billing	Correct workspace rate is used to calculate the time bill.	Passed
POS and Inventory	POS → Inventory	Ordered products are reflected in inventory quantities.	Passed
POS and Billing	POS → Checkout → Billing	Food orders are included in the customer's total bill.	Passed
Session and Transaction	Check-In → Checkout → Transaction	Completed customer sessions are recorded as transactions.	Passed
Sales and Analytics	Transactions → Sales/Analytics	Completed transactions are reflected in sales information.	Passed

Financial Monitoring	Transactions → Receivables/Payables /Debt Collection	Relevant financial records are properly reflected in financial modules.	Passed
Daily Financial Monitoring	Sales/Payments/Expenses → Daily Soft Balancing	Daily financial activities can be reviewed and reconciled.	Passed
Reports	System Records → Reports	Recorded information is reflected in generated reports.	Passed

Table 4.2 presents the results of integration testing.

Table 4.2 presents the results of the integration testing performed on the major interconnected processes of the developed system. The results show that **all identified integration test cases received a Passed status**. This indicates that the tested modules were able to exchange and utilize information according to the expected flow of the system.

The successful result for the **Customer Session Flow** demonstrates that information recorded during customer check-in remained available throughout the customer's active session and could subsequently be used during checkout. This is important because the system's billing process depends on information collected earlier in the customer's visit. The integration therefore reduces the need for staff to manually encode the same customer and session information again during checkout.

The successful **Workspace Billing** result confirms that the selected workspace and corresponding session information were properly connected to the billing process. This supports the system's time-based billing mechanism because the workspace charge is dependent on both the type of workspace used and the duration of the customer's session.

The **POS and Inventory** integration successfully demonstrated the connection between product transactions and stock information. When products were included in completed orders, the corresponding inventory records were updated. This allows the system to maintain a closer relationship between actual sales activity and recorded product quantities.

The successful **POS and Billing** integration also confirms that food and beverage orders recorded during a customer's stay could be incorporated into the final billing process. The system was able to combine the food bill with the workspace time bill, allowing the customer to receive a consolidated total rather than requiring separate calculations for workspace use and café purchases.

The **Session and Transaction** integration demonstrates that completed customer sessions can contribute to the system's historical transaction records. This provides a continuous information flow from the

customer's initial check-in to the completion and storage of the transaction. The resulting records can then be used by other system functions without requiring the transaction information to be manually encoded again.

The successful **Sales and Analytics** integration demonstrates that completed transaction information can be utilized for sales monitoring. This allows the system to move beyond simple transaction recording by providing summarized information that can assist the Owner/Administrator in reviewing business activity.

The **Financial Monitoring** and **Daily Soft Balancing** integration results further demonstrate that financial information maintained within the system can be reviewed through the appropriate financial functions. This provides a more centralized approach to monitoring sales, payments, expenses, payables, receivables, and debt-collection information. Instead of maintaining completely separate records for each financial activity, related information can be accessed through the corresponding system modules.

Finally, the successful **Reports** integration confirms that information generated and stored by the operational modules can be retrieved for reporting purposes. This is important because the usefulness of the reporting function depends on its ability to obtain information from the underlying system records. The successful result indicates that the reporting process can utilize information produced by the system's operational and financial functions.

Overall, the integration testing showed that the major modules of the developed system were able to operate together as an interconnected platform. Information generated from one operational process could be passed to and utilized by other related modules, supporting a continuous workflow from **customer check-in, workspace usage, ordering, billing, payment, inventory updating, transaction recording, financial monitoring, and reporting**. The successful integration results also demonstrate that the system reduces the need for repetitive data entry and supports centralized management of related operational information.

Therefore, the integration testing provides evidence that the developed system functions not only as a collection of individual modules but as an **integrated operations management system** designed around the workflow of IdeaHub Co-working Space + Café.

4.7.3 User Acceptance Testing

User Acceptance Testing (UAT) was conducted to determine whether the developed **IdeaHub Co-working Space + Café Check-In and Billing with POS Integration System** met the operational requirements and expectations of the client. UAT served as the final testing stage in which the intended user was given the opportunity to directly interact with the completed system and assess whether its implemented functions were appropriate for the actual business processes of IdeaHub Co-working Space + Café.

Unlike functional testing, which focused on determining whether individual system functions produced their expected technical results, and integration testing, which examined the interaction among connected modules, UAT focused on the **acceptability and practical suitability of the developed system from the client's perspective**. The purpose was to determine whether the system could support the activities that the client expected to perform during the daily operation of the establishment.

The UAT was conducted through **direct client testing and observation rather than through a formal survey questionnaire**. The development team allowed the client to use the implemented system and test its available functions based on the actual operational requirements identified during the development of the system. The client was able to navigate through the system, perform applicable transactions, review generated information, and observe the resulting outputs.

During the testing, the client reviewed and tested the major functions of the system, including **user login and authentication, dashboard, customer check-in, workspace management, active customer sessions, POS and food ordering, checkout and billing, inventory management, sales and analytics, reports, payables, receivables, debt collection, and daily soft balancing**.

The testing of **Login and Authentication** allowed the client to verify whether authorized users could access the system using their registered credentials. This function was also reviewed in relation to the different user roles implemented in the system.

The **Dashboard** was tested to determine whether the client could obtain an overview of relevant operational and business information through the system. The client was able to access the dashboard and review the information presented by the system.

The **Customer Check-In** and **Workspace Management** functions were tested based on the process of accepting customers into the establishment. The client reviewed whether customer information could be recorded and whether the appropriate workspace could be selected and associated with the customer's session.

The **Active Customer Sessions** function was tested to determine whether currently active customers could be monitored through the system. This allows staff or authorized users to determine which customers are currently using the available workspaces and monitor their ongoing sessions.

The **POS and Food Ordering** function was tested by recording food and beverage orders through the system. The client was able to test the selection of menu items and quantities and observe how the order information was maintained within the customer transaction.

The **Check-Out and Billing** function was also tested to determine whether customer sessions could be completed and whether the system could calculate the corresponding charges. The client was able to review the process of combining workspace usage and food and beverage orders into the customer's total bill.

The **Inventory Management** function was tested to determine whether product information and stock quantities could be monitored through the system. The client reviewed whether inventory information corresponded to the products being processed through the POS.

The client also tested the **Sales and Analytics** and **Reports** functions. These functions were reviewed to determine whether recorded operational information could be presented in a useful form for monitoring and management. The client was able to access the available sales information and generated reports.

The financial functions, including **Payables, Receivables, Debt Collection, and Daily Soft Balancing**, were also included in the UAT. The client reviewed whether the system could support the recording and monitoring of financial information relevant to the establishment. These functions were particularly

important because they extend the system beyond basic customer and POS management and provide additional support for financial monitoring.

The following table summarizes the results of the User Acceptance Testing conducted with the client.

System Feature	Client Testing Result	Status
Login and Authentication	Tested and accepted by the client.	Approved
Dashboard	Tested and accepted by the client.	Approved
Customer Check-In	Tested and accepted by the client.	Approved
Workspace Management	Tested and accepted by the client.	Approved
POS and Food Ordering	Tested and accepted by the client.	Approved
Active Customer Sessions	Tested and accepted by the client.	Approved
Check-Out and Billing	Tested and accepted by the client.	Approved
Inventory Management	Tested and accepted by the client.	Approved
Sales and Analytics	Tested and accepted by the client.	Approved
Reports	Tested and accepted by the client.	Approved
Payables	Tested and accepted by the client.	Approved
Receivables	Tested and accepted by the client.	Approved
Debt Collection	Tested and accepted by the client.	Approved
Daily Soft Balancing	Tested and accepted by the	Approved

	client.	

Table 4.3 presents the User Acceptance Testing Summary

Table 4.3 presents the results of the User Acceptance Testing conducted with the client. The results show that **all identified system features were tested and approved by the client**. The approval indicates that the client was able to access and test the implemented functions and considered the tested features acceptable based on the operational requirements of IdeaHub Co-working Space + Café.

The successful acceptance of the **Customer Check-In, Workspace Management, and Active Customer Sessions** functions indicates that the system was able to support the customer-monitoring process required by the establishment. These functions allow customer information and workspace usage to be recorded and monitored within the system rather than relying solely on manual records.

The approval of the **POS and Food Ordering, Check-Out and Billing, and Inventory Management** functions indicates that the client found the implemented transaction workflow suitable for the café and coworking-space operations. The client was able to test the process from recording customer orders through the completion of the customer's transaction. The integration of these functions is important because the establishment needs to manage both workspace usage and café purchases within the same operational environment.

The approval of **Sales and Analytics and Reports** indicates that the client was able to access information generated from the system for monitoring and management purposes. These functions provide summarized information that can assist the client in reviewing business activities rather than requiring individual transaction records to be examined separately.

The approval of **Payables, Receivables, Debt Collection, and Daily Soft Balancing** further indicates that the system's financial-management functions were acceptable to the client during testing. These features provide additional support for monitoring financial activities beyond the recording of ordinary customer transactions.

Based on the client testing, the developed system received **approval for all tested features**. The client also expressed satisfaction with the implemented system and its functions. No major functional issues were identified during the client testing that would prevent the system from supporting its intended operations.

The feedback and observations provided during the testing process were considered by the development team in reviewing the system and making necessary adjustments and improvements. This allowed the team to address applicable observations before finalizing the system. The UAT therefore served not only as an acceptance procedure but also as an opportunity to verify that the implemented system remained aligned with the actual operational requirements of the client.

Overall, the results of the User Acceptance Testing indicate that the developed **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** was considered acceptable by the client for its intended operational functions. The approval of all tested features, together with the absence of major issues that would prevent intended use, provides supporting evidence that the developed system addressed the operational requirements identified for the project.

4.8 System Evaluation

The evaluation of the developed **IdeaHub Co-working Space + Café Check-In and Billing with POS Integration System** was conducted to determine whether the system addressed the operational needs, requirements, and expectations of the client. The evaluation focused on determining whether the implemented system could support the actual business processes of IdeaHub and whether the developed features were considered appropriate and acceptable by the intended user.

The evaluation process consisted of two related stages. The first stage involved an **initial client interview during the requirements-gathering stage**, while the second stage involved **direct client testing and feedback after the system had been developed and implemented**. The initial interview was used to understand the existing business processes and identify the problems and requirements that the proposed system needed to address. The subsequent testing allowed the client to determine whether the developed system and its implemented functions corresponded to the requirements identified during the initial stage.

During the initial interview, the development team gathered information about the existing operations of IdeaHub Co-working Space + Café. The discussion focused on the processes involved in customer check-in, workspace usage, customer session monitoring, food and beverage ordering, billing and payment, inventory monitoring, sales recording, financial transactions, reservations, and report preparation. The interview also helped the development team identify operational activities that could be improved through the use of a computerized and integrated system.

The information gathered from the client served as a basis for defining the system requirements and determining the modules and functions to be included in the proposed system. These requirements were subsequently translated into the implemented features of the system, including customer session management, POS and order management, billing and payment, inventory management, financial management, reservation management, sales and analytics, reporting, and administrative functions.

After the development and implementation of the system, the client was given the opportunity to directly interact with and test the implemented features. The client performed activities corresponding to the actual operational processes of IdeaHub and reviewed the resulting system outputs. This allowed the client to determine whether the developed system could support the intended workflow and whether the implemented functions were suitable for actual business operations.

The client tested the system's major functions, including **login and authentication, dashboard, customer check-in, workspace management, active customer sessions, POS and food ordering, check-out and billing, inventory management, sales and analytics, reports, payables, receivables, debt collection,**

and daily soft balancing. The client provided feedback and observations during the testing process, which were considered by the development team in reviewing and refining the system.

Based on the client's testing and feedback, the client expressed satisfaction with the developed system and approved the implemented features. The evaluation results therefore provided evidence that the system was aligned with the operational requirements identified during the requirements-gathering stage.

4.8.1 Evaluation Instrument

The evaluation used **two primary approaches: an initial client interview and direct client system testing and feedback**. These approaches were selected because the project required both an understanding of the client's existing operational requirements and an assessment of whether the completed system could address those requirements.

The **initial client interview** served as the primary information-gathering instrument during the requirements analysis stage. The development team interviewed the client to obtain information regarding the existing procedures, operational activities, challenges encountered in the current process, and expected functions of the proposed system.

The interview covered the major areas of IdeaHub's operations, including **customer check-in, workspace selection and usage, active customer monitoring, food and beverage ordering, billing and payment, inventory monitoring, sales recording, reservations, financial transactions, and report preparation**. The discussion also helped identify the information that needed to be recorded, processed, monitored, and retrieved by the proposed system.

The information obtained during the interview was analyzed by the development team and used as a basis for identifying the system requirements. These requirements guided the design and development of the system's major modules and helped determine which business processes needed to be integrated into a centralized system.

The second evaluation approach was **direct client system testing and feedback**. After the system had been implemented, the client was given access to the developed features and was allowed to perform tasks based on the actual business processes of IdeaHub. This approach allowed the client to evaluate the system through actual interaction rather than relying only on a description or demonstration provided by the development team.

During the direct testing, the client reviewed functions related to **customer check-in, workspace management, active customer sessions, time tracking, POS and food ordering, checkout and billing, inventory management, sales and analytics, reports, payables, receivables, debt collection, and daily soft balancing**.

The client's observations, feedback, and approval were then used to determine whether the developed system addressed the requirements identified during the initial interview. This process also allowed the development team to identify areas requiring adjustment before the finalization of the system.

Because the evaluation was based on **direct client testing and feedback rather than a formal survey**

questionnaire, the results were interpreted qualitatively. The evaluation therefore focused on whether the required functions were implemented, whether the client could use and test the functions, whether the functions produced the expected operational results, and whether the client accepted the implemented features.

4.8.2 Evaluation Results

The evaluation results were based on the information gathered during the initial client interview and the feedback obtained after the client tested the completed system. The initial interview established the operational requirements that the system needed to address, while the subsequent client testing provided an opportunity to verify whether the implemented features corresponded to those requirements.

The evaluation considered several areas, including **requirements compliance, functionality, usability, accuracy, efficiency, operational suitability, client satisfaction, and overall acceptance**. These areas were examined through the client's direct interaction with the system and through the feedback provided during the evaluation process.

Evaluation Area	Evaluation Result	Status
Requirements	The system features were developed based on the requirements identified during the client interview.	Met
Functionality	The implemented features performed their intended functions during client testing.	Accepted
Usability	The client was able to navigate and use the major system features.	Accepted
Accuracy	Customer, transaction, billing, and financial information were processed as intended.	Accepted
Efficiency	The system provided a more organized and automated approach to	Accepted

	daily operations.	
Operational Suitability	The system features were considered appropriate for the operational needs of IdeaHub.	Accepted
Client Satisfaction	The client expressed satisfaction with the developed system.	Satisfied
Overall Acceptance	The client approved all of the implemented system features.	Approved

Table 4.4 presents the Client Evaluation Results

Table 4.4 presents the results of the client evaluation. The results indicate that the developed system met the requirements identified during the initial requirements-gathering process and was accepted by the client after direct testing.

For **requirements compliance**, the implemented features were based on the processes and requirements identified during the initial client interview. The developed system therefore incorporated functions that corresponded to the operational activities identified by the client.

For **functionality**, the client was able to test the implemented features and verify their intended operation. This result is consistent with the functional and integration testing presented in the preceding sections, where the major functions and interactions among the modules were successfully tested.

For **usability**, the client was able to navigate and interact with the major features of the system during the testing process. Although no formal usability questionnaire or numerical usability scale was used, the client's successful interaction with the implemented modules provided qualitative evidence that the system could be used to perform the intended operational activities.

For **accuracy**, the evaluation considered the processing of customer information, session information, transactions, billing, inventory, and financial records. The client accepted the resulting system behavior and did not identify any major issue that would prevent the system from supporting its intended operations.

For **efficiency**, the developed system provided a centralized and automated approach to several activities that would otherwise require separate manual recording or calculation. Customer sessions, orders, billing, inventory, transactions, and reports are maintained within the same system, allowing related information to be accessed and processed through interconnected modules.

For **operational suitability**, the client considered the implemented features appropriate for the operational requirements of IdeaHub. The system supports both the coworking-space and café aspects of the business

by integrating customer session management with POS, billing, inventory, financial monitoring, and reporting functions.

The **client satisfaction** result was based on the client's direct feedback after testing the implemented features. The client expressed satisfaction with the developed system and its available functions. The **overall acceptance** result was based on the client's approval of all the implemented features following the testing process.

The client tested and reviewed the following major modules:

- **Login and Authentication**
- **Dashboard**
- **Customer Check-In**
- **Workspace Management**
- **POS and Food Ordering**
- **Active Customer Sessions**
- **Check-Out and Billing**
- **Inventory Management**
- **Sales and Analytics**
- **Reports**
- **Payables**
- **Receivables**
- **Debt Collection**
- **Daily Soft Balancing**

The results indicate that the implemented features successfully addressed the operational requirements identified during the initial interview and were accepted by the client during the final testing. The evaluation also demonstrated that the system provides an integrated environment in which customer, transaction, inventory, financial, and reporting information can be recorded and managed through connected modules.

Overall, the evaluation results indicate that the developed **IdeaHub Co-working Space + Café Check-In and Billing with POS Integration System** was considered suitable and acceptable for its intended operational purpose. The combination of the initial requirements-gathering interview and subsequent direct client testing provided a basis for determining that the implemented system corresponded to the client's identified needs. The client's approval of all implemented features further indicates that the system successfully addressed the major operational requirements within the scope of the study.

4.9 Results and Discussion

This section presents and discusses the results obtained from the development, testing, and client evaluation of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. The discussion focuses on how the implemented modules addressed the operational requirements identified during the requirements-gathering stage.

The results were based on the actual functionality of the developed system, the outcomes of functional and integration testing, and the direct testing and approval provided by the client. The developed system integrated customer session management, workspace monitoring, POS and billing, inventory, reservations, menu management, financial management, reports, analytics, and administrative functions within a centralized web-based platform.

4.9.1 Customer Check-In and Session Management

The developed **Customer Check-In and Session Management** module provided a digital method for recording and monitoring customers using IdeaHub's coworking spaces. Under the previous process, customer information and workspace usage were manually recorded, requiring staff to monitor customer arrival and departure times separately.

Through the developed module, staff can record customer information such as the **customer name, school, course, purpose, selected workspace, and time-in**. Once the check-in process is completed, the system creates an active customer session and begins monitoring the duration of the customer's stay. The **Active Customer Sessions** interface displays the customer's information, selected workspace, time-in, elapsed duration, and current workspace charge. This allows staff to monitor ongoing sessions from a centralized interface.

The system supports the three workspace types offered by IdeaHub: **Regular Lounge, Premium Lounge, and Boardroom**. Regular and Premium Lounge interfaces also provide information regarding current occupancy and available seats. For walk-in Boardroom customers, the system similarly records and monitors the active session based on the corresponding workspace configuration.

The implementation of digital session monitoring reduced dependence on handwritten customer logs and provided staff with a structured method for accessing active customer information. During client testing, the customer check-in and session management functions were tested and approved.

4.9.2 POS, Ordering, and Billing

The developed **Point-of-Sale (POS) and Billing** modules integrated food and beverage transactions with customer workspace sessions. This addressed the need to manage workspace charges and café purchases within the same transaction instead of maintaining them as separate records.

Through the POS interface, staff can select an active customer session, choose menu items and quantities, and associate the resulting order with the corresponding customer. The system maintains the customer's orders throughout the active session and allows staff to review the items before checkout.

During checkout, the system determines the workspace charge based on the recorded session duration and applicable workspace rate. Food and beverage charges associated with the session are then included in the customer's final bill. The billing computation follows:

Total Bill = Time Bill + Food Bill

The system also records the selected payment method and transaction information when the customer completes payment. This integration provides a single transaction record containing both workspace usage and café-related charges.

4.9.3 Checkout Records

The **Checkout Records** module maintains information regarding completed customer transactions. It provides administrators with access to historical checkout information, including the customer, workspace, time-in, time-out, duration, time bill, food bill, payment method, total amount, amount tendered, and change.

The module also provides filtering options that allow records to be searched according to customer, date range, and payment method. Payment summaries for methods such as **Cash and GCash** are presented together with the overall transaction total for the selected records.

By maintaining completed transactions within the system, administrators can retrieve previous customer transactions without relying on separate handwritten records. These records also provide source information for other system functions, including daily balancing, reporting, and analytics.

4.9.4 Inventory Management

The developed **Inventory Management** module provides a centralized interface for maintaining and monitoring ingredients and available stock. Authorized users can add inventory items, update quantities, search records, review stock history, and identify items according to their current stock condition.

The module displays stock information together with defined restock thresholds. Items can be categorized according to conditions such as **Low Stock, Warning, Normal, and Out of Stock**, allowing staff to identify inventory items that may require replenishment.

Inventory information is also connected with the ordering process. When applicable menu items are processed through the POS module, corresponding inventory quantities can be updated based on the

transaction. This provides a more organized approach to maintaining stock information than manually updating inventory records after transactions.

4.9.5 Menu Management

The **Menu Management** module allows authorized users to maintain the food and beverage items offered through the POS system. Menu items can be organized according to categories such as **Main Dish, Snacks, and Beverages**, enabling staff to locate and manage products according to their classification.

Authorized users can add new menu items and categories and maintain the corresponding item information required by the POS module. Search and category-filtering functions are also provided to assist users in locating menu records.

The integration of Menu Management with the POS module ensures that available menu information is maintained through a centralized source. Changes made to menu records can therefore be reflected in the ordering functions used during customer transactions.

4.9.6 Space Booking and Reservation Management

The developed **Space Booking and Reservation Management** module supports advance reservations for IdeaHub's reservable spaces, including the **Boardroom and whole-hub rental arrangements**. The module allows staff to create reservations and review scheduled bookings according to the selected date and time.

The interface displays the availability of time slots and distinguishes between **available, booked, and in-session** periods. Staff can record reservation information and manage changes to existing bookings when necessary. The system also accommodates situations in which a reserved customer arrives later than the originally scheduled time or requires an extension after the session has started.

This functionality provides IdeaHub with a centralized mechanism for reviewing workspace schedules and reservation availability, reducing dependence on manually maintained booking records.

4.9.7 Expenses Management

The **Expenses Management** module provides a structured method for recording expenses incurred during IdeaHub's operations. Authorized users can log an expense together with its category, description, amount, date, and the user responsible for recording the transaction.

Expense records can be reviewed according to a selected date range, while the interface presents the total expenses recorded for that period. This information is subsequently used by the financial monitoring and daily balancing functions of the system.

The module provides IdeaHub with a centralized record of operational expenses and allows these transactions to be incorporated into the broader financial information maintained by the system.

4.9.8 Receivables and Debt Monitoring

The **Receivables and Debt Monitoring** module was implemented to maintain records of amounts expected to be collected from customers. The module displays customer information, contact details, outstanding balances, due dates, payment status, and related transaction actions.

Authorized users can create new receivable records and monitor whether outstanding balances remain unpaid, are partially paid, or have been completely settled. This allows customer debts and subsequent payments to be maintained within the same system rather than through separate records.

The module therefore provides a centralized mechanism for monitoring outstanding customer balances and collection information. The financial management features, including receivables and debt monitoring, were reviewed and approved during client testing.

4.9.9 Payables Management

The **Payables Management** module provides IdeaHub with a centralized method for recording and monitoring financial obligations to suppliers and other service providers. Each payable record may contain the supplier, item or service description, amount owed, amount partially paid, incurred date, due date, and payment status.

The module distinguishes between **Unpaid, Partially Paid, and Paid** obligations and provides summary information regarding the total unpaid amount and number of outstanding payables. Authorized users can create payable records and update their status as payments are made.

This functionality allows management to review outstanding business obligations through a centralized interface and maintain the payment status of each recorded payable.

4.9.10 Daily Sales and Financial Balancing

The **Daily Sales Balance** module consolidates recorded financial information to assist management in reviewing daily business activities. The interface presents information such as collected receivables, current cash flow, outstanding amounts to be collected, projected cash flow, total revenue, total expenses, and net balance.

Payment information can also be grouped according to the recorded payment method, allowing management to review collections from **Cash, GCash, and other configured payment channels**. Date filters provide the ability to examine financial records for selected periods.

The module also supports report generation and export functions, allowing financial summaries to be generated from stored system information. The implementation of this module provides a structured method for reviewing and reconciling the financial activities recorded within the system.

4.9.11 Sales, Business Intelligence, and Analytics

The developed **Business Intelligence and Analytics** module provides management with summarized information derived from recorded system transactions. Administrators can specify a date range and

generate analytical views of available business information.

The module presents selected information through graphical representations, including the **Daily Revenue Flow**, allowing management to examine recorded revenue across different dates. Other available analytics can be used to support the review of sales activity, food-item performance, customer activity, and related operational information.

The analytics functionality complements the system's reporting features by transforming stored transaction information into organized visual summaries. This provides management with an additional means of reviewing business records without manually consolidating individual transactions.

4.9.12 Dashboard and Administrative Management

The **Dashboard and Administrative Management** functions provide administrators with centralized access to major operational and management features. The Admin Panel displays information related to customers, staff accounts, active users, and other operational indicators.

The administrative interface also supports **staff and user management**, allowing authorized administrators to maintain user accounts and assigned roles. Role-based access is used to distinguish administrative functions from operational functions available to staff members.

The centralized navigation interface provides access to customer sessions, checkout records, space bookings, inventory, menu management, financial modules, reports, analytics, and system-related functions. This allows authorized users to access the modules required for their assigned responsibilities from a common system interface.

4.9.13 Overall System Usability and Client Acceptance

The usability and suitability of the developed system were assessed through direct interaction and testing by the client. No formal quantitative usability survey was administered. Instead, the client navigated and tested the implemented modules according to the operational processes performed at IdeaHub.

The client was able to review the system's interfaces and functions, including customer session management, POS and billing, checkout records, workspace management, reservations, inventory, menu management, expenses, receivables, payables, daily balancing, analytics, reporting, and administrative functions. The client subsequently approved the implemented features.

Overall, the results of functional testing, integration testing, and direct client evaluation indicate that the developed system fulfilled the identified functional requirements within the scope of the study. The system provided a centralized and organized platform for maintaining IdeaHub's operational and financial information, while the client's acceptance confirmed that the implemented functionalities corresponded to the requirements established during development.

4.10 Comparison of Existing and Developed System

The **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** was developed to address the operational limitations identified during the requirements-gathering stage. Prior to the development of the proposed system, several business activities relied on manual recording, separate records, and manual computation. These activities included customer check-in, workspace assignment, monitoring of customer sessions, food and beverage ordering, billing, inventory monitoring, transaction recording, sales monitoring, financial record management, and report preparation.

The existing approach required information to be recorded and monitored separately across different operational activities. As a result, related information such as customer session details, food orders, workspace usage, payments, inventory, and financial records was not managed through a single integrated platform. Activities that required information from multiple records could therefore require additional checking, computation, or consolidation by staff.

The developed system addresses these limitations by integrating the major operational and financial activities of IdeaHub into a **centralized web-based management system**. Customer session information, workspace usage, POS orders, billing, transactions, inventory, financial records, sales information, and reports can be maintained within interconnected modules supported by a centralized database.

The comparison between the existing and developed systems focuses on the changes introduced by the proposed system in terms of **data recording, transaction processing, information integration, monitoring, financial management, reporting, data storage, data retrieval, and access control**.

Business Process	Existing System	Developed System
Customer Check-In	Customer information is manually recorded.	Customer information is digitally recorded through the check-in module.
Workspace Selection	Workspace assignment is manually recorded and monitored.	Workspace type is selected and associated with the customer's session.
Time Tracking	Customer usage time is manually monitored.	Customer session duration is automatically tracked from time-in to time-out.
Customer Session Monitoring	Active customers are monitored using manual records.	Active customer sessions are displayed through the system.
Food and Beverage	Orders are manually	Orders are processed

Ordering	recorded.	through the integrated POS module.
Inventory Monitoring	Inventory quantities are manually monitored and updated.	Inventory records are digitally maintained and updated through transactions.
Billing	Charges are manually calculated.	Workspace and food charges are automatically combined and calculated.
Transaction Recording	Transactions may be recorded using separate manual records.	Completed transactions are stored in a centralized database.
Sales Monitoring	Sales information is manually compiled.	Sales information is automatically recorded and presented through the system.
Analytics	Limited or manual analysis of sales information.	Sales information can be presented through dashboards, charts, and analytics.
Reports	Reports require manual compilation.	Reports can be generated from stored system records.
Payables	Financial obligations are manually recorded and monitored.	Payable records are digitally recorded and monitored.
Receivables	Outstanding amounts are manually tracked.	Receivable records and outstanding balances are centrally monitored.
Debt Collection	Debt and collection records are manually maintained.	Outstanding debts and collection information are digitally tracked.
Daily Soft Balancing	Daily financial activities are manually reviewed and reconciled.	Financial activities can be reviewed and reconciled through the Daily Soft Balancing module.

Data Storage	Records are maintained through manual or separate records.	Information is stored in a centralized database.
Data Retrieval	Searching for records may require checking multiple records.	Records can be accessed through the corresponding system modules.
Security	Manual records have limited access control.	User authentication and role-based access help restrict system access.

Table 4.5 presents the Comparison of Existing and Developed System

Table 4.5 presents the comparison between the existing operational processes of IdeaHub and the functions provided by the developed system. The comparison shows that the major difference between the two approaches is the **level of integration and automation applied to the business processes**.

For **customer check-in and workspace management**, the existing process relies on manual recording and monitoring. The developed system provides dedicated modules where customer information and workspace selection can be digitally recorded and associated with an active customer session. This creates a more structured record of the customer's entry into the establishment and selected workspace.

The difference is also evident in **time tracking and customer session monitoring**. Under the existing process, monitoring customer usage time requires manual observation and recording. The developed system maintains the customer's session through recorded time-in and time-out information. Active sessions can also be viewed through the system, providing staff with a centralized view of customers currently using the establishment.

For **food and beverage ordering**, the existing process requires orders to be recorded separately. The developed system introduces an integrated POS module where menu items and quantities can be associated with the customer's active session. This allows food and beverage purchases to remain connected to the customer's overall transaction.

The integration between **POS and Inventory Management** provides another improvement. In the existing process, inventory quantities may need to be monitored and updated separately from sales activities. In the developed system, applicable transactions can be used to update the corresponding inventory records. This provides a closer connection between product sales and available stock information.

The most significant operational difference can be observed in **billing**. Under the existing process, workspace charges and food purchases require manual computation or consolidation. The developed system combines the customer's workspace usage information and recorded food and beverage orders during checkout. This provides a more structured process for determining the customer's total amount due.

For **transaction recording and sales monitoring**, the existing approach may require information to be maintained across separate records. The developed system stores completed transactions in a centralized database. These transaction records can subsequently be used by sales and analytics functions, reducing the need to manually compile sales information from separate sources.

The **analytics and reporting** functions also provide a significant difference. The existing process requires manual compilation and review of information before reports can be prepared. In the developed system, stored transaction and operational information can be retrieved by the reporting and analytics modules. This allows authorized users to view summarized information and generate reports based on available system records.

The developed system also expands support for **financial management** through the inclusion of payables, receivables, debt collection, and Daily Soft Balancing. These functions provide dedicated areas for recording and monitoring financial activities. Rather than relying entirely on separate manual records, the system allows applicable financial information to be maintained within the same environment as the establishment's operational records.

For **data storage**, the existing process uses manual or separate records, while the developed system uses a centralized database. Centralized storage allows related information to be maintained within the same system and accessed by authorized modules. For example, customer session information can be associated with orders and billing records, while completed transaction information can subsequently contribute to sales and reporting functions.

The difference in **data retrieval** is also important. In the existing process, locating historical information may require checking multiple records or sources. In the developed system, information can be retrieved through the appropriate system modules. This provides a more organized method of accessing customer, transaction, inventory, financial, and reporting information.

Finally, the developed system introduces **user authentication and role-based access control**. Rather than relying only on physical or manual restrictions for accessing records, the system requires users to authenticate before accessing the application. The assigned user role determines the functions available to the user. This provides a more structured approach to controlling access to operational and administrative information.

Overall, the comparison demonstrates that the developed system improves the organization and integration of IdeaHub's operational processes by bringing related activities into a centralized web-based platform. The system connects **customer check-in, workspace management, time tracking, POS ordering, billing, inventory, transactions, financial management, sales, analytics, reservations, and reporting** within a single environment.

The developed system therefore does not simply replace individual manual records with digital records. Its primary improvement is the **integration of related business processes**. Information recorded during one stage of an operation can be used by another related stage, such as using customer session information during billing, using POS transactions to update inventory, and using completed transactions for sales analysis and reporting.

This comparison supports the purpose of the proposed system as an integrated operations management solution for IdeaHub Co-working Space + Café. While manual procedures may still be necessary for certain activities outside the system's scope, the developed platform provides a more centralized, organized, and automated approach to the core operational processes included in the study.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

This chapter presents the conclusion and recommendations based on the development, implementation, testing, and evaluation of the **IdeaHub Co-working Space + Café Check-in and Billing with POS Integration System**. It synthesizes the findings presented in the previous chapter and determines how effectively the developed system addressed the operational and financial management needs of IdeaHub Co-working Space + Café.

The chapter discusses the major findings obtained from the development and testing of the system, presents the conclusions drawn from the results, and provides recommendations for future improvements. It also discusses the impact of the developed system on the daily operations of IdeaHub and identifies possible areas for further development and research.

5.2 Summary of Findings

The development and evaluation of the IdeaHub Co-working Space + Café Check-in and Billing with POS Integration System resulted in the implementation of an integrated system designed to support the operational and financial requirements of the business. The system was developed based on the problems and requirements identified during the initial client interview and consultations.

The developed system successfully implemented major operational features, including customer check-in, workspace management, time tracking, active customer session monitoring, POS and food ordering, checkout and billing, inventory management, sales monitoring, analytics, and report generation. These features provide a centralized platform for managing the primary activities involved in IdeaHub's daily operations.

In addition to the operational functions, the system implemented financial management features consisting of Payables, Receivables, Debt Collection, and Daily Soft Balancing. These modules provide the administrator with tools for recording, monitoring, and reviewing the financial activities of the business.

The system underwent functional testing to verify that individual features performed their intended functions. The testing results showed that the implemented features operated successfully. Integration testing was also performed to verify that related modules could properly exchange and utilize information, particularly the integration between customer sessions, POS, inventory, billing, transactions, sales, and financial modules.

For User Acceptance Testing, the client directly tested the developed system rather than completing a formal survey questionnaire. The client reviewed and tested the implemented features based on the actual business processes of IdeaHub. The client expressed satisfaction with the developed system and approved all of the implemented features.

The system evaluation was based on the initial client interview, direct system testing, and client feedback. The results showed that the system addressed the requirements identified during the requirements-gathering stage and provided functions suitable for the client's operational needs.

Overall, the findings demonstrate that the developed system provides a more organized, centralized, and automated approach to IdeaHub's operational and financial management. Compared with the previous manual processes, the developed system reduces manual recording and computation, improves accessibility of business records, supports more accurate transaction processing, and provides the client with centralized tools for monitoring operations and financial activities.

5.3 Conclusion

The **IdeaHub Co-working Space + Café Check-in and Billing with POS Integration System** has proven to be a functional and suitable solution for improving the operational and financial management processes of the business. The system successfully transformed several manual processes into centralized digital workflows, allowing IdeaHub to manage customer sessions, workspace usage, food and beverage orders, billing, inventory, sales, and financial records through an integrated platform.

The implementation of automated customer check-in and time tracking improved the management of workspace usage by providing digital records of customer information and session duration. The integration of POS and billing further improved the transaction process by allowing workspace charges and food and beverage orders to be combined into a single customer transaction.

The system also addressed the need for better inventory and sales monitoring by maintaining centralized records and providing administrators with information that can be used to monitor business activities. The addition of Payables, Receivables, Debt Collection, and Daily Soft Balancing further expanded the system's ability to support the financial management requirements of IdeaHub.

The results of functional and integration testing demonstrated that the implemented modules operated as intended and were able to work together as part of a unified system. Furthermore, the direct testing conducted with the client confirmed that the system met the client's expectations. The client's satisfaction and approval of all implemented features provide evidence that the developed system was acceptable and suitable for the intended operations of IdeaHub.

Therefore, the study concludes that the developed system successfully addressed the operational challenges identified during the initial client interview. It provides a more organized, centralized, automated, and efficient approach to managing IdeaHub's daily operations and financial activities.

5.4 Recommendations

To ensure the continued effectiveness, reliability, and usefulness of the IdeaHub system, the following recommendations are proposed.

5.4.1 System Improvements

Based on the results of the system development, testing, and evaluation, several improvements may be considered for future versions of the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café**. Although the developed system successfully implemented the major operational and financial functions within the scope of the study, further enhancements may improve its functionality, efficiency, security, and compatibility with the actual hardware used in daily business operations. Future improvements may focus on strengthening inventory monitoring, financial management, reporting, security, analytics, notifications, and integration with physical POS and cashier equipment. These improvements are intended to extend the capabilities of the system and provide additional support as the operational requirements of IdeaHub increase.

The following improvements are recommended for future development:

- **Enhance inventory notifications.** Future versions may include automated notifications for low-stock products and other inventory conditions to help staff identify items that require immediate attention.
- **Improve financial monitoring.** The Payables, Receivables, Debt Collection, and Daily Soft Balancing modules may be expanded with additional financial summaries, filtering options, and reconciliation features.
- **Enhance report generation.** The reporting module may be improved by supporting additional export formats such as **PDF, Excel, and CSV**, making reports easier to share, print, and process.
- **Strengthen system security.** Additional security features, such as stronger password policies, session management, and two-factor authentication, may be considered to provide additional protection for operational and financial information.
- **Enhance dashboard analytics.** Future versions may provide more advanced visualizations and customizable dashboards to allow the administrator to examine sales, workspace usage, inventory, and financial performance in greater detail.
- **Improve automated notifications.** The system may be enhanced with notifications for outstanding receivables, upcoming payable due dates, pending debt collections, and other important business activities.
- **Integrate the system with POS and cashier hardware.** Future versions may integrate the system with physical cashier equipment such as a **cash drawer, barcode scanner, receipt printer, and customer display**, depending on the operational requirements of IdeaHub. Integration with a receipt printer would allow completed transactions to be printed directly after payment, while a cash drawer

could provide a more organized method of handling cash transactions.

5.4.2 Future Research Directions

The results of the study also provide opportunities for further research and development. While the developed system addressed the identified operational requirements of IdeaHub Co-working Space + Café, future researchers may explore additional technologies, integrations, and research methods that could further improve the system's capabilities. Future studies may focus on the use of accumulated operational data, mobile technologies, digital payment systems, accounting integration, and hardware-based POS solutions. These directions may help determine how the system can be expanded beyond its current scope and adapted to changing business requirements.

● **Explore predictive analytics and machine learning.** Future researchers may investigate the use of predictive analytics or machine learning techniques to analyze accumulated historical data and forecast **sales, inventory requirements, customer demand, and workspace utilization**. Such capabilities could assist the administrator in making more informed operational and business decisions.

● **Conduct a longitudinal study.** A longitudinal study may be conducted to examine the system's long-term effects on **operational efficiency, transaction accuracy, inventory management, customer service, and financial performance**. Monitoring the system over an extended period could provide more comprehensive evidence of its benefits and identify operational issues that may not be observable during the initial implementation.

● **Develop a mobile application.** Future researchers may explore the development of a **mobile application** that would allow administrators and staff to access selected system functions through smartphones and tablets. Mobile access could be particularly useful for monitoring active customer sessions, checking sales information, reviewing inventory, and performing other administrative tasks while away from a desktop computer.

● **Integrate online payment platforms and digital wallets.** Future studies may investigate the integration of the system with **online payment platforms and digital wallets** to provide customers with additional payment options. Such integration could allow digital payments to be recorded automatically within the customer's transaction and reduce the need for manual encoding of payment information.

● **Integrate accounting software.** Researchers may investigate integration with existing **accounting software** to further automate the transfer and management of financial records. This could reduce repetitive encoding between systems and provide a more continuous connection between operational transactions and formal accounting processes.

● **Investigate POS hardware integration.** Future researchers may explore the integration of the system with physical **POS and cashier hardware**, including receipt printers, cash drawers, barcode scanners, and customer displays. The study could examine the compatibility, reliability, transaction speed, and usability of these devices when connected to the developed POS module. Researchers may also evaluate whether hardware integration can reduce manual encoding, improve receipt processing, and provide a more efficient cashier workflow for café operations.

● **Evaluate hardware and software integration in actual business conditions.** Future research may examine the performance of the integrated system under actual operating conditions, particularly during periods of high customer volume. This could help determine whether the system and connected POS hardware can maintain reliable transaction processing when multiple customer sessions, orders, payments, and inventory updates are being handled simultaneously.

Overall, these future research directions may provide opportunities to extend the current system beyond its existing functionality. Further investigation of predictive technologies, mobile access, digital payments, accounting integration, and POS hardware could contribute to the development of a more comprehensive and scalable **operations management system for coworking spaces and café-based businesses**.

5.4.3 Implementation Strategies

The successful implementation of the developed system does not depend solely on the availability of the software but also on how effectively it is adopted and maintained by the users and management of IdeaHub Co-working Space + Café. Proper implementation strategies are therefore necessary to ensure that the system can be used consistently, that the information stored in the system remains reliable, and that operational and financial records are adequately protected. The following strategies are recommended to support the effective adoption, maintenance, and continued use of the developed system.

● **Provide proper staff training and orientation.** IdeaHub management should provide appropriate training and orientation to staff members before the full implementation of the system. The training should cover commonly performed activities such as customer check-in, workspace management, POS ordering, checkout and billing, inventory updates, and other functions relevant to each user's assigned role. Staff should also be given an opportunity to practice using the system before it is fully adopted for daily operations. This can help reduce user errors and allow staff members to become familiar with the system's workflow.

● **Maintain accurate and complete data entry.** Staff should consistently enter **accurate, complete, and updated information** into the system. Customer information, workspace sessions, orders, inventory quantities, payments, and financial records should be properly recorded to maintain the reliability of the system's outputs. Since sales reports, analytics, inventory information, and financial monitoring depend on the data stored in the system, inaccurate or incomplete entries may affect the accuracy of the information presented to management.

● **Perform regular database backups.** Regular database backups should be performed to protect important operational and financial information from accidental deletion, hardware failure, software problems, or other unexpected incidents. Backup procedures should be scheduled according to the volume and importance of the data being generated. Backup copies should also be stored in a secure location and periodically checked to ensure that the data can be restored when necessary.

● **Conduct regular system maintenance and updates.** The system should undergo regular maintenance to identify and address errors, performance issues, security concerns, and compatibility problems. Software dependencies, database components, and other technologies used by the system should be reviewed and updated when necessary. Maintenance should preferably be performed during an appropriate period to minimize disruption to the daily operations of IdeaHub.

● **Apply appropriate user access controls.** Access to sensitive financial, administrative, and management functions should be limited to **authorized personnel** through appropriate user roles and permissions. Staff members should only be given access to the functions necessary for their assigned responsibilities, while administrative functions should remain available only to authorized management personnel. User accounts should also be reviewed periodically, particularly when staff responsibilities change or when an employee is no longer authorized to access the system.

● **Establish a system usage and support procedure.** IdeaHub management should establish a simple procedure for reporting system problems, user concerns, and operational issues encountered during daily use. Designated personnel should be responsible for documenting and coordinating these concerns so that technical problems can be addressed promptly. This can help prevent recurring issues from affecting the continued operation of the system.

● **Monitor system performance after implementation.** After the system is deployed for regular use, management should periodically review its performance and usage. Areas such as transaction processing, inventory updates, billing, report generation, and system availability may be monitored to identify functions that require improvement. Feedback from staff members may also be collected informally to identify difficulties that were not observed during the initial testing period.

● **Implement a transition period before full adoption.** If the system will replace existing manual procedures, IdeaHub may consider a short transition period in which the new system is used alongside the existing process. This can provide staff with time to become familiar with the system and allow management to verify that important records are being properly transferred and maintained. Once the system has demonstrated stable operation, the organization may gradually reduce its dependence on the previous manual process.

Overall, these implementation strategies are intended to support the **successful adoption, reliable operation, and long-term sustainability** of the developed system. Proper training, accurate data entry, regular backups, system maintenance, controlled access, and continuous monitoring can help ensure that the system remains useful to IdeaHub beyond the initial implementation and continues to support its operational and management requirements.

5.5 Impact of the Study

The developed **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** contributes to the practical application of Information Technology in the digital management of small-business operations. The study demonstrates how a customized information system can be designed and implemented to address operational challenges involving customer management, workspace utilization, time-based billing, point-of-sale transactions, inventory monitoring, sales tracking, financial management, and reporting. By integrating these activities into a centralized platform, the study demonstrates the potential of information technology to improve the organization and management of business processes.

For **IdeaHub Co-working Space + Café**, the developed system provides a centralized platform for managing its major operational and financial activities. The integration of customer check-in, workspace management, active session monitoring, POS ordering, billing, inventory, sales, reporting, and financial-management functions provides the establishment with a more organized method of managing its daily operations. Instead of relying primarily on separate manual records and calculations, related information can be recorded, processed, stored, and retrieved through interconnected system modules. This can support more consistent record keeping and provide management with easier access to important business information.

For **staff members**, the system can reduce the amount of repetitive manual work involved in daily operations. Functions such as digital customer registration, workspace session monitoring, POS ordering, automated billing computation, and inventory updating provide staff with structured processes for handling customer transactions. By reducing the need to repeatedly record or calculate related information manually, the system can allow staff to spend more time attending to customers and performing other operational responsibilities. The system also provides a consistent workflow that can help staff follow the required procedures when processing customer sessions and transactions.

For **management and the owner**, the system provides improved access to operational and financial information needed for monitoring the business. Customer sessions, workspace usage, sales transactions, inventory information, payables, receivables, debt collection records, and daily financial activities can be accessed through their respective modules. The availability of centralized information can help management review business activities more efficiently and support decision-making based on recorded operational data. Sales and analytics functions can also provide summarized information that may assist management in identifying business trends and reviewing the performance of the establishment.

For **customers**, the system may provide a more organized transaction experience through improved handling of customer check-in, workspace usage, food and beverage ordering, and billing. The integration of workspace charges and food purchases into the billing process can provide a more consistent method of determining the customer's total amount due. More organized transaction processing may also help reduce delays caused by repetitive manual recording and computation.

For the **field of Information Technology**, the study demonstrates the practical application of software development concepts in addressing the requirements of an actual business establishment. The system

integrates technologies and concepts such as **web-based application development, relational database management, centralized data storage, automated transaction processing, role-based access control, POS integration, and data visualization** into a single business-oriented solution. The study therefore demonstrates how information technology can be applied not only to individual business functions but also to the integration of multiple operational processes within a small-business environment.

For **future researchers and developers**, the study may serve as a reference for the development of similar information systems for coworking spaces, cafés, small service businesses, or other establishments that require integrated customer, transaction, inventory, billing, and financial-management functions. The system's architecture, implemented modules, testing procedures, and identified areas for improvement may provide a foundation for future studies involving mobile applications, digital payments, accounting integration, POS hardware, predictive analytics, and other technologies.

The study also demonstrates the potential contribution of **digital transformation to small businesses**. Many small establishments may still rely on manual procedures because of limited resources, simple operational structures, or the absence of systems specifically designed for their needs. The developed IdeaHub system demonstrates that a customized information system can be designed around the actual workflow of a small business rather than requiring the business to completely adapt its operations to a generic system.

Although the developed system has limitations and opportunities for further improvement, its successful implementation, testing, and client approval demonstrate its practical value within the scope of the study. Overall, the study shows that a customized and integrated information system can contribute to the **organization, automation, monitoring, and management of small-business operations**, particularly for businesses such as IdeaHub Co-working Space + Café that combine workspace services with café and point-of-sale activities.

5.6 Summary

In summary, the **Check-In and Billing with POS Integration: A Smart Operations Management System for IdeaHub Co-working Space + Café** was successfully developed to address the operational and financial management requirements identified during the study. The project applied software engineering and information technology concepts to create a centralized web-based system designed around the actual workflow of IdeaHub Co-working Space + Café.

The developed system integrated the major operational processes of the establishment, including **customer check-in, workspace management, active customer session monitoring, time tracking, POS and food ordering, checkout and billing, inventory management, sales and analytics, reporting, payables, receivables, debt collection, and daily soft balancing**. By connecting these functions within a centralized system, the project provided an organized approach to recording, processing, storing, and retrieving operational and financial information.

The system underwent **functional testing** to verify whether individual features performed their intended functions and **integration testing** to determine whether the different modules could properly communicate and exchange information. The results of both testing activities showed that the identified test cases successfully produced their expected results.

The system was also subjected to **User Acceptance Testing**, during which the client directly tested the implemented features based on the actual operational activities of IdeaHub. The client reviewed the major system functions and expressed satisfaction with the developed system. All implemented features were approved by the client, indicating that the system was considered acceptable for its intended operational requirements.

The system evaluation further showed that the implemented features addressed the requirements identified during the initial client interview. The developed system provided a more centralized and organized approach to managing customer sessions, transactions, inventory, sales, reporting, and financial information compared with the existing processes. It also demonstrated the potential of integrating multiple business activities into a single platform rather than managing them through separate manual records and procedures.

The study further identified several opportunities for future improvement, including **enhanced inventory notifications, expanded financial monitoring, additional report formats, stronger security features, improved dashboard analytics, automated notifications, POS and cashier hardware integration, mobile access, digital payment integration, accounting software integration, and predictive analytics**. These improvements may further expand the capabilities of the system as the operational requirements of IdeaHub develop.

Overall, the study successfully achieved its purpose of developing a **practical and integrated operations management system for IdeaHub Co-working Space + Café**. The completed system demonstrates how customized information technology solutions can be applied to support the digital transformation of small-business operations. Through the integration of customer management, workspace services, POS transactions, billing, inventory, sales, reporting, and financial management, the system provides a

foundation for more organized business operations while also allowing opportunities for continued development and future enhancement.